Lecture notes. Lecture 13

Learning quantum states and Hamiltonians from measurements
tomography, classical shadows, neural networks and neural differential equations

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These notes are written for bachelor students of physics, mathematics and computer science. A quantum device returns nothing but bit strings, one per run, and every run destroys the state it measured. From many such bit strings, recorded in different measurement bases, we ask how much can be learned about the device itself. We first reconstruct the full state that the device prepares, a task called quantum state tomography, and find that its cost grows exponentially with the number of qubits. Classical shadows give up the full state and predict many local properties from one set of randomised measurements at a cost that does not depend on the size of the system. A neural network trained on simulated data corrects the systematic error of tomography with few measurements, following the attention-based method of Ref. [1]. Finally we learn the law that moves the states, the Hamiltonian of the device, by fitting a differentiable simulation of the experiment to the bit strings, and we extend the fit with neural differential equations when the landscape of the loss becomes rugged. All numerical results come from simulated measurement records with fixed random seeds. Each result states whether it is our simulation or a published result, and which quantities only a simulation can compute.

Prerequisites

Qubits and registers of qubits as unit vectors of complex amplitudes, with qubit 0 as the most significant bit of the index and $|0\rangle$ as the eigenstate of $\hat{Z}$ with eigenvalue $+1$. Pauli matrices, Pauli strings, the Hadamard gate $\hat{H}$ and the phase gate $\hat{S}$. The Schrödinger equation and time evolution $e^{-i\hat{\mathcal{H}}t}$ with $\hbar = 1$. Measurements in the computational basis and estimates from a finite number of shots with their standard errors. Gradient descent, the Adam optimiser and automatic differentiation. Elementary probability, the multinomial distribution and the logarithm of a likelihood. Neural networks as parametrised functions trained on a loss. The density matrix and the Lindblad equation are introduced where they are needed.

The same objects carry different names in physics, mathematics and computer science. Table 1 collects the names used in these notes.

physics	mathematics	computer science
density matrix $\hat{\rho}$	positive semidefinite matrix of unit trace	model of a data source
measurement setting	orthonormal basis, set of projectors	query, readout configuration
shots, bit strings	independent samples of a categorical distribution	data set
state tomography	inverse problem for a linear map	supervised reconstruction
estimate $\tilde{\rho}$	estimator, a function of the data	model prediction
fidelity, trace distance	similarity and distance of matrices	evaluation metric
classical shadow	unbiased random matrix	sketch of the data
Cholesky factor C	triangular factor of a positive matrix	unconstrained output of a network
Hamiltonian learning	parameter estimation for a matrix exponential	system identification
negative log-likelihood	minus the logarithm of a probability	cross-entropy loss
Fisher information	curvature of the expected log-likelihood	sensitivity of the data to a parameter
neural differential equation	differential equation with a learned right-hand side	neural ODE

Table 1: Physics, mathematics and computer-science names for the same objects.

1 What bit strings can teach

1.1 States, measurements and the Born rule

A quantum device prepares a state of N qubits, rotates every qubit by a chosen single-qubit gate and measures all qubits in the computational basis. One run, called a shot, returns one bit string $\mathbf{b} = b_0 b_1 \dots b_{N-1}$ and destroys the state, so we repeat the preparation many times. If the device prepared the same pure state $|\psi\rangle$ in every shot, the probability of a bit string would be $|\langle \mathbf{b}|\psi\rangle|^2$. A real device fluctuates and prepares the state $|\psi_a\rangle$ with an unknown probability w_a, so every prediction about the measured statistics follows from the density matrix

$$\hat{\rho} = \sum_a w_a |\psi_a\rangle\langle\psi_a|. \tag{1.1}$$

For N qubits $\hat{\rho}$ is a $d \times d$ matrix with $d = 2^N$. It is Hermitian, it has unit trace, $\mathrm{Tr}\hat{\rho} = \sum_a w_a = 1$, and it has non-negative eigenvalues, since $\langle\phi|\hat{\rho}|\phi\rangle = \sum_a w_a |\langle\phi|\psi_a\rangle|^2 \geq 0$ for every vector $|\phi\rangle$. These three properties define a physical state, and every matrix with them can be written in the form (1.1). The purity $\mathrm{Tr}\hat{\rho}^2$ equals one for a pure state $\hat{\rho} = |\psi\rangle\langle\psi|$ and takes its smallest value $1/d$ for the maximally mixed state $\hat{I}/d$, which describes a device that outputs a completely random state.

A measurement with outcomes $\mathbf{b}$ is described by operators $\hat{\pi}_{\mathbf{b}}$ that are positive semidefinite and sum to the identity, $\sum_{\mathbf{b}} \hat{\pi}_{\mathbf{b}} = \hat{I}$. The Born rule gives the probability of outcome $\mathbf{b}$,

$$p(\mathbf{b}) = \mathrm{Tr}\big[\hat{\rho}\,\hat{\pi}_{\mathbf{b}}\big]. \tag{1.2}$$

It lies between zero and one because both $\hat{\rho}$ and $\hat{\pi}_{\mathbf{b}}$ are positive semidefinite. For a pure state and a measurement in the computational basis, $\hat{\pi}_{\mathbf{b}} = |\mathbf{b}\rangle\langle\mathbf{b}|$, it reduces to $|\langle\mathbf{b}|\psi\rangle|^2$. Any set of positive operators that sums to the identity is called a positive operator-valued measure (POVM). Most measurements in these notes are projective, with orthogonal projectors $\hat{\pi}_{\mathbf{b}}$.

Bases other than the computational one are reached by a rotation before the measurement. The Pauli matrix $\hat{Z}$ is measured directly, with bit 0 for the eigenvalue $+1$ and bit 1 for -1. To measure $\hat{X}$ we apply the Hadamard gate $\hat{H}$ first, and to measure $\hat{Y}$ we apply $\hat{S}^\dagger$ and then $\hat{H}$, where $\hat{S} = \mathrm{diag}(1, i)$. With the rotation $\hat{u}_P$ of basis P,

$$\hat{u}_X = \hat{H}, \qquad \hat{u}_Y = \hat{H}\hat{S}^\dagger, \qquad \hat{u}_Z = \hat{I}, \qquad \hat{u}_P^\dagger \hat{Z} \hat{u}_P = \hat{P}, \tag{1.3}$$

the projector of outcome b in basis P is

$$\hat{\pi}_{P,b} = \hat{u}_P^\dagger |b\rangle\langle b| \hat{u}_P = \frac{\hat{I} + (-1)^b \hat{P}}{2}. \tag{1.4}$$

A measurement setting assigns one of the bases X, Y or Z to every qubit, and in one shot of the setting every qubit q returns the value $(-1)^{b_q}$ of its measured Pauli matrix. Figure 1 shows the rotation on the Bloch sphere.

1.2 Questions of increasing ambition

The simplest question about the device asks for one expectation value $\langle\hat{O}\rangle = \mathrm{Tr}(\hat{\rho}\hat{O})$, for instance the energy of the prepared state. If $\hat{O}$ is a Pauli string, every shot of a setting that measures it gives a value ± 1, and the mean of M such values has the standard error $\sqrt{(1 - \langle\hat{O}\rangle^2)/M}$. An accuracy ε therefore requires $M \propto 1/\varepsilon^2$ shots, independently of the number of qubits and without any assumption about the state. A more ambitious question asks for the full state $\hat{\rho}$, from which every expectation value, the fidelity (the overlap) with an intended state and the entanglement follow. State tomography, which answers it, needs measurements in many settings and a number of real parameters that grows as 4^N. Between these two extremes lies the prediction of many properties at once, for example all two-point correlations of a spin chain, without reconstructing the state. Beyond the state lies the dynamics, the Hamiltonian $\hat{\mathcal{H}}$ that generates the evolution of every state the device can prepare.

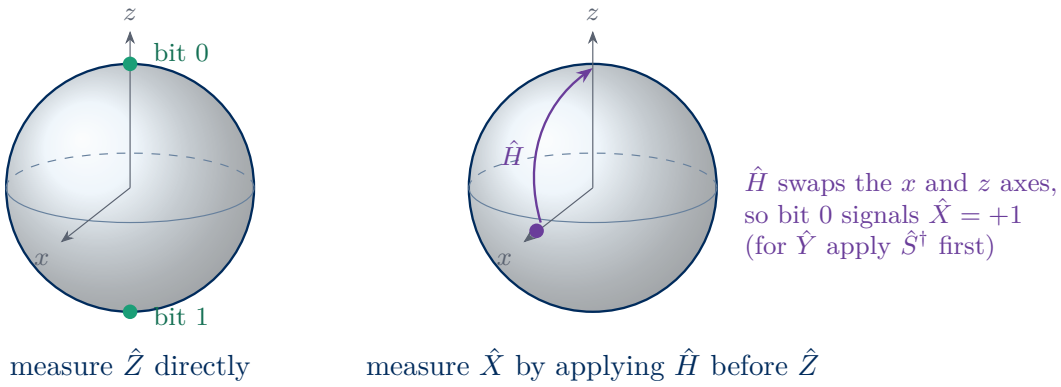


Figure 1: Measurement in a rotated basis. The device measures $\hat{Z}$ directly, with bit 0 for the eigenvalue $+1$. The Hadamard gate $\hat{H}$ applied before the measurement swaps the x and z axes of the Bloch sphere, so bit 0 then signals $\hat{X} = +1$. For $\hat{Y}$ the phase gate $\hat{S}^\dagger$ is applied first.

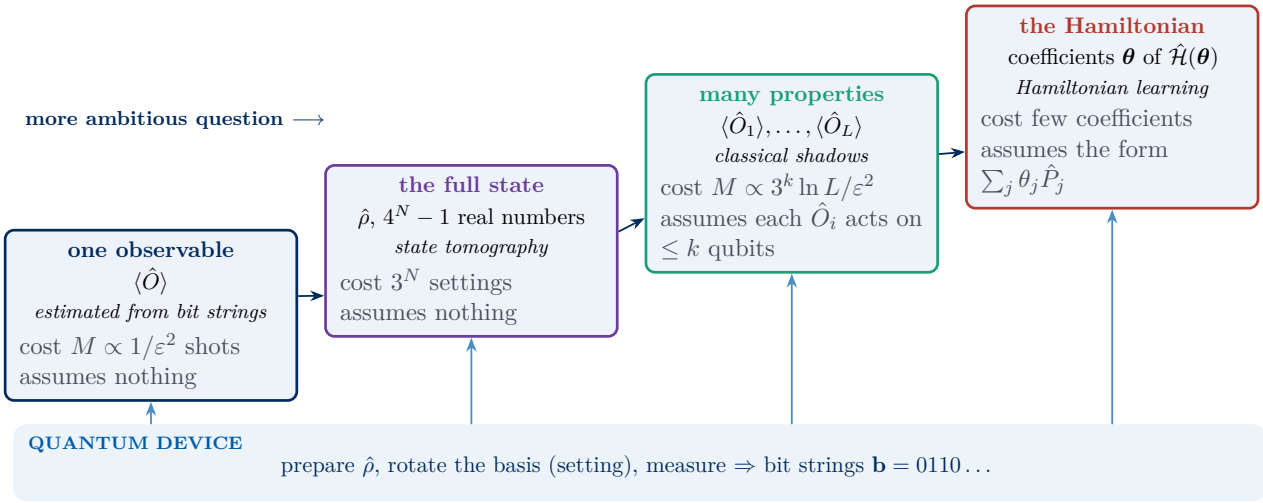


Figure 2: Questions answered from the same measured bit strings, from one observable to the Hamiltonian, with the cost of each and its assumptions. One observable and the full state require no assumption about the state. Many properties are predicted efficiently when every observable acts on few qubits, and the Hamiltonian is learned when its form is known and only its coefficients are unknown.

In Figure 2 all rungs use the same raw data, bit strings from chosen settings, and differ in what is computed from them and in the assumptions that make the computation affordable. Throughout these notes, exact quantities, such as fidelities with the true state, are computed from the density matrix or the state vector, which a simulation knows and an experimenter does not. Estimates are computed from bit strings alone and carry a tilde, as in $\tilde{\rho}$, whereas sample means carry a bar.

2 Quantum state tomography

Quantum state tomography reconstructs the density matrix of a device from the frequencies of measured bit strings [2,3]. The name comes from medical imaging, which reconstructs a three-dimensional body from many two-dimensional projections, and each measurement setting plays the role of one projection. Our running example is a two-qubit device that may prepare three states. The Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \tag{2.1}$$

is pure and maximally entangled. The Werner state [4]

$$\hat{\rho}_W = v\,|\Phi^+\rangle\langle\Phi^+| + (1 - v)\,\frac{\hat{I}}{4}, \qquad v = 0.8, \tag{2.2}$$

is the Bell state mixed with white noise, and it is entangled for a visibility $v > 1/3$ [5,6]. Its eigenvalues are $v + (1-v)/4 = 0.85$ and three times $(1-v)/4 = 0.05$. The third state is a random mixed state

$$\hat{\rho}_{\text{rand}} = \frac{GG^\dagger}{\text{Tr}(GG^\dagger)}, \tag{2.3}$$

where G is a 4×4 matrix of independent complex Gaussian entries, a construction that draws states uniformly with respect to the Hilbert–Schmidt measure [7]. The drawn state has four distinct eigenvalues, the smallest equal to 0.049, and no symmetry between the two qubits. Figure 3 shows the three density matrices.

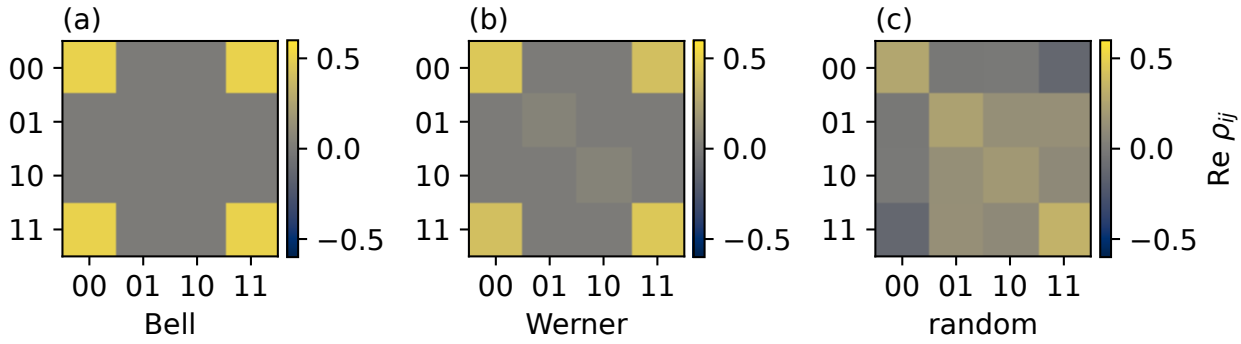


Figure 3: Real parts of the three test states in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. (a) The Bell state has four equal entries in the corners, the populations of $|00\rangle$ and $|11\rangle$ and the coherence between them. (b) The Werner state keeps this pattern with a reduced coherence on top of a uniform diagonal. (c) The random state has nonzero entries everywhere and an imaginary part not shown here.

2.1 Informationally complete measurements

Every tomography experiment follows the same chain (Figure 4). The device prepares the state, rotates every qubit into the basis of a measurement setting and returns bit strings, and the classical computer turns the counts into an estimate, repairs it when it is unphysical and, in a simulation, compares it with the true state.

A single qubit already contains the whole logic of tomography. Its four matrices $\hat{I}$, $\hat{X}$, $\hat{Y}$, $\hat{Z}$ form a basis of the 2×2 matrices and are orthogonal, $\text{Tr}(\hat{P}\hat{P}') = 2\delta_{PP'}$, so every one-qubit density matrix can be written as

$$\hat{\rho} = \frac{1}{2}(\hat{I} + r_x\hat{X} + r_y\hat{Y} + r_z\hat{Z}), \qquad r_P = \langle \hat{P} \rangle = \text{Tr}(\hat{\rho}\hat{P}). \tag{2.4}$$

The Bloch vector $\mathbf{r} = (r_x, r_y, r_z)$ fixes the state. The eigenvalues of (2.4) are $(1 \pm |\mathbf{r}|)/2$, so the physical states fill the unit ball $|\mathbf{r}| \le 1$, and the pure states lie on its surface. By (1.4) the probability of bit 1 in basis P is $(1 - r_P)/2$. After M shots in basis P, of which n_1 return bit 1 and $n_0 = M - n_1$ return bit 0, the estimate and its variance are

$$\tilde{r}_P = \frac{n_0 - n_1}{M}, \qquad \text{Var}(\tilde{r}_P) = \frac{1 - r_P^2}{M}, \tag{2.5}$$

because every shot gives an independent value ± 1 with mean r_P and variance $1 - r_P^2$. The three settings X, Y and Z thus determine the state (Figure 5).

The estimate (2.5) is unbiased, i.e. its mean over many repetitions of the experiment equals the true value, but it ignores the constraint $|\mathbf{r}| \le 1$. We repeat the one-qubit experiment 2000 times for a mixed state with $\mathbf{r} = (0.54, 0, 0.72)$, $|\mathbf{r}| = 0.9$, and $M = 50$ shots per setting. The spread of the components agrees with (2.5), and in 23% of the runs the estimated vector lies outside the unit ball (Figure 6). Such an estimate has a negative eigenvalue and is no density matrix.

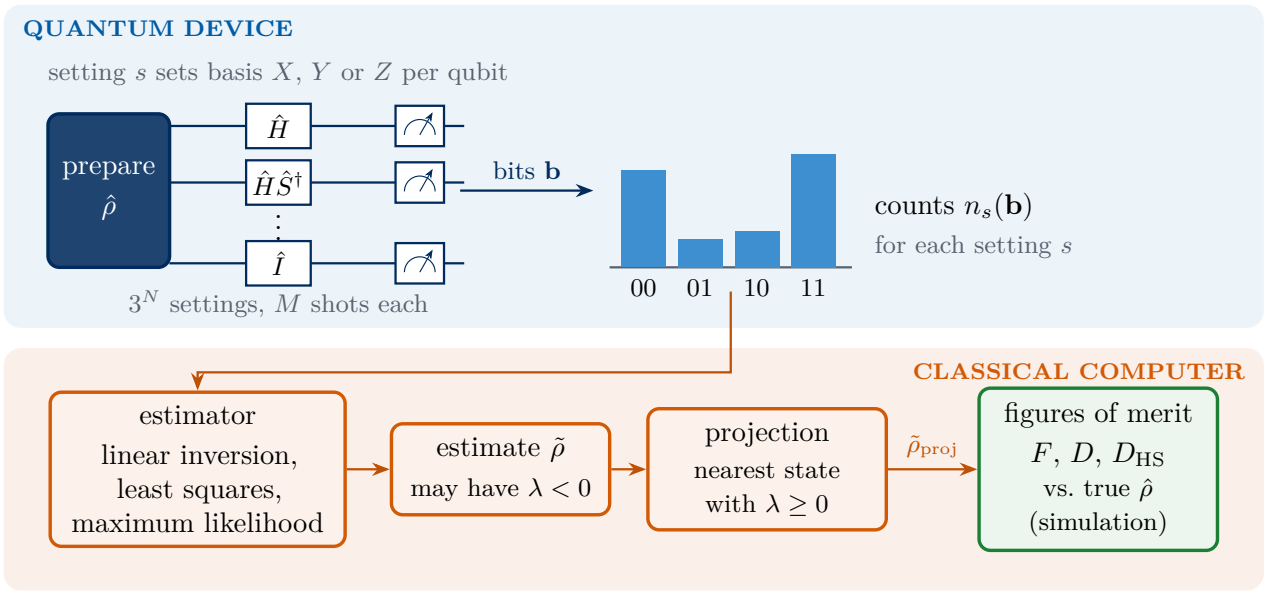


Figure 4: State tomography from the device to the figures of merit. The device rotates every qubit into the basis of setting s and returns M bit strings per setting. The classical computer turns the counts into an estimate, projects it onto the physical states when an eigenvalue is negative and, in a simulation, compares it with the true state.

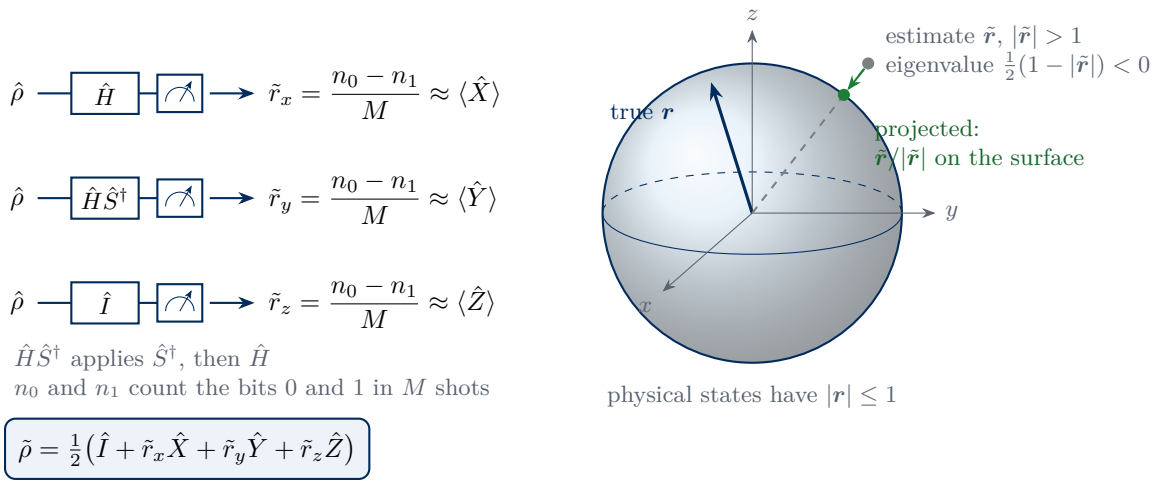


Figure 5: Tomography of one qubit. Each of three settings gives one component of the Bloch vector from the counts n_0 and n_1 of M shots. An estimate $\tilde{\mathbf r}$ outside the unit ball has the negative eigenvalue $(1 - |\tilde{\mathbf r}|)/2$, and the closest physical state is the unit vector in the same direction.

For N qubits the role of $\hat I, \hat X, \hat Y, \hat Z$ is taken by the 4^N Pauli strings $\hat P = \hat\sigma_{p_0} \otimes \hat\sigma_{p_1} \otimes \cdots \otimes \hat\sigma_{p_{N-1}}$ with letters $p_q \in \{I, X, Y, Z\}$, qubit 0 being the leftmost factor. They are orthogonal, $\mathrm{Tr}(\hat P \hat P') = 2^N \delta_{PP'}$, and every density matrix has the expansion

$$\hat\rho = \frac{1}{2^N} \sum_P \langle \hat P \rangle \, \hat P, \tag{2.6}$$

which generalises (2.4). The coefficient of the identity string equals one by normalisation, so a general state of N qubits has $4^N - 1$ real parameters, fifteen for two qubits. The number of non-identity letters of $\hat P$ is its weight k. A measurement setting $s = (s_0, \ldots, s_{N-1})$ with $s_q \in \{X, Y, Z\}$ rotates every qubit by $\hat u_{s_q}$ of (1.3) and measures all of them, and there are 3^N settings. The projector of outcome $\mathbf b$ in setting s is the product of single-qubit projectors (1.4),

$$\hat\pi_{s,\mathbf b} = \bigotimes_{q=0}^{N-1} \frac{\hat I + (-1)^{b_q} \hat\sigma_{s_q}}{2}, \tag{2.7}$$

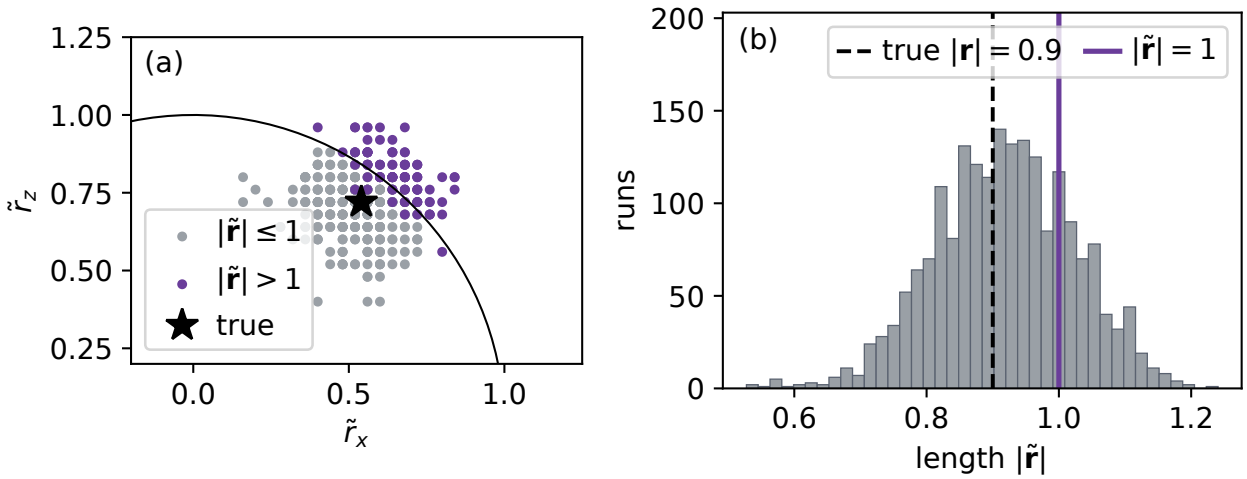


Figure 6: Tomography of one qubit with $\mathbf{r} = (0.54, 0, 0.72)$ and $M = 50$ shots per setting, simulated 2000 times. (a) 400 estimated Bloch vectors projected on the xz plane, the true vector as a star and the unit circle as the boundary of the physical states. Dark points lie outside the ball. (b) Histogram of the length $|\tilde{\mathbf{r}}|$ with the true length dashed and the physical limit $|\tilde{\mathbf{r}}| = 1$ as a solid line.

and the setting returns M bit strings with probabilities $p_s(\mathbf{b}) = \mathrm{Tr}[\hat{\rho}\,\hat{\pi}_{s,\mathbf{b}}]$, of which we record the counts $n_s(\mathbf{b})$ and the frequencies $f_s(\mathbf{b}) = n_s(\mathbf{b})/M$.

One setting measures every Pauli string whose letters agree with s on the support of the string, the set of qubits where the letter is not I. The product of the values $(-1)^{b_q}$ over the support is the eigenvalue of the string in that shot. The string $\hat{X}\hat{I}$, for example, is measured by the settings XX, XY and XZ, and in general a string of weight k is measured by 3^{N-k} settings. Together the 3^N settings determine all 4^N expectation values in (2.6). A measurement whose outcome probabilities determine every state uniquely is called informationally complete (IC).

A single POVM with at least d^2 outcomes can also be informationally complete. For one qubit the symmetric informationally complete POVM [8] has four elements built from the vectors $\mathbf{n}_a$ that point to the corners of a regular tetrahedron inscribed in the Bloch sphere,

$$\hat{\pi}_a = \frac{1}{4}(\hat{I} + \mathbf{n}_a \cdot \hat{\boldsymbol{\sigma}}), \qquad \sum_{a=1}^{4} \mathbf{n}_a = 0, \qquad \mathbf{n}_a \cdot \mathbf{n}_b = -\frac{1}{3}\ (a \neq b), \tag{2.8}$$

where $\hat{\boldsymbol{\sigma}} = (\hat{X}, \hat{Y}, \hat{Z})$. The elements sum to the identity because the four vectors sum to zero. The probabilities $p_a = (1 + \mathbf{r} \cdot \mathbf{n}_a)/4$ are linear in the Bloch vector, and with $\sum_a \mathbf{n}_a \mathbf{n}_a^{\mathsf{T}} = \frac{4}{3}\mathbb{1}_3$, where $\mathbb{1}_3$ is the 3×3 unit matrix, they invert to $\mathbf{r} = 3\sum_a p_a \mathbf{n}_a$, or

$$\hat{\rho} = \sum_{a=1}^{4} p_a\,(6\hat{\pi}_a - \hat{I}). \tag{2.9}$$

A product of such POVMs on N qubits has 4^N outcomes and is informationally complete with a single setting. It requires an ancilla or an environment coupling for each qubit, so Pauli settings are the common choice on qubit devices. Figure 7 shows both kinds of informationally complete measurement.

We test the simulated device by comparing the sampled counts with the multinomial distribution through Pearson's statistic $\chi^2 = \sum_{s,\mathbf{b}} [n_s(\mathbf{b}) - M p_s(\mathbf{b})]^2 / [M p_s(\mathbf{b})]$ [9]. Its mean over many data sets equals the number of degrees of freedom, three per setting of two qubits and 27 in total. The correct sampler gives a mean of 26.4, and a sampler with the reversed bit order, qubit 0 read as the least significant bit, gives 2709 on the random state and is rejected. In the Bell-state data set of Figure 8, the settings XX and ZZ give only the two outcomes of even parity and YY only those of odd parity, since $\langle \hat{X}\hat{X} \rangle = \langle \hat{Z}\hat{Z} \rangle = 1$ and $\langle \hat{Y}\hat{Y} \rangle = -1$. In the six mixed settings all four outcomes are equally likely.

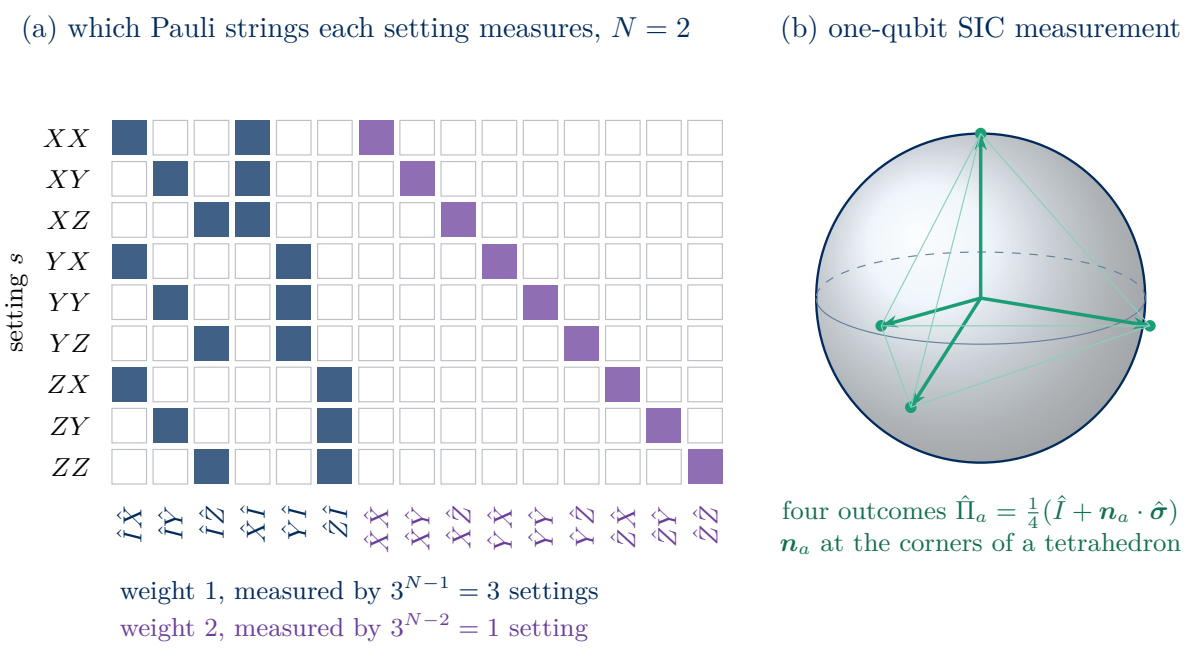


Figure 7: Informationally complete measurements. (a) For two qubits, each of the nine settings (rows) measures the Pauli strings (columns) whose letters agree with it on their support. A weight-one string is measured by three settings, a weight-two string by one. (b) The tetrahedral POVM of one qubit (2.8), four outcomes whose Bloch vectors point to the corners of a regular tetrahedron.

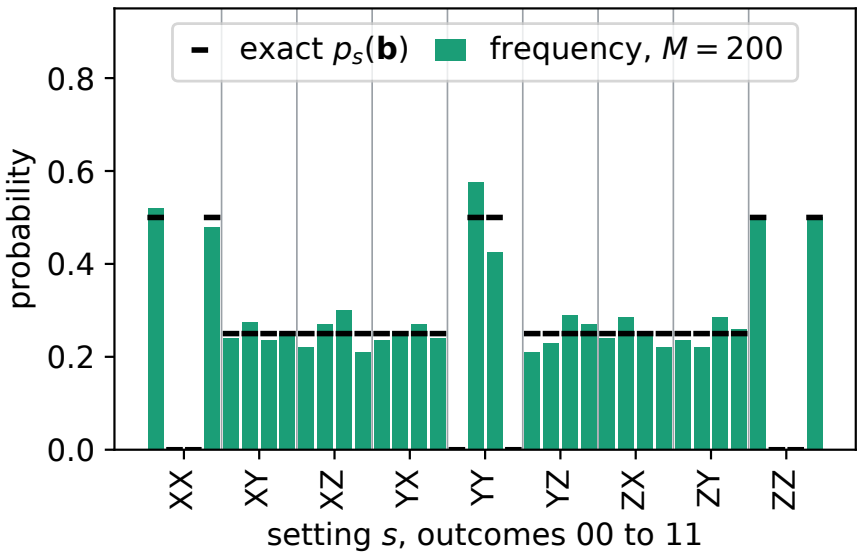


Figure 8: One simulated data set of the Bell state with $M = 200$ shots in each of the nine Pauli settings. Bars show the frequencies of the four outcomes $00, 01, 10, 11$ of every setting, and black marks show the exact probabilities.

2.2 Linear inversion

The Born rule (1.2) is linear in $\hat{\rho}$, and with the expansion (2.6) the probabilities become linear functions of the Pauli coefficients $r_P = \langle \hat{P} \rangle$,

$$p_s(\mathbf{b}) = \sum_P A_{(s,\mathbf{b}),P}\, r_P, \qquad A_{(s,\mathbf{b}),P} = \frac{\mathrm{Tr}[\hat{\pi}_{s,\mathbf{b}} \hat{P}]}{2^N}. \tag{2.10}$$

The design matrix A has one row for every pair of setting and outcome, $9 \times 4 = 36$ rows for two qubits, and one column for each of the 16 Pauli strings. By (2.7) its entries are $\pm 2^{-N}$ when setting s measures $\hat{P}$, with the sign given by the parity of the bits on the support, and zero otherwise

(Figure 9). Replacing the probabilities by the frequencies $\mathbf{f}$ gives an overdetermined linear system, whose least-squares solution

$$\tilde{\mathbf{r}} = \arg\min_{\mathbf{r}} \|A\mathbf{r} - \mathbf{f}\|^2 = (A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}\mathbf{f} \tag{2.11}$$

defines the linear-inversion estimate $\tilde{\rho}_{\mathrm{LI}}$ through (2.6). The matrix $A^{\mathsf{T}}A$ is the Gram matrix of the columns of A, also called the frame operator of the measurement, and the measurement is informationally complete exactly when it is invertible, i.e. when A has full column rank. With exact probabilities in place of frequencies, (2.11) returns the true state. The same construction gives the dual operators $6\hat{\pi}_a - \hat{I}$ of the tetrahedral POVM in (2.9).

For Pauli settings the columns of A are orthogonal. Two different strings $\hat{P}$ and $\hat{P}'$ that are both measured by a setting give orthogonal sign patterns over the outcomes, and a string of weight k is measured by 3^{N-k} settings, each contributing 2^N entries of size 2^{-2N} to the diagonal. Hence

$$A^{\mathsf{T}}A = \mathrm{diag}\Big(\frac{3^{N-k}}{2^N}\Big), \tag{2.12}$$

and (2.11) reduces to an average of parities over the settings that measure each string,

$$\tilde{r}_P = \frac{1}{3^{N-k}} \sum_{s \text{ measures } P} \sum_{\mathbf{b}} (-1)^{\sum_{q\in \mathrm{supp}P} b_q} f_s(\mathbf{b}), \qquad \mathrm{Var}(\tilde{r}_P) = \frac{1 - r_P^2}{3^{N-k}M}. \tag{2.13}$$

The variance follows as in (2.5), since $3^{N-k}M$ independent shots contribute to the string. Strings of low weight are measured by more settings and are therefore more precise. Inserting (2.13) into (2.6) gives the linear-inversion estimate as a weighted sum of the frequencies,

$$\tilde{\rho}_{\mathrm{LI}} = \sum_s \sum_{\mathbf{b}} f_s(\mathbf{b})\, \hat{D}_{s,\mathbf{b}}, \qquad \hat{D}_{s,\mathbf{b}} = \frac{1}{3^N} \bigotimes_{q=0}^{N-1} (3\hat{\pi}_{s_q,b_q} - \hat{I}), \tag{2.14}$$

where $\hat{\pi}_{s_q,b_q}$ is the single-qubit projector (1.4). For one qubit this is the identity $\hat{\rho} = \frac{1}{3}\sum_{P,b} p_P(b)(3\hat{\pi}_{P,b} - \hat{I})$, which follows from $\hat{\pi}_{P,0} + \hat{\pi}_{P,1} = \hat{I}$ and $\hat{\pi}_{P,0} - \hat{\pi}_{P,1} = \hat{P}$. The estimate is linear in the data and unbiased, and its fluctuations decrease as $M^{-1/2}$.

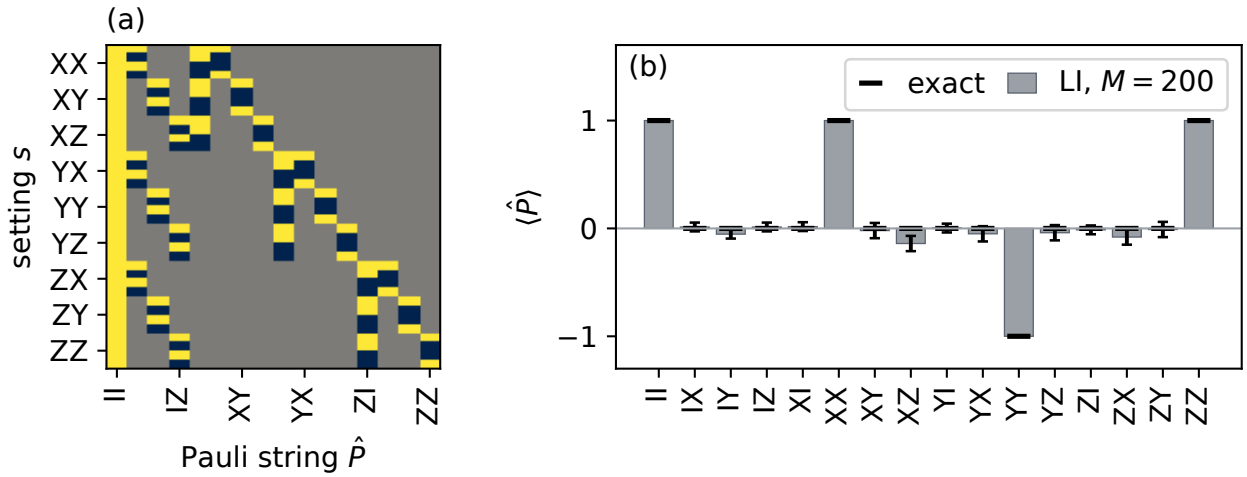


Figure 9: Linear inversion for two qubits. (a) The design matrix $2^N A$ of (2.10), rows labelled by setting and outcome, columns by Pauli string, with entries $+1$ (yellow), -1 (dark) and 0 (grey). (b) Pauli coefficients of the Bell state estimated from the data of Figure 8, with error bars from (2.13) and exact values as black marks. Look at the three coefficients ± 1, which carry no error, and at the weight-one strings, which fluctuate less than the weight-two strings.

Among the estimated Pauli coefficients of the Bell state in Figure 9(b), the nonzero values $\langle \hat{X}\hat{X} \rangle = \langle \hat{Z}\hat{Z} \rangle = 1$ and $\langle \hat{Y}\hat{Y} \rangle = -1$ are recovered without error, because every shot of the corresponding setting gives the same parity and the variance in (2.13) vanishes for $r_P = \pm 1$. The vanishing coefficients fluctuate around zero.

2.3 Least squares and weighted least squares

The least-squares problem (2.11) treats all frequencies as equally precise. A frequency estimates a probability p with the variance $p(1-p)/M$, so frequencies close to zero or one are more precise than those near one half. Weighted least squares divides every squared residual by its variance,

$$\tilde{\mathbf{r}}_{\mathrm{WLS}} = \arg\min_{\mathbf{r}} \sum_{s,\mathbf{b}} \frac{\left[(A\mathbf{r})_{(s,\mathbf{b})} - f_s(\mathbf{b})\right]^2}{v_s(\mathbf{b})}, \qquad v_s(\mathbf{b}) = \frac{f_s^{\mathrm{sh}}(\mathbf{b})\left(1 - f_s^{\mathrm{sh}}(\mathbf{b})\right)}{M}, \qquad f_s^{\mathrm{sh}}(\mathbf{b}) = \frac{n_s(\mathbf{b}) + 1}{M + 2}, \tag{2.15}$$

where the shifted frequency f^{sh} avoids a zero variance for outcomes that never occurred, and the identity coefficient is kept at one. We measure the error of an estimate by the Hilbert–Schmidt distance, the Euclidean distance between two matrices viewed as vectors of their entries,

$$\|\hat{O}\|_{\mathrm{HS}} = \sqrt{\mathrm{Tr}(\hat{O}^\dagger \hat{O})}, \qquad D_{\mathrm{HS}}(\hat{\rho}, \hat{\sigma}) = \|\hat{\rho} - \hat{\sigma}\|_{\mathrm{HS}}. \tag{2.16}$$

On 300 data sets of the Werner state with $M = 200$ the mean Hilbert–Schmidt distance to the true state is 0.104 for least squares and 0.102 for weighted least squares. For a complete and balanced set of Pauli settings the weighting therefore changes little, and it matters only when some settings receive more shots than others or when the detectors have unequal efficiencies. Neither estimator enforces non-negative eigenvalues, and both have errors that decrease as $M^{-1/2}$.

2.4 Unphysical estimates and the projection onto physical states

The one-qubit estimates of Figure 6 left the Bloch ball in a quarter of the runs, and for two qubits the same happens when $\tilde{\rho}_{\mathrm{LI}}$ has a negative eigenvalue. The statistical noise of (2.13) is spread over all fifteen coefficients and shifts every eigenvalue of $\tilde{\rho}_{\mathrm{LI}}$, so true eigenvalues close to zero, which every pure state has, are pushed below zero. For each test state and each budget of $M = 20$, 200 and 2000 shots per setting we draw 400 independent data sets and record the smallest eigenvalue $\lambda_{\min}$ of every linear-inversion estimate (Figure 10). For the Bell state every estimate at every budget has a negative eigenvalue, because the three true eigenvalues zero are surrounded by noise of both signs. For the Werner state, whose small eigenvalues equal 0.05, the fraction of unphysical estimates falls from 100% at $M = 20$ to 51% at $M = 200$ and to 0% at $M = 2000$, once the noise becomes smaller than the smallest true eigenvalue. A linear-inversion estimate of a nearly pure state is thus unphysical in practice, and its negative eigenvalue predicts negative probabilities for some measurements.

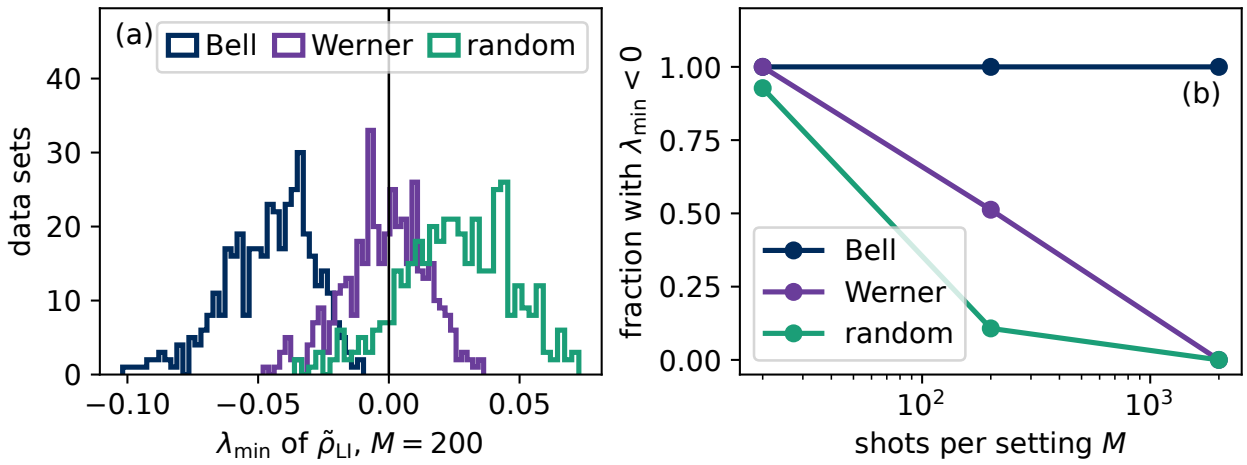


Figure 10: Unphysical linear-inversion estimates, 400 simulated data sets per state and budget. (a) Distribution of the smallest eigenvalue $\lambda_{\min}$ of $\tilde{\rho}_{\mathrm{LI}}$ at $M = 200$ for the three test states. (b) Fraction of estimates with $\lambda_{\min} < 0$ against the number of shots per setting. The fraction stays at one for the pure Bell state and falls to zero for the full-rank states.

A first repair replaces $\tilde{\rho}_{\mathrm{LI}}$ by the physical state closest to it in the Hilbert–Schmidt norm (2.16),

$$\tilde{\rho}_{\mathrm{proj}} = \operatorname*{arg\,min}_{\hat{\sigma}\geq 0,\ \mathrm{Tr}\hat{\sigma}=1} \left\|\tilde{\rho}_{\mathrm{LI}} - \hat{\sigma}\right\|_{\mathrm{HS}}. \tag{2.17}$$

Smolin, Gambetta and Smith showed that the minimiser keeps the eigenvectors of $\tilde{\rho}_{\mathrm{LI}}$ and changes only the eigenvalues [10]. With the eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d$ of $\tilde{\rho}_{\mathrm{LI}}$ the new eigenvalues are

$$\mu_i = \max(\lambda_i - \tau,\, 0), \tag{2.18}$$

where the common shift τ is fixed by $\sum_i \mu_i = 1$. In practice one sets the negative eigenvalues to zero, subtracts their total weight equally from the remaining ones and repeats while an eigenvalue is negative (Figure 11), at the cost of one diagonalisation of a $d \times d$ matrix.

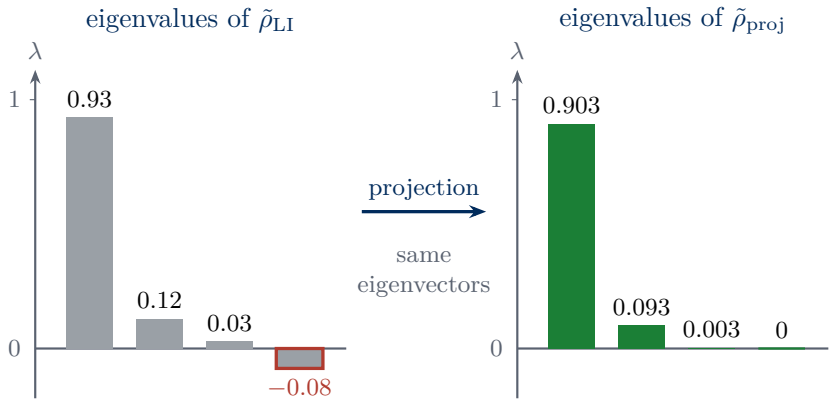


Figure 11: Projection onto the physical states (2.18). The negative eigenvalue of the linear-inversion estimate (left) is set to zero, and its weight is subtracted equally from the other eigenvalues, so that the trace stays one (right). The eigenvectors do not change.

Equation (2.17) asks for the point of the convex set of physical states closest to a given matrix. A physical $\tilde{\rho}$ is this point exactly when the angle between $\tilde{\rho}_{\mathrm{LI}} - \tilde{\rho}$ and every direction into the set is obtuse, $\mathrm{Tr}[(\tilde{\rho}_{\mathrm{LI}} - \tilde{\rho})(\hat{\sigma} - \tilde{\rho})] \leq 0$ for every physical $\hat{\sigma}$. The largest value of $\mathrm{Tr}(\hat{\Delta}\hat{\sigma})$ over physical states is the largest eigenvalue of $\hat{\Delta}$, attained at its eigenvector, so the condition is equivalent to the certificate

$$\lambda_{\max}(\hat{\Delta}) - \mathrm{Tr}(\hat{\Delta}\tilde{\rho}) \leq 0, \qquad \hat{\Delta} = \tilde{\rho}_{\mathrm{LI}} - \tilde{\rho}. \tag{2.19}$$

The projection (2.18) satisfies (2.19) to numerical precision on all data sets, whereas the common shortcut that sets negative eigenvalues to zero and rescales the others violates it. The same geometry shows that the projection never moves the estimate away from the true state, because the true state lies in the convex set onto which we project. All 400 Bell-state data sets at $M = 200$ end closer to the true state after the projection (Figure 12).

2.5 Maximum likelihood

The projection uses the data only through $\tilde{\rho}_{\mathrm{LI}}$, which weights all frequencies equally. Maximum-likelihood estimation searches directly among physical states for the one under which the observed counts are most probable [11,12]. The shots are independent, so the probability of the record is a product of Born probabilities, and its logarithm is, up to a state-independent constant,

$$\ell(\hat{\rho}) = \sum_{s} \sum_{\mathbf{b}} n_s(\mathbf{b}) \ln \mathrm{Tr}\left[\hat{\rho}\, \hat{\pi}_{s,\mathbf{b}}\right], \tag{2.20}$$

measured in nats. The estimate is $\tilde{\rho}_{\mathrm{MLE}} = \arg\max \ell(\hat{\rho})$ over physical states.

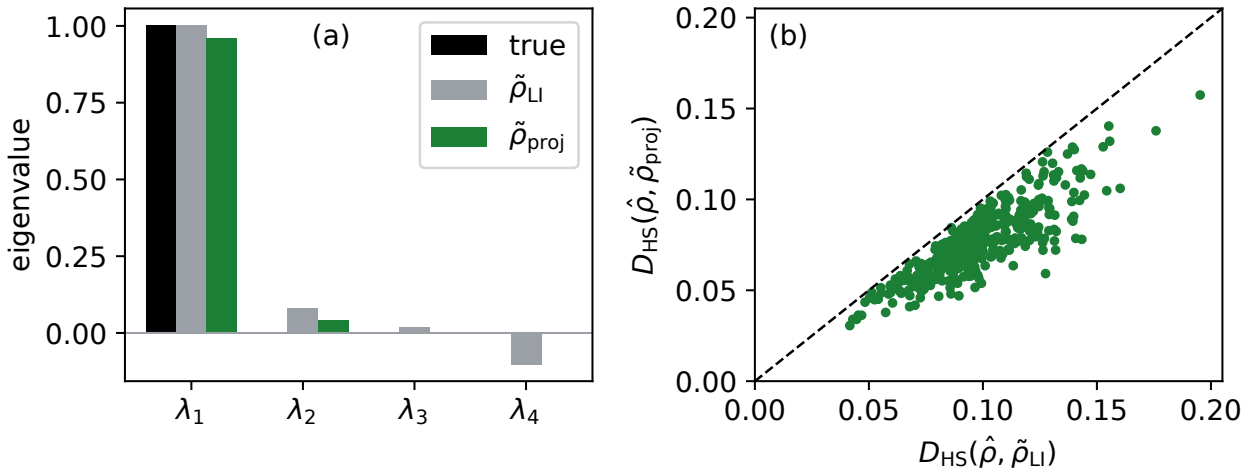


Figure 12: Projection of Bell-state estimates at $M = 200$. (a) Eigenvalues of the true state, of the estimate with the most negative eigenvalue and of its projection. (b) Hilbert–Schmidt distance to the true state after the projection against the distance before it, for 400 data sets. Every point lies below the diagonal.

To find the condition for the maximum, we move from $\hat{\rho}$ a small step ϵ towards another physical state $\hat{\sigma}$, $\hat{\rho}_\epsilon = (1-\epsilon)\hat{\rho} + \epsilon\hat{\sigma}$, which stays physical by convexity. With $p_s(\mathbf{b}|\hat{\rho}) = \mathrm{Tr}[\hat{\rho}\,\hat{\pi}_{s,\mathbf{b}}]$ the derivative of (2.20) is

$$\frac{d\ell(\hat{\rho}_\epsilon)}{d\epsilon}\Big|_{\epsilon=0} = \sum_{s,\mathbf{b}} n_s(\mathbf{b})\,\frac{\mathrm{Tr}\big[(\hat{\sigma}-\hat{\rho})\hat{\pi}_{s,\mathbf{b}}\big]}{p_s(\mathbf{b}|\hat{\rho})} = 3^N M\Big[\mathrm{Tr}(\hat{R}\hat{\sigma}) - \mathrm{Tr}(\hat{R}\hat{\rho})\Big], \tag{2.21}$$

where we used $n_s(\mathbf{b}) = M f_s(\mathbf{b})$ and defined

$$\hat{R}(\hat{\rho}) = \frac{1}{3^N}\sum_{s,\mathbf{b}} \frac{f_s(\mathbf{b})}{p_s(\mathbf{b}|\hat{\rho})}\,\hat{\pi}_{s,\mathbf{b}}. \tag{2.22}$$

Since $\mathrm{Tr}(\hat{R}\hat{\rho}) = 3^{-N}\sum_{s,\mathbf{b}} f_s(\mathbf{b}) = 1$, no direction increases ℓ when $\mathrm{Tr}(\hat{R}\hat{\sigma}) \le 1$ for every physical $\hat{\sigma}$. At the maximum the largest eigenvalue of $\hat{R}$ therefore equals one, and $\hat{R}\tilde{\rho}_{\mathrm{MLE}} = \tilde{\rho}_{\mathrm{MLE}}$ on the support of the estimate. If the model reproduced the frequencies exactly, every ratio f/p would equal one and $\hat{R} = \hat{I}$, because the projectors of all 3^N settings sum to $3^N\hat{I}$.

The fixed-point condition suggests the iteration

$$\hat{\rho}_{t+1} = \frac{\hat{R}(\hat{\rho}_t)\,\hat{\rho}_t\,\hat{R}(\hat{\rho}_t)}{\mathrm{Tr}\big[\hat{R}(\hat{\rho}_t)\,\hat{\rho}_t\,\hat{R}(\hat{\rho}_t)\big]}, \tag{2.23}$$

started at $\hat{I}/d$, where t counts the iterations. In this $R\rho R$ algorithm [11] every iterate is positive semidefinite with unit trace by construction, but ℓ may decrease in some steps. The normalised diluted variant $\hat{\rho} \to (\hat{I} + \epsilon\hat{R})\hat{\rho}(\hat{I} + \epsilon\hat{R})$ with a small ϵ increases ℓ in every step [13].

A second route uses the Cholesky decomposition, which writes every positive semidefinite matrix as $CC^\dagger$ with a lower-triangular factor C. We write the state through a complex lower-triangular matrix C with real diagonal,

$$\hat{\rho}(C) = \frac{CC^\dagger}{\mathrm{Tr}(CC^\dagger)}, \tag{2.24}$$

a Cholesky-type parametrisation [12,3]. The matrix $CC^\dagger$ is Hermitian and positive semidefinite for every C, so (2.24) is physical for all values of the d^2 real parameters of C, sixteen for two qubits. The constrained maximisation of ℓ thus becomes an unconstrained minimisation of $-\ell(\hat{\rho}(C))$, which we perform with Adam [14] and gradients from automatic differentiation [15]. Figure 13 contrasts the two routes.

On one Werner data set with $M = 2000$ both algorithms reach the same maximum to numerical precision (Figure 14), and at this maximum the largest eigenvalue of $\hat{R}$ is one. The projected estimate

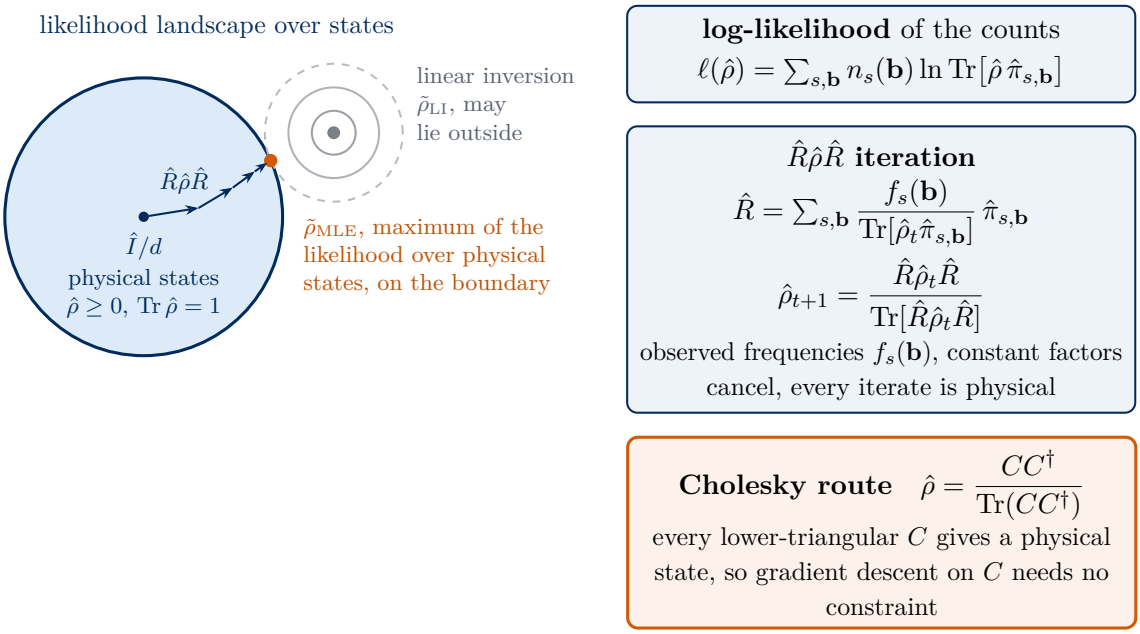


Figure 13: Maximum likelihood over the physical states. The $R\rho R$ iteration (2.23) starts at $\hat{I}/d$ and moves towards the most likely physical state, which lies on the boundary when the unconstrained optimum, near $\tilde{\rho}_{\mathrm{LI}}$, is unphysical. The Cholesky form (2.24) gives a physical state for every C, so gradient descent on T needs no constraint. The sketch writes $\hat{R}$ without the factor 3^{-N} of (2.22), which cancels in the normalisation.

of the same data is physical, but its largest eigenvalue of $\hat{R}$ is 1.027, so a physical state with a higher likelihood exists. When the unconstrained maximum lies outside the physical states, the maximum-likelihood estimate lies on their boundary and has at least one eigenvalue zero. For pure states this is the correct structure, whereas for full-rank states with few shots the zero eigenvalue is spurious. The estimate is then biased, i.e. its mean over repeated experiments differs from the true state.

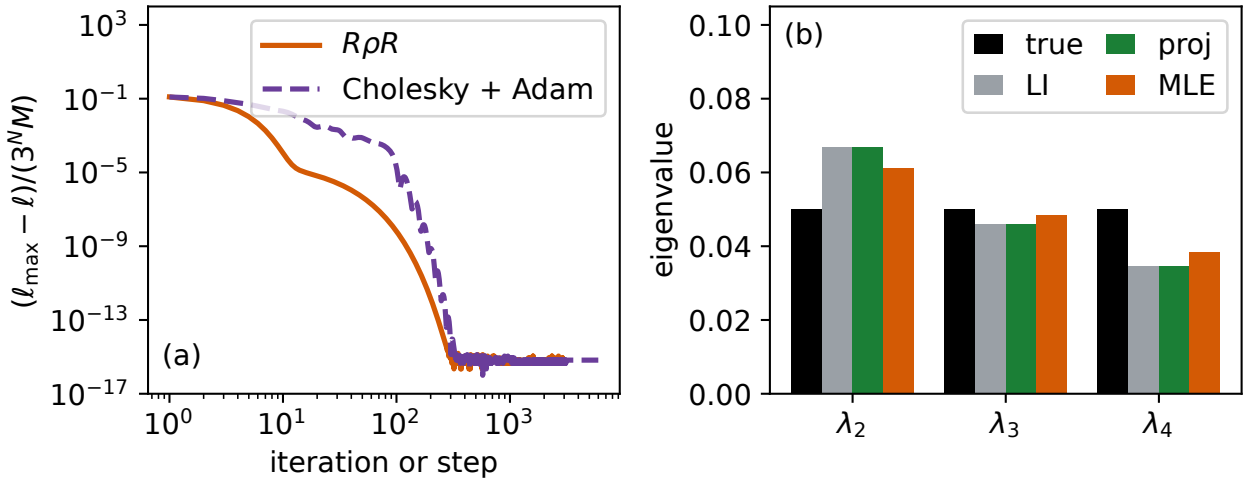


Figure 14: Maximum likelihood for one Werner data set with $M = 2000$. (a) Distance of the log-likelihood per shot from its maximum along the $R\rho R$ iteration and along Adam on the Cholesky parameters. (b) The three small eigenvalues of the true state, all equal to 0.05, and of the linear-inversion, projected and maximum-likelihood estimates of the same data.

2.6 Figures of merit

To compare estimators we need distances between the estimate $\tilde\rho$ and the true state $\hat\rho$, which only a simulation can evaluate. The fidelity [16,17]

$$F(\hat\rho,\tilde\rho)=\left(\mathrm{Tr}\sqrt{\sqrt{\hat\rho}\,\tilde\rho\,\sqrt{\hat\rho}}\right)^2 \tag{2.25}$$

equals one for identical states and lies between zero and one for physical states. For a pure true state $\hat\rho=|\psi\rangle\langle\psi|$ it reduces to $\langle\psi|\tilde\rho|\psi\rangle$, the probability that the estimate passes a test for $|\psi\rangle$. The fidelity is defined only for physical $\tilde\rho$, a further reason to project unphysical estimates. The trace distance

$$D(\hat\rho,\tilde\rho)=\frac{1}{2}\|\hat\rho-\tilde\rho\|_1=\frac{1}{2}\sum_i|\lambda_i|, \tag{2.26}$$

where $\|\hat O\|_1=\mathrm{Tr}\sqrt{\hat O^\dagger\hat O}$ is the trace norm, the sum of the singular values, and λ_i are the eigenvalues of $\hat\rho-\tilde\rho$, is the largest difference between the probabilities that the two states assign to an outcome of any measurement [18]. The Hilbert–Schmidt distance (2.16) is the easiest to compute and enters least squares. The Bures distance D_B follows from the fidelity,

$$D_B^2(\hat\rho,\tilde\rho)=2-2\sqrt{F(\hat\rho,\tilde\rho)}. \tag{2.27}$$

For the Bell state and the Werner state (2.2), which share their eigenvectors, the fidelity is $F=v+(1-v)/4=0.85$ and the trace distance $D=\frac{3}{4}(1-v)=0.15$. For two pure states the two measures are tied by $D=\sqrt{1-F}$, so a fidelity of 0.99 corresponds to a trace distance of 0.1. An infidelity of one per cent therefore allows the outcome probabilities of some measurement to differ by ten per cent, and infidelities of a few per cent describe clearly different states. Figure 15 shows the three measures for two states of one qubit.

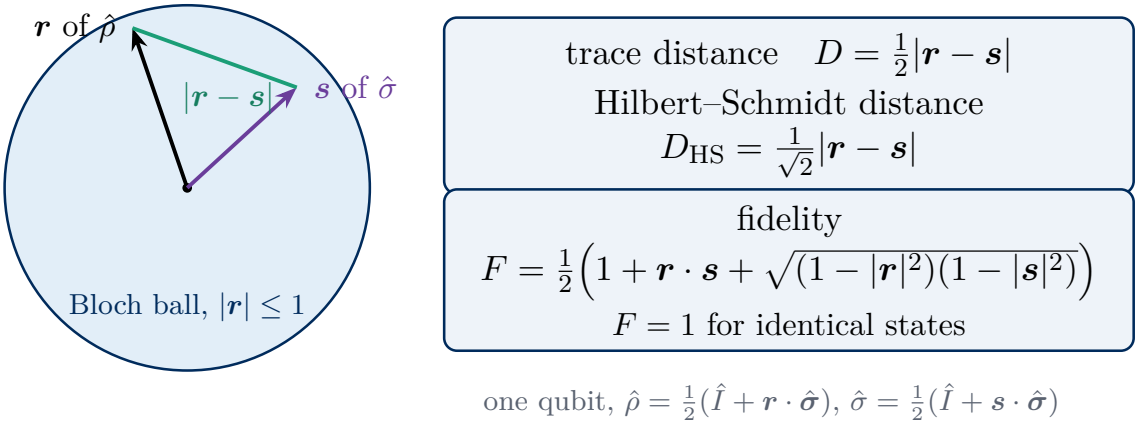


Figure 15: Figures of merit for two one-qubit states with Bloch vectors $\mathbf{r}$ and $\mathbf{s}$. The trace distance is half and the Hilbert–Schmidt distance $1/\sqrt{2}$ of the Euclidean distance of the Bloch vectors, whereas the fidelity also depends on their lengths and reduces to $(1+\mathbf{r}\cdot\mathbf{s})/2$ when one state is pure.

2.7 Error and the number of shots

Each frequency has a statistical error proportional to $M^{-1/2}$, and so have the linear-inversion coefficients (2.13). Every distance that is linear in the error of the estimate, such as the trace distance and the Hilbert–Schmidt distance, therefore decreases as

$$D\propto M^{-1/2}. \tag{2.28}$$

The infidelity of a physical estimate of a pure state behaves differently. An estimate close to $|\psi\rangle$ can be written as a slightly rotated pure state $|\psi'\rangle=\sqrt{1-\delta^2}|\psi\rangle+\delta|\psi_\perp\rangle$, with $\langle\psi|\psi_\perp\rangle=0$, carrying most of the weight, plus small eigenvalues of total weight η on states orthogonal to $|\psi'\rangle$,

$$\tilde\rho\approx(1-\eta)|\psi'\rangle\langle\psi'|+\eta\,\hat\tau,\qquad 1-F=1-\langle\psi|\tilde\rho|\psi\rangle\approx\delta^2+\eta(1-\langle\psi|\hat\tau|\psi\rangle), \tag{2.29}$$

where $\hat{\tau}$ is a state of unit trace. The rotation amplitude δ is linear in the statistical noise, so $\delta^2 \propto M^{-1}$. The Bell state is an eigenstate of three of the nine measured settings, in which outcomes of zero probability never occur. Maximum likelihood then keeps η of order M^{-1} as well, and its infidelity decreases as M^{-1}. For a generic pure state measured in fixed bases the small eigenvalues of the estimate fluctuate at order $M^{-1/2}$, and the infidelity decreases more slowly unless the bases are adapted to the state [19]. For Haar-random pure states of two qubits in the nine Pauli settings our simulation gives the slope -0.53 for maximum likelihood and -0.52 for the projection. The projection lowers the large eigenvalue by the weight of the clipped negative eigenvalues, which is of the order of the noise itself, $\eta \propto M^{-1/2}$, and its infidelity decreases only as $M^{-1/2}$. For full-rank states the infidelity is asymptotically, at large M, a quadratic form in the deviation $\tilde{\rho} - \hat{\rho}$ and then scales as M^{-1}.

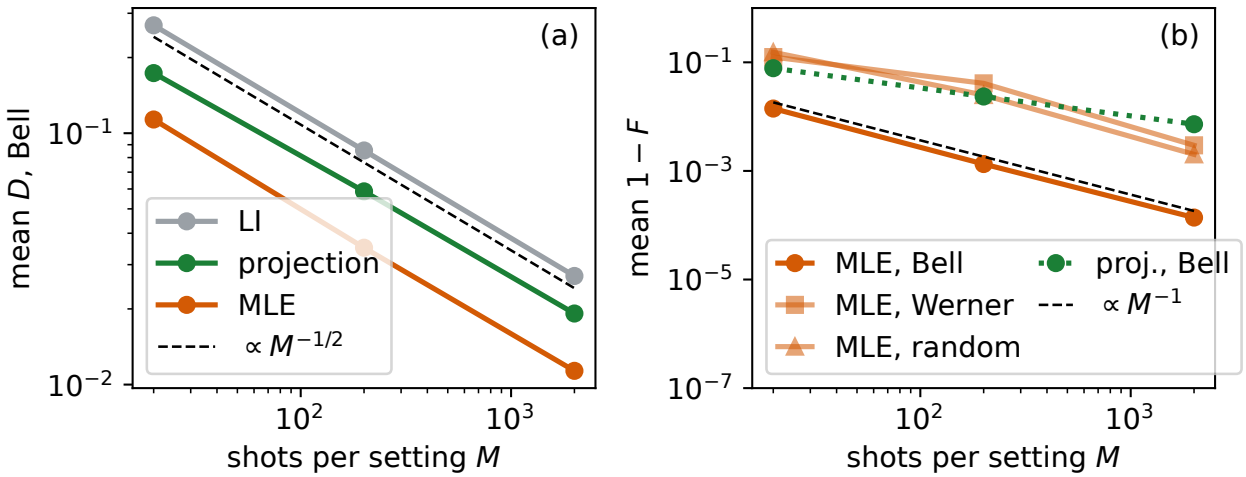


Figure 16: Errors against the number of shots per setting, means over 400 simulated data sets per point. (a) Trace distance (2.26) of the linear-inversion, projected and maximum-likelihood estimates of the Bell state, with the slope $M^{-1/2}$ dashed. (b) Infidelity of maximum likelihood for the three states and of the projection for the Bell state, with the slope M^{-1} dashed. Compare the slopes of the two estimators in (b).

In Figure 16 all three estimators of the Bell state follow the slope -0.50 in the trace distance and differ by a constant factor, the projection below linear inversion and maximum likelihood lowest. The infidelity of maximum likelihood for the Bell state follows the fitted slope -1.00 and reaches about 1×10^{-4} at $M = 2000$, so ten times more shots reduce the infidelity tenfold and the trace distance only about threefold. For the Werner and random states most maximum-likelihood estimates at the smallest budget lie on the boundary of the physical states, with a spurious zero eigenvalue. This bias flattens the slopes fitted over the three budgets to -0.81 for the Werner state and -0.94 for the random state, below the value -1 expected at large M, which these data do not reach.

2.8 The cost wall

The resources of tomography grow quickly with the number of qubits, since there are 3^N settings with 2^N outcomes each, and the density matrix has $4^N - 1$ real parameters, which take $16 \cdot 4^N$ bytes in double precision. Every $R\rho R$ step evaluates the 6^N probabilities and assembles $\hat{R}$ from as many projectors, which costs of order $3^N \cdot 4^N = 12^N$ operations when the product structure (2.7) is used. The matrix products of (2.23) add only of order 8^N. The number of shots grows even faster at fixed accuracy. By (2.6) the squared Hilbert–Schmidt error of linear inversion is $D_{\mathrm{HS}}^2 = 2^{-N} \sum_{P \neq I} (\tilde{r}_P - r_P)^2$, and its mean follows from (2.13),

$$\mathbb{E}\, D_{\mathrm{HS}}^2 = \frac{1}{2^N} \sum_{P \neq I} \frac{1 - r_P^2}{3^{N-k} M} \leq \frac{1}{2^N M} \sum_{k=1}^{N} \binom{N}{k} \frac{3^k}{3^{N-k}} = \frac{10^N - 1}{6^N M}. \tag{2.30}$$

The inequality counts the $\binom{N}{k}3^k$ strings of weight k and becomes an equality for the maximally mixed state, where all r_P with $P \neq I$ vanish. The last step uses $\sum_{k=1}^{N}\binom{N}{k}9^k = 10^N - 1$. With the total budget $M_{\mathrm{tot}} = 3^N M$ the error becomes $\mathbb{E}\, D_{\mathrm{HS}}^2 \approx 5^N / M_{\mathrm{tot}}$ for large N, so at a fixed total number of shots every additional qubit multiplies the squared error by about five. Simulations for one, two and three qubits agree with (2.30) and disagree with a count that ignores the factor 3^{N-k}, as if every string were measured by a single setting.

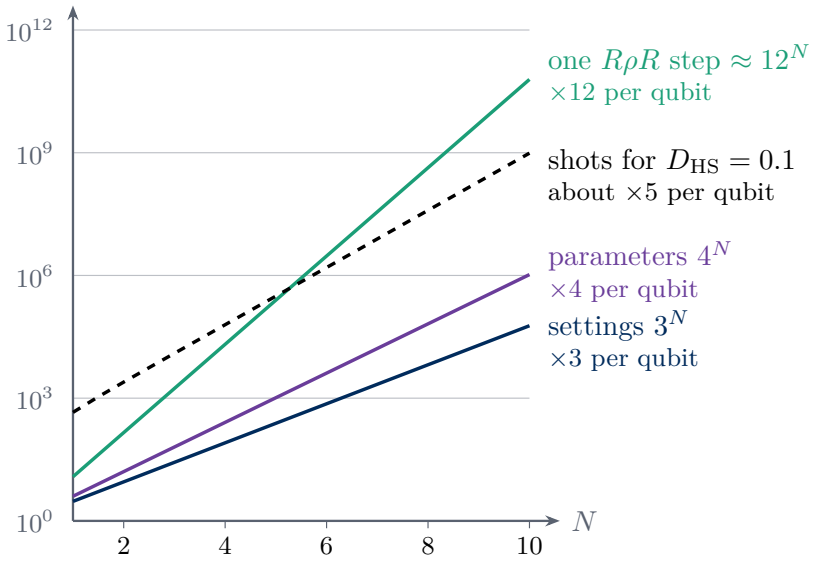


Figure 17: Resources of full tomography against the number of qubits on a logarithmic scale. The number of settings grows by a factor three per qubit, the number of parameters by four and the cost of one $R\rho R$ step by twelve. At a fixed total number of shots each setting receives a fraction 3^{-N} of them.

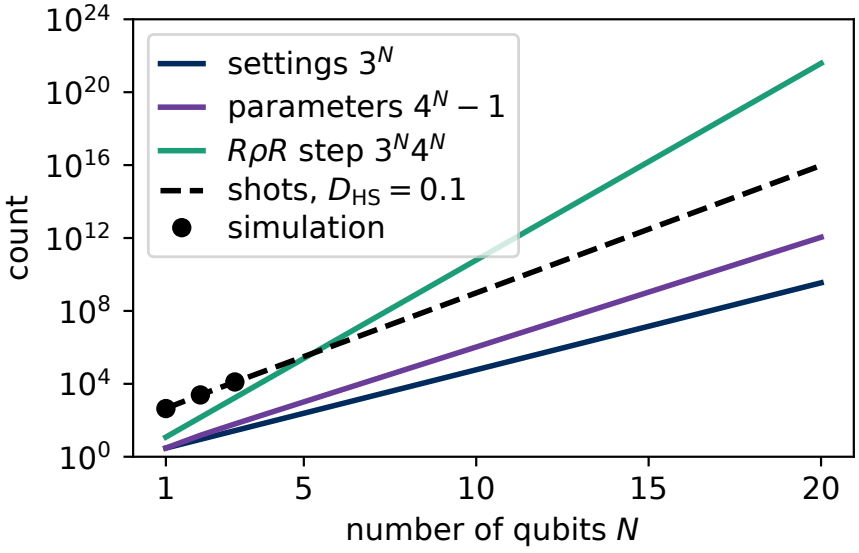


Figure 18: Resources of full tomography from (2.30) and the counts in the text, namely settings 3^N, parameters $4^N - 1$, operations $3^N 4^N$ of one $R\rho R$ step (dashed) and the total number of shots that gives $D_{\mathrm{HS}} = 0.1$ for the maximally mixed state. Circles mark the simulated values for $N = 1, 2, 3$.

As Figures 17 and 18 show, ten qubits already need about 1×10^9 shots for a Hilbert–Schmidt error of 0.1, and fifty qubits lie beyond any experiment. Full tomography is therefore limited to a few qubits. Larger systems require either assumptions about the state, such as a low rank in compressed sensing [20,21], or a smaller goal, the prediction of selected properties without the full state.

3 Classical shadows

Many questions about a device of many qubits need much less than the full state, such as its energy, the correlation between two spins or the magnetisation of each site. Classical shadows answer such questions from one set of randomised measurements without reconstructing the state [22,23], and they belong to a wider family of protocols with randomised measurements [24]. Each qubit is measured in a randomly drawn basis, and from the recorded bases and outcomes one formula predicts the expectation value of any Pauli string. The number of runs needed for a string acting on k qubits grows as 3^k, independently of the total number of qubits.

3.1 Random measurements and snapshots

In every run each qubit j is measured in a basis X, Y or Z drawn at random with equal probability, by the rotation $\hat{u}_j$ of (1.3) followed by a measurement in the computational basis. The record of run m consists of the N bases and the N bits of its bit string (Figure 19), and each run uses a fresh random setting.

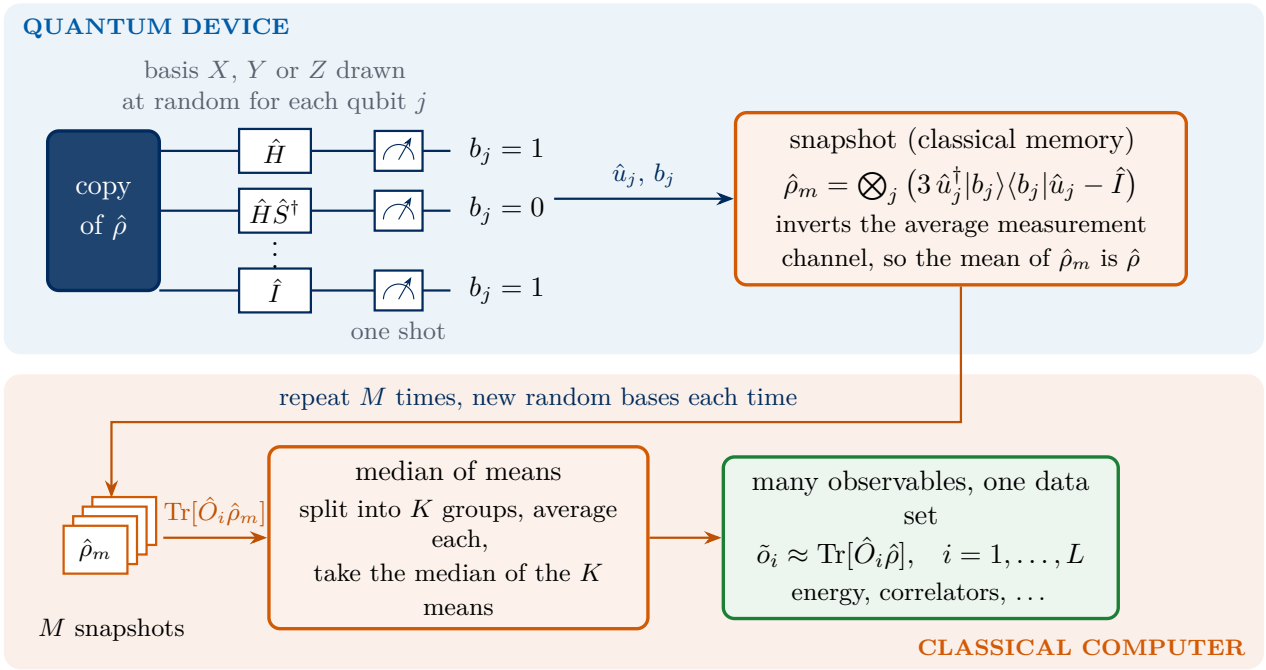


Figure 19: Classical shadows. Every run measures each qubit in a random basis and records the bases $\hat{u}_j$ and the bits b_j. Each record defines a snapshot $\hat{\rho}_m$ whose mean over the random bases and outcomes is the state, and many observables are estimated from the same records with the median of means.

To see how a record represents the state, we take one qubit in the state $\hat{\rho} = \frac{1}{2}(\hat{I} + \sum_P r_P \hat{P})$ and average the post-measurement state $\hat{u}^\dagger|b\rangle\langle b|\hat{u} = \hat{\pi}_{P,b}$ of (1.4) over the random basis and the outcome. Basis P is drawn with probability $1/3$ and outcome b follows with probability $\mathrm{Tr}(\hat{\rho}\,\hat{\pi}_{P,b}) = [1 + (-1)^b r_P]/2$. The average is a linear map of $\hat{\rho}$, called the measurement channel,

$$\mathcal{M}(\hat{\rho}) = \frac{1}{3}\sum_{P=X,Y,Z}\sum_{b=0,1}\mathrm{Tr}(\hat{\rho}\,\hat{\pi}_{P,b})\,\hat{\pi}_{P,b} = \frac{1}{3}\sum_P \frac{\hat{I} + r_P\hat{P}}{2} = \frac{\hat{\rho} + \hat{I}}{3}. \tag{3.1}$$

The middle step sums over the outcome,

$$\sum_{b=0,1}\frac{1 + (-1)^b r_P}{2}\,\frac{\hat{I} + (-1)^b\hat{P}}{2} = \frac{\hat{I} + r_P\hat{P}}{2}, \tag{3.2}$$

because the cross terms with $(-1)^b$ cancel and $[(-1)^b]^2 = 1$. The last step uses $\sum_P r_P\hat{P} = 2\hat{\rho} - \hat{I}$. The averaged post-measurement state is thus the true state mixed with the identity in a known way,

which can be undone. Solving (3.1) for $\hat\rho$ gives the inverse channel

$$\mathcal{M}^{-1}(\hat A) = 3\hat A - \mathrm{Tr}(\hat A)\,\hat I. \tag{3.3}$$

For N qubits with independent random bases the channel is a product of one-qubit channels, and applying its inverse to a single record gives the snapshot of run m,

$$\hat\rho_m = \bigotimes_{j=0}^{N-1} \left(3\,\hat u_j^\dagger |b_j\rangle\langle b_j| \hat u_j - \hat I\right), \qquad \mathbb{E}[\hat\rho_m] = \hat\rho, \tag{3.4}$$

where the mean runs over the random bases and the measurement outcomes. A snapshot is no density matrix, since each factor has the eigenvalues 2 and -1, but its mean is the state. It is stored as the record itself, never as a $2^N \times 2^N$ matrix. Figure 20 shows the channel and its inverse on the Bloch ball. Each one-qubit factor has $\mathrm{Tr}(3\hat\pi - \hat I)^2 = 4 + 1 = 5$, so the mean $\bar\rho_M = \frac{1}{M}\sum_{m=1}^{M} \hat\rho_m$ of M one-qubit snapshots has the mean squared Hilbert–Schmidt error

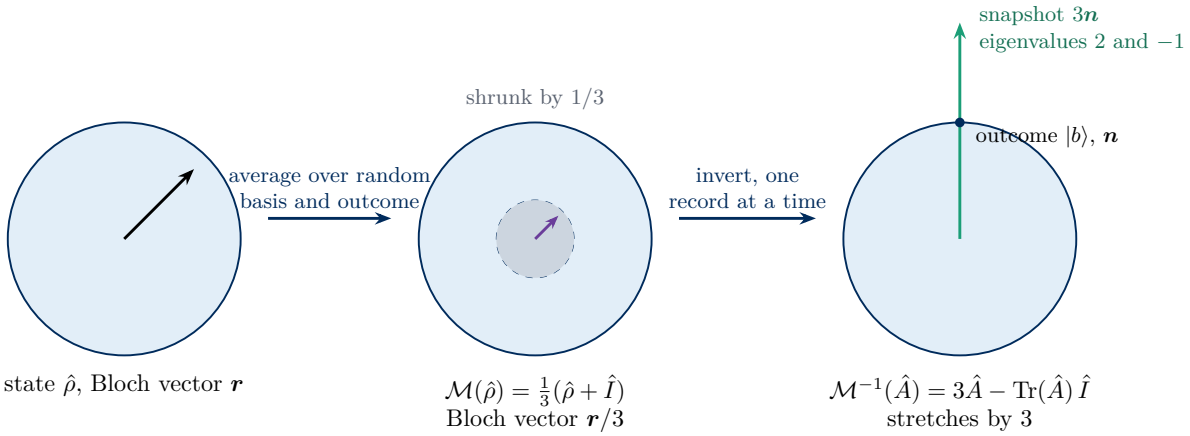


Figure 20: The measurement channel and its inverse for one qubit. Averaged over the random basis and the outcome, the post-measurement state has the Bloch vector $\mathbf{r}/3$ (3.1). The inverse channel (3.3) stretches each single record by three, so a snapshot has the Bloch vector $3\mathbf{n}$ of length three and the eigenvalues 2 and -1, and only the mean over many records is the state.

$$\mathbb{E}\,\|\bar\rho_M - \hat\rho\|_{\mathrm{HS}}^2 = \frac{\mathbb{E}\,\mathrm{Tr}\hat\rho_m^2 - \mathrm{Tr}\hat\rho^2}{M} = \frac{5 - \mathrm{Tr}\hat\rho^2}{M}, \tag{3.5}$$

the variance of a mean of M independent random matrices. Figure 21 confirms (3.5) for a qubit with Bloch vector $(0.3, -0.5, 0.6)$. Without the inverse channel the mean of the post-measurement states converges to $(\hat\rho + \hat I)/3$ and stays at the distance 0.39 from the state for every M.

3.2 Pauli strings from snapshots

A Pauli string $\hat P$ of weight k has non-identity letters P_j on the k qubits of its support. Its value in the snapshot (3.4) is a product of one-qubit traces. Outside the support $\mathrm{Tr}[\hat I(3\hat\pi - \hat I)] = 3 - 2 = 1$. On the support, by (1.4), $\mathrm{Tr}[\hat P_j(3\hat\pi_{u_j,b_j} - \hat I)] = 3(-1)^{b_j}$ when the measured basis u_j equals P_j, and zero otherwise, because different Pauli matrices are orthogonal. The value of run m is therefore

$$o_m(\hat P) = \mathrm{Tr}[\hat P\hat\rho_m] = \prod_{j\in\mathrm{supp}\,\hat P} 3\,\delta_{u_j,P_j}\,(-1)^{b_j}, \tag{3.6}$$

and the estimate from M runs is the sample mean $\bar o(\hat P) = \frac{1}{M}\sum_m o_m(\hat P)$. The value is $\pm 3^k$ when the random bases agree with the string on all k qubits of its support, which happens with probability 3^{-k}, and zero otherwise (Figure 22). Its second moment and variance are therefore

$$\mathbb{E}[o_m^2] = 3^{-k}\,9^k = 3^k, \qquad \mathrm{Var}(o_m) = 3^k - \langle\hat P\rangle^2, \qquad \sigma_{\bar o} = \sqrt{\frac{3^k - \langle\hat P\rangle^2}{M}}, \tag{3.7}$$

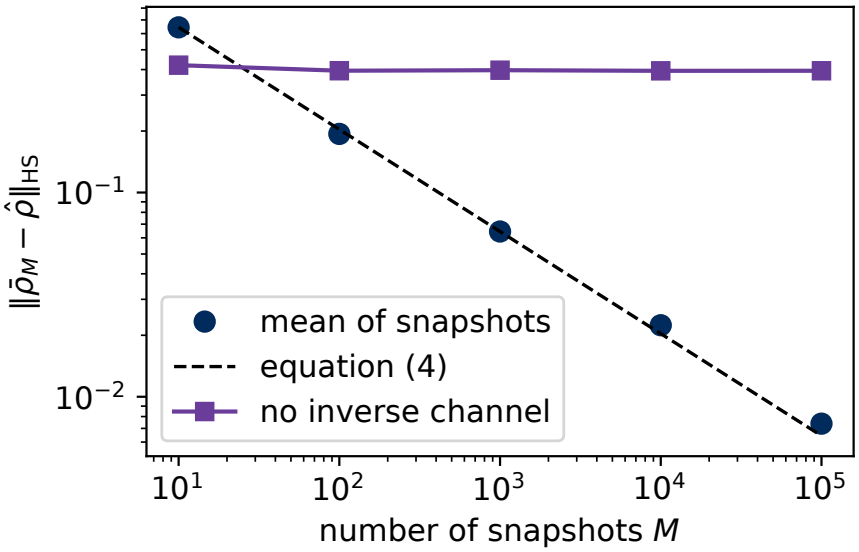
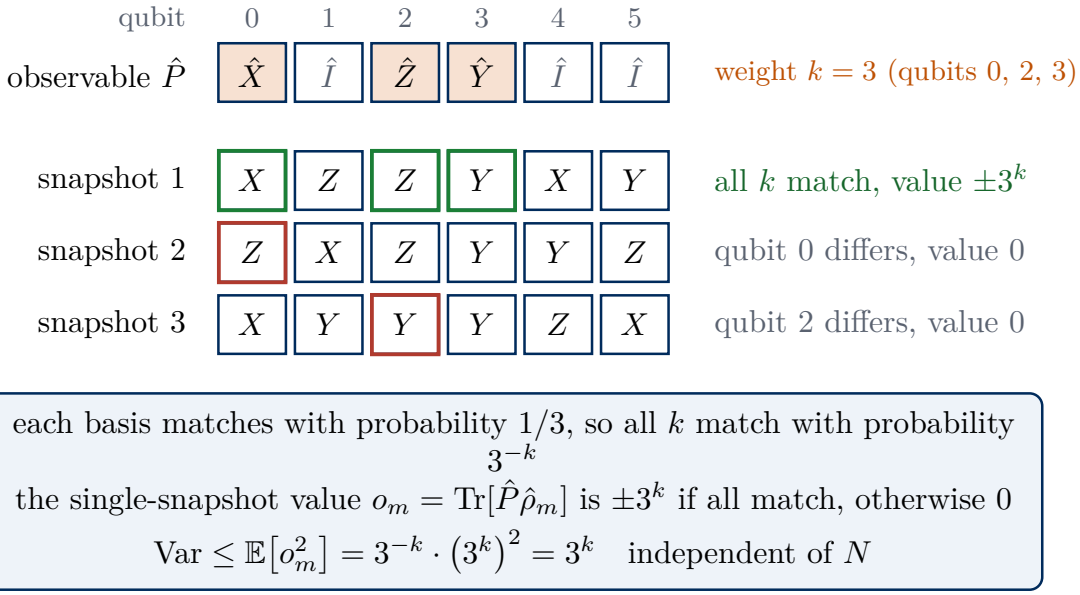


Figure 21: One qubit with Bloch vector $(0.3, -0.5, 0.6)$. Root-mean-square Hilbert–Schmidt distance between the mean of M snapshots (3.4) and the true state over 40 simulated data sets (circles), the prediction (3.5) (dashed) and the mean of the post-measurement states without the inverse channel (squares), which stops at the distance between $(\hat{\rho} + \hat{I})/3$ and $\hat{\rho}$.



each basis matches with probability $1/3$, so all k match with probability 3^{-k}

the single-snapshot value $o_m = \mathrm{Tr}[\hat{P}\hat{\rho}_m]$ is $\pm 3^k$ if all match, otherwise 0

$$\mathrm{Var} \le \mathbb{E}\left[o_m^2\right] = 3^{-k} \cdot \left(3^k\right)^2 = 3^k \quad \text{independent of } N$$

Figure 22: A string of weight k is seen only by the runs whose random bases agree with it on all k qubits of its support. Such a run contributes $\pm 3^k$, all others contribute zero, and the second moment 3^k does not depend on the number of qubits.

and none of them depends on the number of qubits N.

We check (3.6) and (3.7) on the ground state of the transverse-field Ising chain with open ends [25],

$$\hat{\mathcal{H}}_{\mathrm{TFIM}} = -J \sum_{i=0}^{N-2} \hat{Z}_i \hat{Z}_{i+1} - h \sum_{i=0}^{N-1} \hat{X}_i, \tag{3.8}$$

with $N = 8$ and $J = h = 1$, the critical point of the infinite chain. The coupling J favours neighbouring spins aligned along z, the field h turns every spin along x, and at $h = J$ their competition produces correlations that decay slowly with distance. The exact ground state, with energy $E_0 = -9.8380J$, comes from sparse diagonalisation [26] and serves for comparison. The simulated device prepares this state in each of a pool of 100 000 runs, draws the bases, rotates the amplitudes and samples one bit string per run. For the strings $\hat{Z}_0\hat{Z}_1 \cdots \hat{Z}_{k-1}$ in Figure 24 the measured second moment follows

3^k over almost four decades, and the estimates agree with the exact values. The ground state is an eigenstate of the parity $\hat{\Pi} = \prod_i \hat{X}_i$, which flips every $\hat{Z}_i$, so the strings of odd weight have expectation value zero. For $k = 8$ only 15 runs of the pool have matching bases, and the standard error 0.25 is comparable to the expectation value. Figure 23 shows the chain.

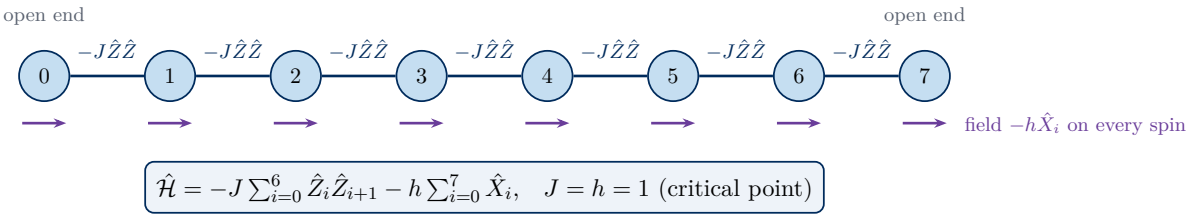


Figure 23: The open transverse-field Ising chain (3.8) of eight spins used as the test state of classical shadows. Neighbouring spins are coupled by $-J\hat{Z}_i\hat{Z}_{i+1}$, and a field $-h\hat{X}_i$ acts on every spin, at the critical point $J = h = 1$.

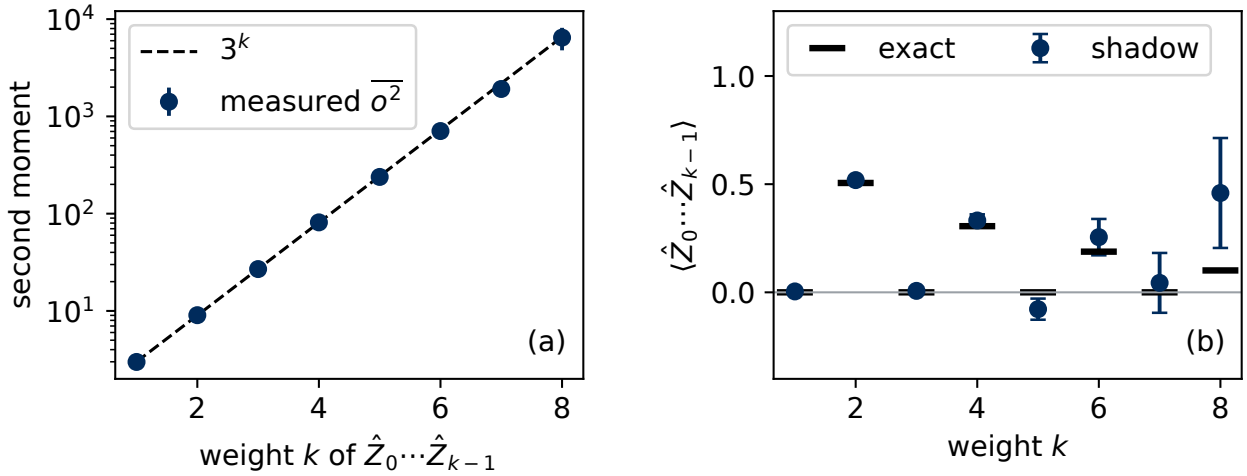


Figure 24: Snapshots of the critical Ising ground state, $N = 8$, from a pool of 10^5 simulated runs. (a) Second moment of the single-run values (3.6) of $\hat{Z}_0 \cdots \hat{Z}_{k-1}$ against the weight k, with 3^k dashed. (b) Estimates of $\langle \hat{Z}_0 \cdots \hat{Z}_{k-1} \rangle$ with the standard errors (3.7) and the exact values as black marks. The equation numbers in the legends are those of the notebook. The error bars grow by $\sqrt{3}$ per added qubit of the support.

3.3 Many observables and the median of means

Since one data set serves all observables at once, we need the number of runs that keeps L estimates all within ε of their exact values with probability at least $1 - \delta$. For a single sample mean with variance σ^2/M the Chebyshev inequality bounds the probability of an error larger than ε by $\sigma^2/(M\varepsilon^2)$. This bound decreases only as $1/M$, and a union over L observables would require $M \propto L$. The median of means [27,28] converts it into an exponential bound. The M runs are split into K groups of M/K runs, the sample mean $\bar{o}^{(g)}$ is formed in every group g, and the estimate is the median of the K group means,

$$\tilde{o}_{\mathrm{MoM}} = \mathrm{median}\left(\bar{o}^{(1)}, \ldots, \bar{o}^{(K)}\right). \tag{3.9}$$

If every group has $M/K \geq 4\sigma^2/\varepsilon^2$ runs, Chebyshev's inequality makes each group mean wrong by more than ε with probability at most $1/4$. The median is wrong only if at least half of the K independent groups are wrong, and by Hoeffding's inequality [29] for K independent events of probability at most $1/4$ this happens with probability at most $e^{-2K(1/4)^2} = e^{-K/8}$. A union over L observables bounds the probability that any of them fails by $Le^{-K/8}$, which is at most δ for $K = 8\ln(L/\delta)$. With $\sigma^2 \leq 3^k$ from (3.7) the total number of runs becomes

$$M = K\,\frac{4 \cdot 3^k}{\varepsilon^2} = \frac{32 \cdot 3^k\,\ln(L/\delta)}{\varepsilon^2} = O\Big(\frac{3^k \ln(L/\delta)}{\varepsilon^2}\Big) \tag{3.10}$$

for L Pauli strings of weight at most k [22]. The number of observables enters only through its logarithm, and the number of qubits does not enter at all.

We compare the plain mean with the median of means on all $L = 276$ Pauli strings of weight one and two of the Ising chain, with 50 independent data sets of 2000 runs and $K = 10$ groups. The largest error over all strings is on average 0.21 for the mean and 0.24 for the median of means (Figure 25). For Pauli strings the plain mean is as accurate or slightly more accurate, because the single-run values are bounded by 3^k and their means are close to Gaussian already for 2000 runs, whereas each group of the median of means has only 200 runs. The median of means helps when single-run values have rare large outliers, as for general observables, and it makes the logarithmic bound (3.10) provable. We therefore estimate Pauli strings with the plain mean and its standard error.

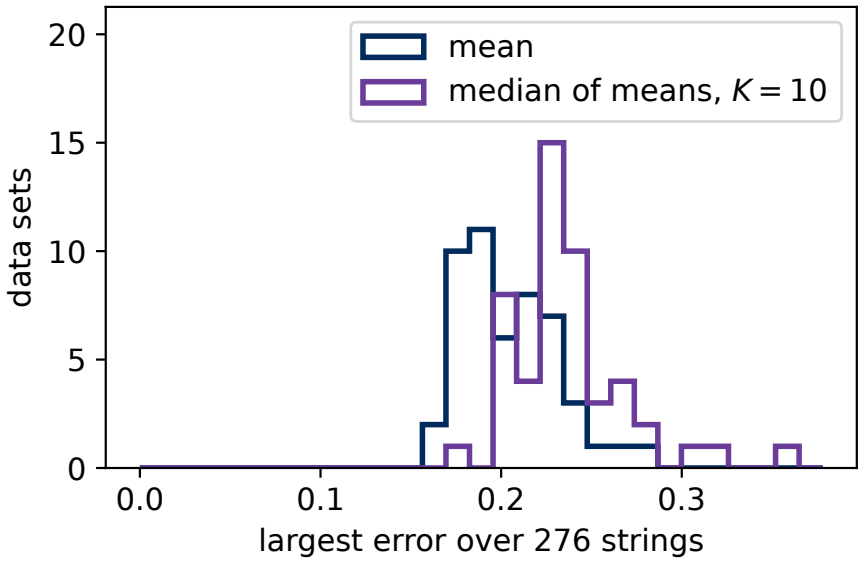


Figure 25: Histogram over 50 simulated data sets of 2000 runs of the largest error among the 276 Pauli strings of weight one and two of the critical Ising chain, $N = 8$, for the plain mean and for the median of means with $K = 10$ groups.

3.4 Energy and correlations of the Ising chain

We take the first 20 000 runs of the pool as one experiment and estimate from it the energy, the magnetisation $\langle \hat{X}_i \rangle$ of every site and the correlations $\langle \hat{Z}_i \hat{Z}_j \rangle$ of all 28 pairs of sites. By (3.8) and (3.6) every run gives an energy value

$$e_m = -J \sum_{i=0}^{N-2} o_m(\hat{Z}_i \hat{Z}_{i+1}) - h \sum_{i=0}^{N-1} o_m(\hat{X}_i), \tag{3.11}$$

and the energy estimate is the sample mean of e_m with the standard error $\mathrm{std}(e_m)/\sqrt{M}$, which includes the correlations between the terms of one run. The result is $\tilde{E} = -9.943 \pm 0.065$, within two standard errors of the exact $E_0 = -9.8380$. Without the inverse channel the same data give -2.43, many standard errors away. The record was not planned for these observables and serves any other Pauli string as well.

In Figure 26 the correlations decay with distance and are smaller, at equal distance, for pairs that include an open end, where the spin has only one neighbour. The transverse magnetisation is enhanced at the ends, where the field competes with only one $\hat{Z}\hat{Z}$ bond. All 36 estimates agree with the exact values within two standard errors, and the mean square of the deviations in units of the standard error is 1.00, as expected for correct error bars. The error bars are equal for all pairs, because every string of weight two has the variance $9 - \langle \hat{P} \rangle^2$ regardless of the distance between its

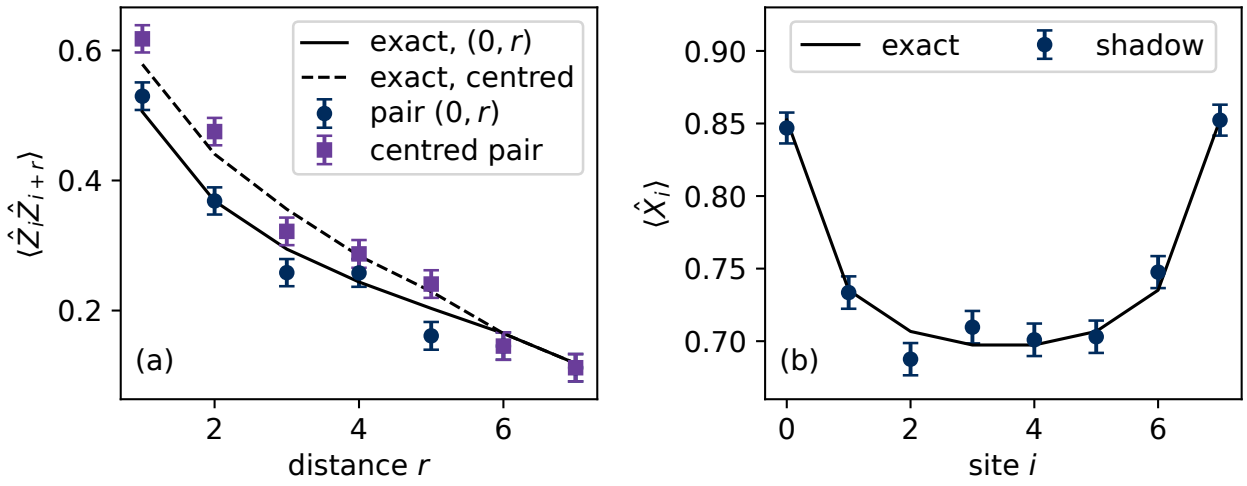


Figure 26: Properties of the critical Ising ground state, $N = 8$, from one simulated data set of 20 000 snapshots, with exact values as lines. (a) Correlations $\langle \hat{Z}_i \hat{Z}_{i+r} \rangle$ against the distance r for pairs that start at the edge of the chain (circles) and for pairs centred in the chain (squares). (b) Transverse magnetisation $\langle \hat{X}_i \rangle$ of every site.

qubits. By (3.7) every error decreases as $M^{-1/2}$. In independent blocks of runs (Figure 27) the fitted slopes, -0.52 for the energy and -0.47 for a correlation, agree with $-1/2$. The energy error is about three times larger than that of a single correlation, because the energy sums fifteen terms.

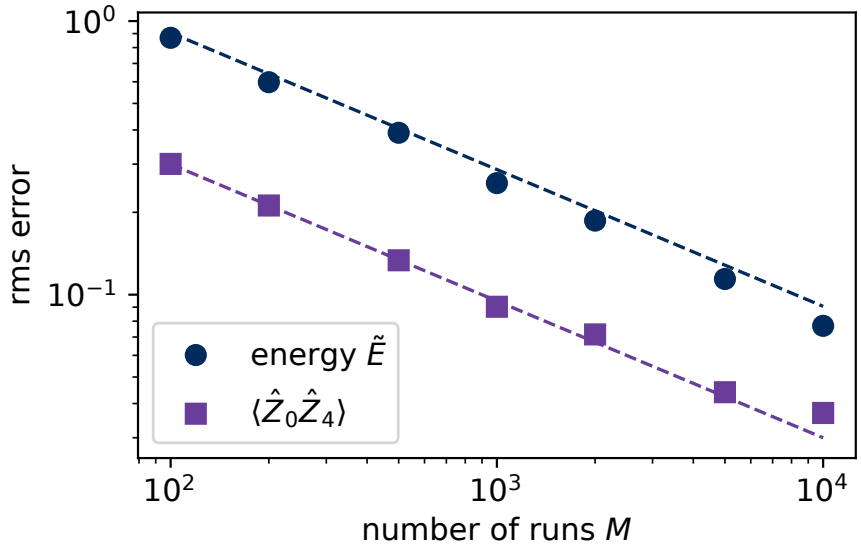


Figure 27: Root-mean-square error of the shadow estimates of the energy and of $\langle \hat{Z}_0 \hat{Z}_4 \rangle$ against the number of runs M, from independent blocks of the simulated pool, with the predictions $\mathrm{std}(o_m)/\sqrt{M}$ dashed.

3.5 Independence of the system size

The variance (3.7) depends on the weight of the string alone. A local observable such as the nearest-neighbour correlation $\langle \hat{Z}_0 \hat{Z}_1 \rangle$ is therefore estimated from M runs with the same error for every chain length, whereas a string acting on all qubits, such as the parity $\hat{\Pi}$, has the variance $3^N - 1$ in the ground state, an eigenstate of $\hat{\Pi}$ with eigenvalue $+1$. With $M = 5000$ runs for the ground states of chains of $N = 2, 4, \ldots, 10$ qubits, the standard error of $\langle \hat{Z}_0 \hat{Z}_1 \rangle$ stays between 0.041 and 0.042 for all lengths, whereas the predicted error of the parity, $\sqrt{(3^N - 1)/M}$, grows by $\sqrt{3}$ per qubit and reaches 3.4 at $N = 10$ (Figure 28). For $N \geq 8$ the 5000 runs contain no run with all bases equal to X, so the parity cannot be estimated and only the prediction is shown.

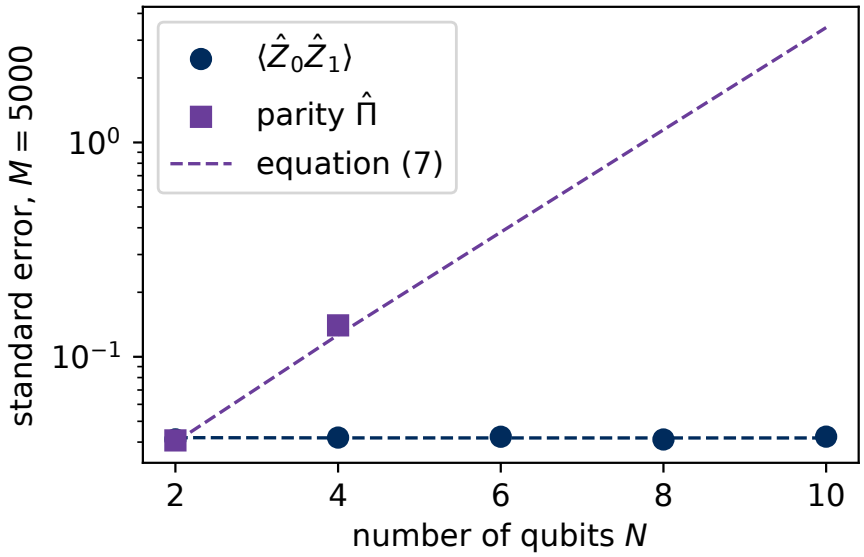


Figure 28: Standard error of shadow estimates from $M = 5000$ simulated runs against the number of qubits of the critical Ising chain, for the local correlation $\langle \hat{Z}_0 \hat{Z}_1 \rangle$ (circles) and for the parity $\hat{\Pi} = \prod_i \hat{X}_i$ (squares), with the predictions (3.7) dashed. Parity points are shown only where enough runs match all bases.

The same exponential growth makes the fidelity with a Greenberger–Horne–Zeilinger (GHZ) state $(|0 \ldots 0\rangle + |1 \ldots 1\rangle)/\sqrt{2}$ [30] hard to estimate from random single-qubit bases, since its projector contains strings of weight N. Random global Clifford circuits in place of single-qubit rotations make such fidelities cheap, at the price of a deep entangling circuit before every measurement [22].

3.6 Shadows and tomography at an equal number of runs

For two qubits we compare shadows with full tomography of the Bell state at the same total number of runs M_{tot}. Tomography measures each of the nine settings $M_{\text{tot}}/9$ times and estimates each string from the 3^{N-k} settings that measure it, so a string of weight k receives a fixed fraction 3^{-k} of the runs and by (2.13) $\text{Var}(\tilde{r}_P) = 3^k(1 - r_P^2)/M_{\text{tot}}$. Shadows draw a random setting for every run, and the number of matching runs is itself random, which gives $\text{Var}(\tilde{r}_P) = (3^k - r_P^2)/M_{\text{tot}}$ by (3.7). The two variances agree for strings with $r_P = 0$, whereas tomography is more precise for strings with $r_P = \pm 1$, such as $\hat{X}\hat{X}$, $\hat{Y}\hat{Y}$ and $\hat{Z}\hat{Z}$ of the Bell state, whose fixed settings return the same parity in every run. Summed over the fifteen strings, M_{tot} times the variances give $6 \cdot 3 + 9 \cdot 9 - 3 = 96$ for shadows and $6 \cdot 3 + 6 \cdot 9 = 72$ for tomography. The mean squared Hilbert–Schmidt error of shadows is therefore larger by $96/72 = 4/3$, and the simulated ratios at the three budgets $M_{\text{tot}} = 90$, 900 and 9000 are 1.24, 1.42 and 1.31, scattered around 4/3 by the finite number of 200 data sets (Figure 29).

For two qubits shadows are therefore a randomised form of tomography without a gain, and their advantage appears for many qubits. Tomography needs all 3^N settings and a reconstruction of 4^N numbers, whereas shadows predict any set of local observables from the same records with an error independent of N.

4 Neural networks for tomography

Full tomography of a few qubits remains the method of choice when the complete state is needed, for instance to certify a gate or to quantify entanglement. Its weak point is the regime of few shots per setting, where linear inversion is noisy and unphysical and the physical estimators are systematically displaced from the true state. A neural network trained on simulated pairs of a noisy estimate and the true state can learn to correct this displacement. We follow the attention-based method of Ref. [1] in a reduced form, with our own two-qubit and four-qubit simulations reported separately from the published results.

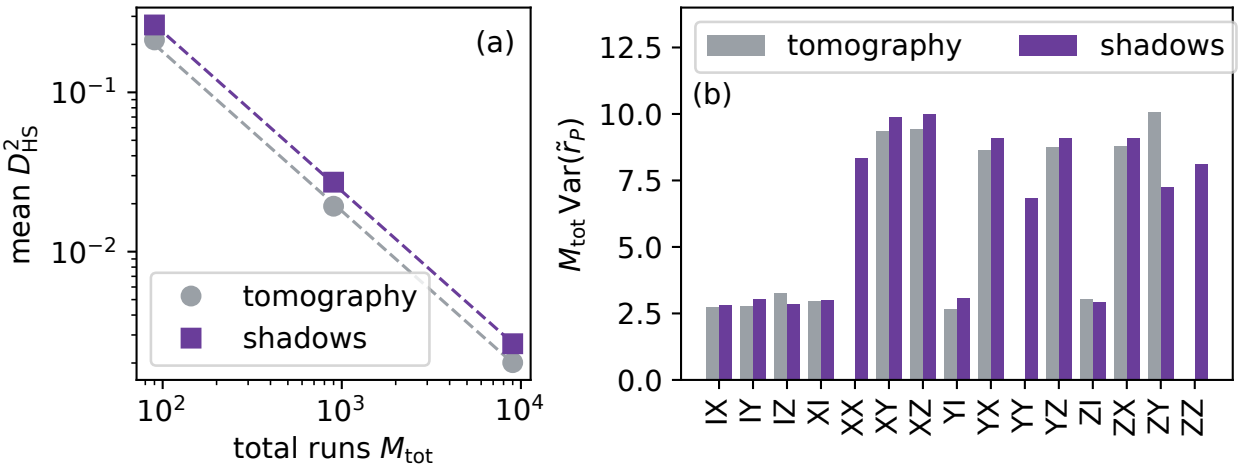


Figure 29: Shadows and tomography of the Bell state at an equal total number of simulated runs. (a) Mean squared Hilbert–Schmidt error over 200 data sets per budget, with the predictions dashed. (b) Variance of every Pauli coefficient times M_{tot} at the largest budget. Look at $\hat{X}\hat{X}$, $\hat{Y}\hat{Y}$ and $\hat{Z}\hat{Z}$, which tomography measures without error.

4.1 The bias of physical estimates

Linear inversion (2.14) is unbiased, since it is linear in the frequencies and the mean frequency is the probability. For a pure state it has a negative eigenvalue in almost every run. The projection (2.18) sets the negative and the smallest positive eigenvalues to zero and lowers the remaining ones, and its result is pure only in a minority of runs. A pure state is an extreme point of the convex set of density matrices, i.e. it cannot be written as a mixture $\sum_m w_m \hat{\sigma}_m$ of other physical states. A mean of physical estimates therefore equals a pure state only if every estimate equals it, and any spread of the estimates moves their mean towards mixed states with a purity below one. The errors of many repetitions do not average out, and this systematic error is called a bias. It is largest for states on the boundary of the state space, such as pure states, and every estimator that always returns a physical state, maximum likelihood included, has such a bias near the boundary [1]. Figure 30 illustrates the argument on the Bloch ball.

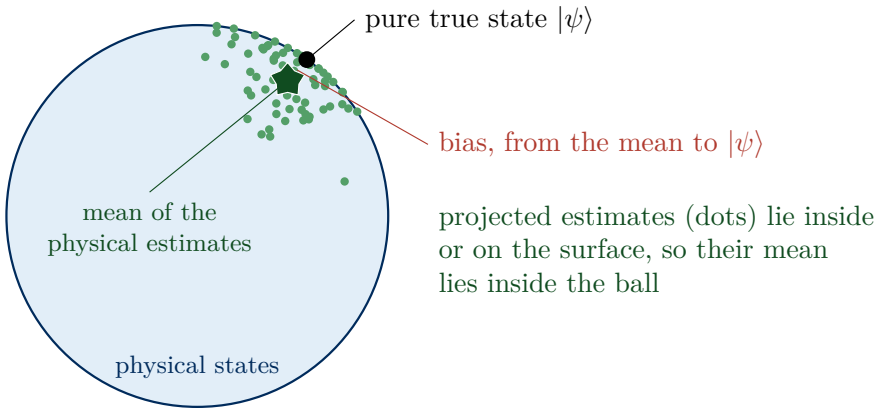


Figure 30: Bias of physical estimates of a pure state. The pure state lies on the surface of the Bloch ball. Projected estimates (dots) lie inside or on the surface, so their mean (star) lies inside, displaced from the pure state by the bias.

For 1000 random pure states of two qubits every linear-inversion estimate is unphysical at all three budgets, and after the projection only a fraction 0.21, 0.11 and 0.08 of the estimates is pure for $M = 10$, 100 and 1000. The mean purity of the projected estimates stays below one at every budget, from 0.88 at $M = 10$ to 0.979 at $M = 1000$ (Figure 31), so the displacement shrinks with M and always has the same sign. Repeating the measurement of one fixed pure state 1000 times at $M = 100$

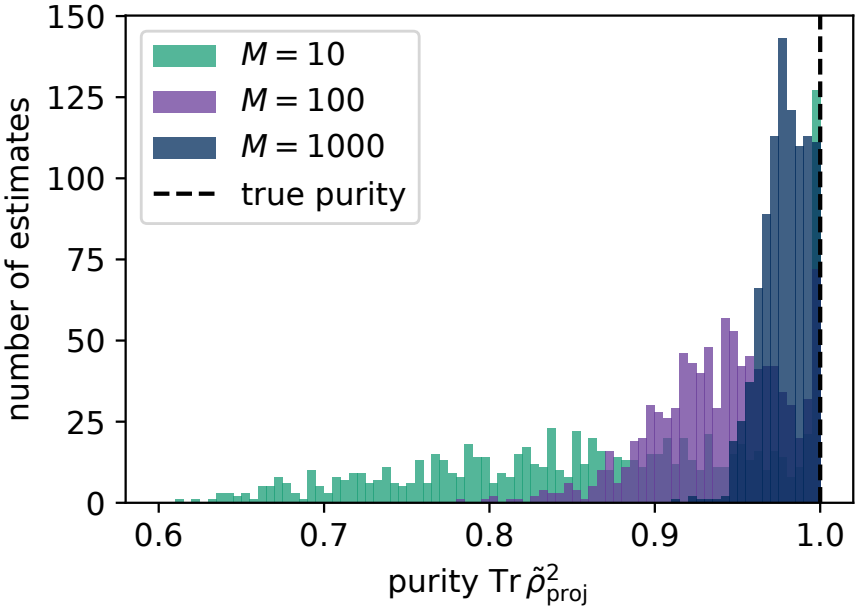


Figure 31: Purity $\mathrm{Tr}\tilde{\rho}^2_{\mathrm{proj}}$ of the projected linear-inversion estimates of 1000 simulated uniformly distributed (Haar-random) pure states of two qubits for $M = 10$, 100 and 1000 shots per setting. The true purity is one (dashed). The distributions move towards one with M, always from below.

shows the bias as defined above. The mean of the linear-inversion estimates lies at a Hilbert–Schmidt distance of 0.006 from the true state, within the statistical error of 0.005, whereas the means of the projected and of the maximum-likelihood estimates lie at 0.043 and 0.033, with purities 0.929 and 0.948.

The bias depends on the unknown true state, about which the estimate carries information. Suppose that the true states $\hat{\rho}$ are drawn from a known distribution and that every state produces an estimate $\tilde{\rho}$ through the simulated measurement. A function f that maps the estimate to a corrected state and minimises the mean squared error $\mathbb{E}\,\|f(\tilde{\rho}) - \hat{\rho}\|^2$ over this distribution is the conditional mean,

$$f^\star(\tilde{\rho}) = \mathbb{E}[\hat{\rho}\,|\,\tilde{\rho}], \tag{4.1}$$

because for every fixed $\tilde{\rho}$ the mean squared distance to a random $\hat{\rho}$ is smallest at its mean. A network trained with a squared-error loss on many pairs $(\tilde{\rho}, \hat{\rho})$ approximates $f^\star$. It returns the average of the true states that produce estimates similar to the given one, a correction called a conditional debiaser in Ref. [1]. The mean in (4.1) runs over the true states that are compatible with one given estimate. For an informative estimate these states lie close together, so their mean stays close to a pure state when the true states are pure. The bias above is a mean over the noise for one fixed true state, which is a different average. The same argument holds for any representation of the state in which the squared error is taken, and the method of this section takes it between the entries of a triangular factor of the state. The correction depends on the distribution of the training states, which acts as a prior, so tests on states from other families are needed. Figure 32 shows the protocol.

4.2 Distances between quantum states

The loss of the network compares its output with the true state for every training state at every training step, and its gradient drives the training. A suitable loss has a physical meaning, so that a small value implies that the two states are hard to distinguish by any measurement. It costs little more than one pass over the d^2 entries of a state, and its derivative with respect to the output stays bounded when the output approaches the target. We examine three distances between states against these requirements.

The Bures distance follows from the geometry of the Hilbert space. The vectors $|\psi\rangle$ and $e^{i\alpha}|\psi\rangle$

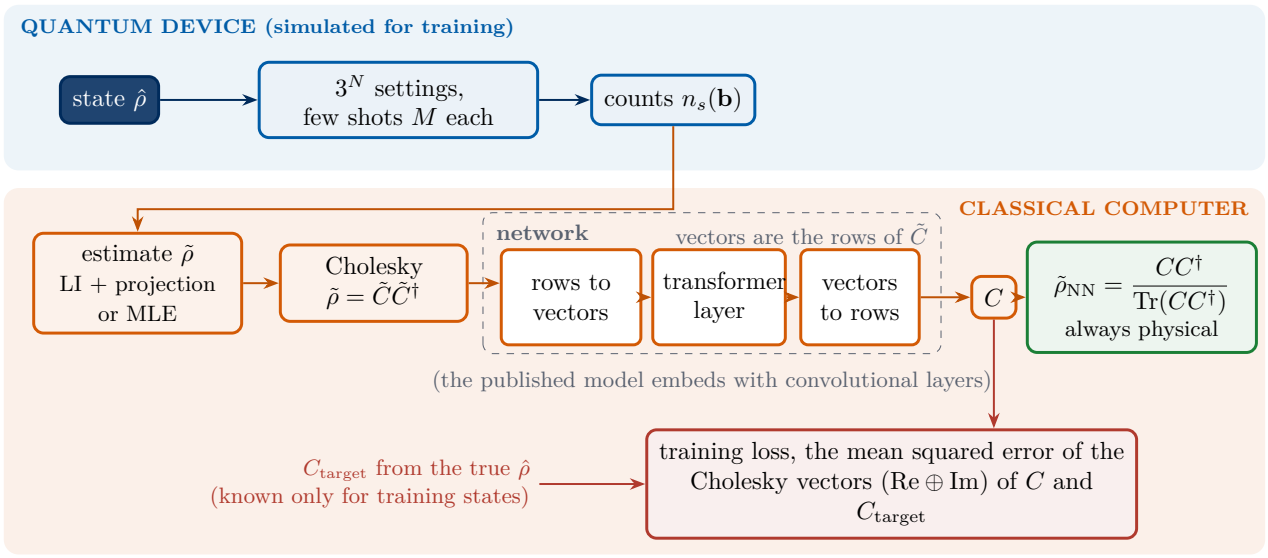


Figure 32: Neural-network tomography following Ref. [1], with the layers of our reduced network inside the dashed box. Counts from the 3^N settings give an estimate $\tilde{\rho}$ by linear inversion with projection or by maximum likelihood. The network maps its Cholesky factor $\tilde{C}$ to a factor C, which always defines a physical state $\tilde{\rho}_{\mathrm{NN}}$. Training compares C with the factor C_{target} of the true state, known only for simulated training states.

describe the same state, so a pure state is a ray, the set of all vectors that differ by a phase. The distance between two pure states is the shortest distance between vectors of the two rays,

$$D^2_{\mathrm{ray}}(\psi,\phi) = \min_\alpha \big\| |\psi\rangle - e^{i\alpha}|\phi\rangle \big\|^2 = 2 - 2\max_\alpha \mathrm{Re}\big(e^{i\alpha}\langle\psi|\phi\rangle\big) = 2 - 2\,|\langle\psi|\phi\rangle|. \tag{4.2}$$

It vanishes only for identical states, reaches its largest value $D^2_{\mathrm{ray}} = 2$ for orthogonal states and depends on the states only through the overlap $|\langle\psi|\phi\rangle|$, whose square is the fidelity (2.25) of two pure states. The angle $\arccos|\langle\psi|\phi\rangle|$ between the rays is called the Fubini–Study distance, and for one qubit it is half the angle between the Bloch vectors [31].

A mixed state has no state vector, but it can be written as $\hat{\rho} = AA^\dagger$ with a $d \times d$ matrix A, called an amplitude of $\hat{\rho}$. For a pure state $|\psi\rangle\langle\psi|$ the matrix with $|\psi\rangle$ in its first column and zeros elsewhere is an amplitude. An amplitude is fixed up to a unitary matrix $\hat{U}$, since $A\hat{U}$ gives the same state, and this freedom replaces the phase of a state vector. The Bures distance (2.27) is again the shortest Euclidean distance between the representatives of the two states [16,31],

$$D^2_B(\hat{\rho},\hat{\sigma}) = \min_{\hat{\rho}=AA^\dagger,\ \hat{\sigma}=BB^\dagger} \mathrm{Tr}\big[(A-B)(A-B)^\dagger\big] = 2 - 2\sqrt{F(\hat{\rho},\hat{\sigma})}, \tag{4.3}$$

and for pure states it reduces to (4.2).

To derive the second equality, let A and B be amplitudes of two states of unit trace. Every square matrix is the product of a positive semidefinite matrix and a unitary, $A = \sqrt{AA^\dagger}\,\hat{V}$, which is its polar decomposition. Hence $A = \sqrt{\hat{\rho}}\,\hat{V}$ and $B = \sqrt{\hat{\sigma}}\,\hat{W}$ with unitaries $\hat{V}$ and $\hat{W}$, so that

$$\mathrm{Tr}AB^\dagger = \mathrm{Tr}\big(\sqrt{\hat{\rho}}\,\hat{V}\hat{W}^\dagger\sqrt{\hat{\sigma}}\big) = \mathrm{Tr}\big(\sqrt{\hat{\sigma}}\sqrt{\hat{\rho}}\,\hat{V}\hat{W}^\dagger\big). \tag{4.4}$$

For any matrix Q and any unitary $\hat{U}$ the modulus $|\mathrm{Tr}(Q\hat{U})|$ is at most the sum of the singular values of Q, the trace norm $\mathrm{Tr}\sqrt{Q^\dagger Q}$. To see this, write the singular value decomposition $Q = \sum_k s_k|a_k\rangle\langle b_k|$ with singular values $s_k \geq 0$ and orthonormal vectors $|a_k\rangle$ and $|b_k\rangle$. Then $\mathrm{Tr}(Q\hat{U}) = \sum_k s_k\langle b_k|\hat{U}|a_k\rangle$, and every matrix element of a unitary between normalised vectors has modulus at most one. Equality holds for the unitary that maps every $|a_k\rangle$ to $|b_k\rangle$. For $Q = \sqrt{\hat{\sigma}}\sqrt{\hat{\rho}}$ the trace norm is $\mathrm{Tr}\sqrt{\sqrt{\hat{\rho}}\,\hat{\sigma}\sqrt{\hat{\rho}}} = \sqrt{F}$ by (2.25), hence

$$\mathrm{Re}\,\mathrm{Tr}AB^\dagger \leq |\mathrm{Tr}AB^\dagger| \leq \sqrt{F(\hat{\rho},\hat{\sigma})}, \tag{4.5}$$

with equality for the best choice of $\hat{V}\hat{W}^\dagger$, which is Uhlmann's form of the fidelity [16,31]. Expanding the squared distance of the amplitudes and using the unit traces gives

$$\mathrm{Tr}\big[(A-B)(A-B)^\dagger\big] = 2 - 2\,\mathrm{Re}\,\mathrm{Tr}AB^\dagger \ \geq\ 2 - 2\sqrt{F(\hat{\rho},\hat{\sigma})}, \tag{4.6}$$

and the best unitaries reach the equality, which proves (4.3).

The Bures distance is a metric, and no physical operation applied to both states, such as a unitary evolution or a noise process, increases it. As a loss it is costly, because the fidelity (2.25) needs two matrix square roots, each a diagonalisation with about d^3 operations. The derivative of a matrix square root also diverges for eigenvalues near zero, which occur for nearly pure states.

The trace distance (2.26) equals the largest difference between the probabilities that the two states assign to one measurement outcome,

$$D(\hat{\rho}, \hat{\sigma}) = \max_{\hat{E}} \left| \mathrm{Tr}[\hat{E}\hat{\rho}] - \mathrm{Tr}[\hat{E}\hat{\sigma}] \right|, \tag{4.7}$$

where the maximum runs over all operators $\hat{E}$ with eigenvalues between zero and one, each of which describes one outcome of a measurement. Given one copy of either state with equal prior probabilities, the best measurement identifies the state with probability $\frac{1}{2}(1 + D)$ [32]. No physical operation increases the trace distance, and the fidelity bounds it from both sides [33],

$$1 - \sqrt{F} \leq D \leq \sqrt{1 - F}. \tag{4.8}$$

Its evaluation needs the eigenvalues of $\hat{\rho} - \hat{\sigma}$, again about d^3 operations for every training state at every step. The modulus of an eigenvalue has a kink at zero, the value that a good reconstruction approaches, and derivatives of eigenvalues become unstable when eigenvalues coincide.

The Hilbert–Schmidt distance (2.16) is the Euclidean distance of the two matrices,

$$D_{\mathrm{HS}}^2(\hat{\rho}, \hat{\sigma}) = \mathrm{Tr}\left[(\hat{\rho} - \hat{\sigma})^2\right] = \sum_{a,b=1}^{d} \left|\rho_{ab} - \sigma_{ab}\right|^2 = \frac{1}{2^N} \sum_{\hat{P}} \left(\langle \hat{P} \rangle_\rho - \langle \hat{P} \rangle_\sigma\right)^2, \tag{4.9}$$

where the last form holds for N qubits and follows from the expansion (2.6) of both states in the 4^N Pauli strings $\hat{P}$. It sums the squared differences of all Pauli expectation values, so it measures how much the two states differ for an average observable, whereas the trace distance measures the difference for the best one. A physical operation on both states can increase the Hilbert–Schmidt distance [34], and it controls the trace distance only up to a factor that grows with the dimension, $D_{\mathrm{HS}} \leq 2D \leq \sqrt{d}\, D_{\mathrm{HS}}$. Its evaluation is one sum over the d^2 entries, and it is a polynomial in the entries with a smooth gradient, the squared error that neural-network libraries provide. Table 2 collects the three distances.

	Bures	trace	Hilbert–Schmidt
meaning	distance of rays, overlap of the states	difference for the best measurement	difference for an average observable
increased by a physical operation	never	never	possible
cost per state	two diagonalisations, about d^3	one diagonalisation, about d^3	one sum, about d^2
gradient	diverges near pure states	kinks at zero eigenvalues	smooth

Table 2: The three distances between quantum states as candidates for a training loss.

Applied to the states themselves, each distance fails at least one of the requirements. The loss of Ref. [1] applies the Euclidean distance to one fixed pair of amplitudes, the Cholesky factors of the two states. A fixed pair is at least as far apart as the closest pair in (4.3), so this distance is an upper bound on the squared Bures distance, with the cost and the smoothness of the Hilbert–Schmidt distance. Ref. [1] writes D_B for $2 - 2\sqrt{F}$, which in our notation is D_B^2.

4.3 Cholesky representation and the loss

A network outputs unconstrained real numbers, whereas a density matrix must be Hermitian, of unit trace and with non-negative eigenvalues, and the Cholesky decomposition removes these constraints. Every positive definite matrix can be written uniquely as

$$\hat{\rho} = CC^\dagger, \tag{4.10}$$

with a lower-triangular matrix C whose diagonal is real and positive. For a state with zero eigenvalues, such as a pure state, we factorise $\hat{\rho} + \epsilon\hat{I}$ with $\epsilon = 10^{-6}$ and divide by $\sqrt{1 + d\epsilon}$, so that $\mathrm{Tr}CC^\dagger = 1$. In the opposite direction any complex matrix C defines a physical state,

$$\tilde{\rho}_{\mathrm{NN}} = \frac{CC^\dagger}{\mathrm{Tr}CC^\dagger}, \tag{4.11}$$

because $CC^\dagger$ is Hermitian with non-negative eigenvalues, as in (2.24). The free entries of C are the real parts of the lower triangle including the diagonal and the imaginary parts of the strict lower triangle. These $d(d+1)/2 + d(d-1)/2 = d^2$ real numbers form the Cholesky vector

$$\mathbf{c} = (\mathrm{Re}\, C_{ij})_{i \geq j} \oplus (\mathrm{Im}\, C_{ij})_{i > j}, \tag{4.12}$$

sixteen numbers for two qubits and 256 for four. Let $\mathbf{c}_{\boldsymbol{\theta}}(\tilde{C})$ be the output of the network with parameters $\boldsymbol{\theta}$ for the input factor $\tilde{C}$ of an estimate, normalised to $\|\mathbf{c}_{\boldsymbol{\theta}}\| = 1$, which is $\mathrm{Tr}CC^\dagger = 1$, and let $\mathbf{c}_{\mathrm{target}}$ be the vector of the true state. Figure 33 shows the factor, the vector and the state for two qubits. Training minimises the squared distance of the two vectors averaged over a batch of N_{b} training states,

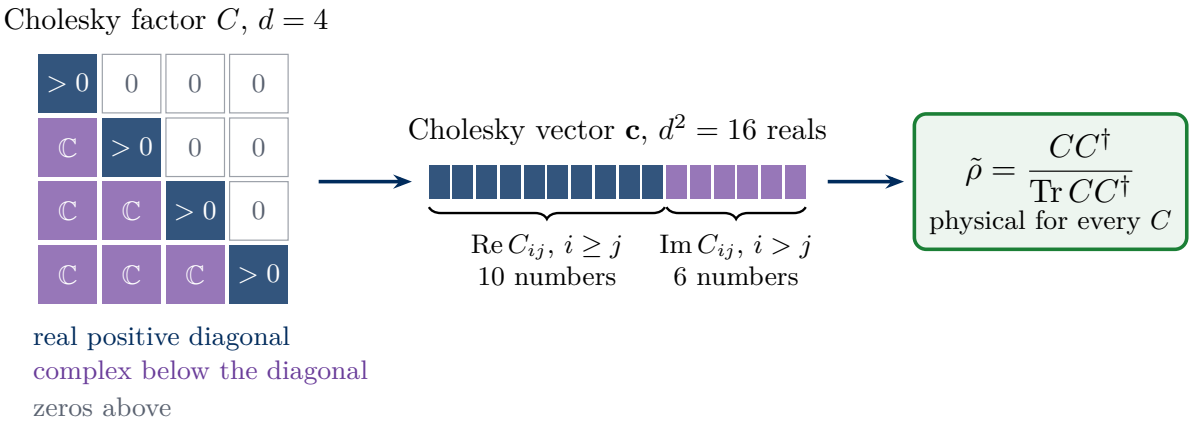


Figure 33: The Cholesky representation for $d = 4$. The factor C has a real positive diagonal, complex entries below it and zeros above. Its free entries form the Cholesky vector $\mathbf{c}$ of $d^2 = 16$ real numbers (4.12), and every C defines a physical state (4.11).

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N_{\mathrm{b}}} \sum_{m=1}^{N_{\mathrm{b}}} \left\| \mathbf{c}_{\boldsymbol{\theta}}(\tilde{C}_m) - \mathbf{c}_{\mathrm{target},m} \right\|^2. \tag{4.13}$$

The loss (4.13) is the Euclidean distance of one fixed pair of amplitudes. For two states $\hat{\rho} = C_1 C_1^\dagger$ and $\hat{\sigma} = C_2 C_2^\dagger$ written with factors of unit trace, the inequality (4.6) gives

$$\|C_1 - C_2\|^2 = \mathrm{Tr}\left[(C_1 - C_2)(C_1 - C_2)^\dagger\right] \;\geq\; 2 - 2\sqrt{F(\hat{\rho}, \hat{\sigma})} = D_B^2(\hat{\rho}, \hat{\sigma}). \tag{4.14}$$

For Cholesky factors $\|C_1 - C_2\|^2$ equals $\|\mathbf{c}_1 - \mathbf{c}_2\|^2$, the sum of the squared differences of the real and imaginary parts of all entries. The loss (4.13) is therefore an upper bound on the mean squared Bures distance between the output and the true state, and a single term guarantees

$$F \geq \left(1 - \tfrac{1}{2}\|\mathbf{c}_1 - \mathbf{c}_2\|^2\right)^2 \tag{4.15}$$

whenever $\|\mathbf{c}_1 - \mathbf{c}_2\|^2 \leq 2$. The bound holds for any amplitudes of unit trace, so it applies to the normalised output of the network also when its diagonal entries are negative. For an output C with $\mathrm{Tr}CC^\dagger = n^2 \neq 1$ the same steps give $\|C - C_2\|^2 \geq n^2 + 1 - 2n\sqrt{F} \geq 1 - F \geq \frac{1}{2}D_B^2$, a bound that is weaker by a factor of two, and we therefore normalise the output inside the loss. The loss of Ref. [1] adds the term $\mathrm{Tr}[C_{\boldsymbol{\theta}} C_{\boldsymbol{\theta}}^\dagger]$ as a regulariser and normalises the state after the network. On 600 random pairs of two-qubit states the bound holds in every case, with D_B^2 on average 0.67 of the squared distance of the Cholesky vectors.

4.4 Convolutional layers

A convolutional layer applies K kernels to its input. A kernel is a vector of w trained weights κ_q^k that slides along the input, and at every position p it returns one number,

$$F_p^k = \gamma\Big(\sum_{q=1}^{w} \kappa_q^k \, \tilde{c}_{p+q} + b^k\Big), \qquad k = 1, \ldots, K, \tag{4.16}$$

with a trained offset b^k and the Gaussian error linear unit (GELU) γ [35], a smooth version of the rectifier. The K kernels thus produce K feature vectors $\mathbf{F}^1, \ldots, \mathbf{F}^K$ (Figure 34). Each entry of a feature vector depends on a window of w neighbouring entries of the input, and the weights are the same at every position and for every input.

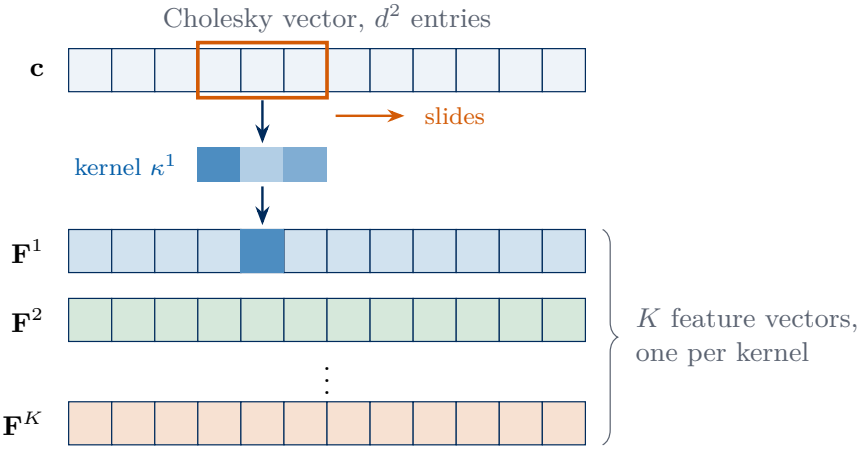


Figure 34: A convolutional layer on the Cholesky vector. A kernel of w trained weights slides along the vector and returns one number per position (4.16), and K kernels give K feature vectors.

4.5 The attention mechanism

The input of an attention layer is a set of K vectors $\mathbf{x}_1, \ldots, \mathbf{x}_K$ of length d_{model} that describe parts of one object. In the network of Ref. [1] they are the feature vectors of a convolutional layer, one per kernel, and in our reduced version they are the rows of the Cholesky factor. A linear layer would mix them with fixed couplings, $\mathbf{y}_i = \sum_j J_{ij}\mathbf{x}_j$, where the trained numbers J_{ij} are the same for every input. Attention [36,37] mixes them with couplings that are computed from the vectors themselves, in three operations that compare the vectors, normalise the result and mix.

The comparison uses two trained matrices W_{L} and W_{R} with d_{h} rows and d_{model} columns, named after their positions as the left and right factor of the comparison. The score of the pair (i, j) is the dot product of the images of the two vectors,

$$s_{ij} = (W_{\mathrm{L}}\mathbf{x}_i) \cdot (W_{\mathrm{R}}\mathbf{x}_j) = \mathbf{x}_i^{\mathsf{T}} g \, \mathbf{x}_j, \qquad g = W_{\mathrm{L}}^{\mathsf{T}} W_{\mathrm{R}}. \tag{4.17}$$

The matrix g is a trained bilinear form. For g equal to the identity the score is the ordinary dot product, which measures the alignment of the two vectors. A large score marks vector j as relevant for the update of vector i.

The normalisation turns the scores of each i into positive weights that sum to one,

$$a_{ij} = \frac{\exp\left(s_{ij}/\sqrt{d_\mathrm{h}}\right)}{Z_i}, \qquad Z_i = \sum_{j'=1}^{K} \exp\left(s_{ij'}/\sqrt{d_\mathrm{h}}\right). \tag{4.18}$$

The weights form a canonical distribution over the vectors j with energies $E_{ij} = -s_{ij}$, temperature $\sqrt{d_\mathrm{h}}$ and partition function Z_i. Scores that differ by much more than $\sqrt{d_\mathrm{h}}$ concentrate the weight on the best-matching vector, as at low temperature, and nearly equal scores give a uniform average, as at high temperature. Through Z_i the weight of vector j depends on all other vectors of the same input. The division by $\sqrt{d_\mathrm{h}}$ keeps the arguments of the exponential of order one at initialisation, since a dot product of d_h components of order one has a spread of order $\sqrt{d_\mathrm{h}}$.

The mixing uses a third trained matrix W_V with d_h rows and d_model columns, which selects the content that each vector passes on,

$$\mathbf{y}_i = \sum_{j=1}^{K} a_{ij}\, W_\mathrm{V}\, \mathbf{x}_j, \tag{4.19}$$

a thermal average of the transformed vectors (Figure 35). The three matrices are trained together with all other weights of the network by gradient descent on the loss (4.13). After training they are the same for every input, whereas the scores, the weights and the outputs are computed anew for every input.

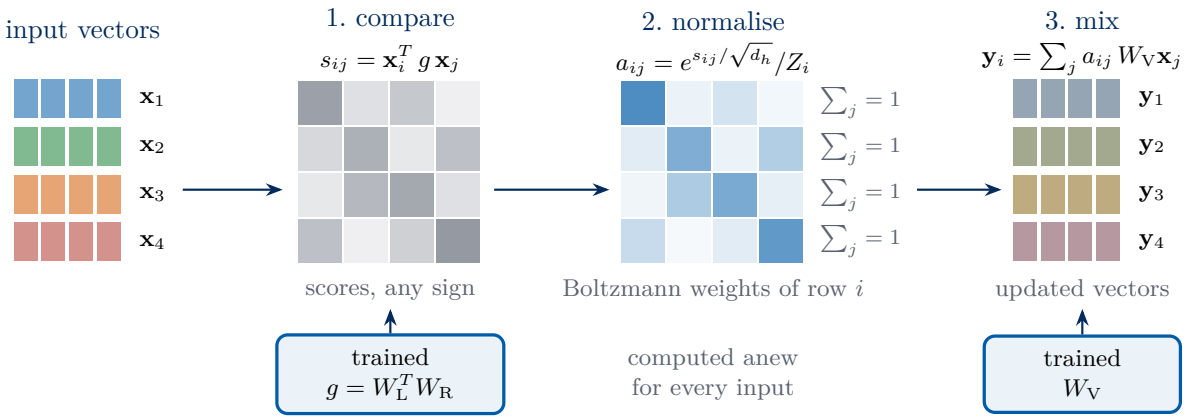


Figure 35: Attention in three operations. The trained bilinear form g compares all pairs of input vectors (4.17), the scores of each row are normalised to Boltzmann weights (4.18), and the output vectors are averages of the transformed inputs with these weights (4.19).

The singular value decomposition $g = \sum_{a=1}^{d_\mathrm{h}} \sigma_a\, \mathbf{p}_a \mathbf{q}_a^\mathsf{T}$ writes the score as a sum over d_h pairs of directions,

$$s_{ij} = \sum_{a=1}^{d_\mathrm{h}} \sigma_a\, (\mathbf{p}_a \cdot \mathbf{x}_i)(\mathbf{q}_a \cdot \mathbf{x}_j). \tag{4.20}$$

The form g thus decides which feature $\mathbf{p}_a \cdot \mathbf{x}_i$ of the receiving vector is matched with which feature $\mathbf{q}_a \cdot \mathbf{x}_j$ of the source vector. The identity matches every direction only with itself, which is plain similarity. Directions $\mathbf{p}_a \neq \mathbf{q}_a$ match different features, so that a vector can draw on vectors unlike itself. Training chooses the directions that lower the loss, and here they encode which features make one part of the Cholesky factor useful for correcting another.

For equal matrices, $W_\mathrm{L} = W_\mathrm{R} = W$, the form $g = W^\mathsf{T} W$ is symmetric and positive semidefinite. With $\mathbf{u}_i = W\mathbf{x}_i$ the score becomes

$$s_{ij} = \mathbf{u}_i \cdot \mathbf{u}_j = \tfrac{1}{2}|\mathbf{u}_i|^2 + \tfrac{1}{2}|\mathbf{u}_j|^2 - \tfrac{1}{2} d_g^2(\mathbf{x}_i, \mathbf{x}_j), \qquad d_g^2(\mathbf{x}, \mathbf{x}') = (\mathbf{x} - \mathbf{x}')^\mathsf{T} g\, (\mathbf{x} - \mathbf{x}'), \tag{4.21}$$

and the term $|\mathbf{u}_i|^2$ cancels in the normalisation over j, which leaves

$$a_{ij} \propto \exp\left[\frac{-d_g^2(\mathbf{x}_i, \mathbf{x}_j) + |\mathbf{u}_j|^2}{2\sqrt{d_\mathrm{h}}}\right]. \tag{4.22}$$

The weights are then Gaussian in a trained distance d_g, multiplied by a factor that depends only on the source vector j, and vectors that are close in this distance mix strongly. Since g has rank at most d_h, the distance d_g vanishes along the directions that W removes. In general W_L and W_R differ, and g is neither symmetric nor positive. Vector i can then use vector j strongly while j ignores i, since $s_{ij} - s_{ji} = \mathbf{x}_i^\mathsf{T}(g - g^\mathsf{T})\mathbf{x}_j$. Attention thus trains a rule of comparison, which for equal matrices reduces to a trained distance.

The training data fix g and W_V, which encode the features that make two vectors relevant to each other across the whole training set. The present input fixes the weights a_{ij}, so the same trained g produces different couplings for different estimates. With several layers the second layer compares vectors that already contain the others, and the effective comparison of two original vectors then depends on the whole input.

Several heads apply independent triples W_L, W_R and W_V in parallel, each with its own rule of comparison. With H heads of width $d_\mathrm{h} = d_\mathrm{model}/H$ the concatenated outputs again have the length d_model, and one more trained matrix mixes them into the attention output $\mathbf{y}_i$. A transformer layer [37] combines attention with a small network that acts on every vector separately,

$$\mathbf{x}'_i = \mathbf{x}_i + \mathbf{y}_i, \qquad \mathbf{x}''_i = \mathbf{x}'_i + h_\mathrm{MLP}(\mathbf{x}'_i), \tag{4.23}$$

where h_MLP is a two-layer network with the GELU activation. Both additions keep the input and let each part of the layer learn only a correction. In our version layer normalisation [38] rescales each vector to zero mean and unit variance before the attention and before h_MLP, which keeps the scores (4.17) of order one during training.

4.6 The published network

The network of Ref. [1] maps the Cholesky vector $\tilde{\mathbf{c}}$ of the estimate to a corrected Cholesky vector by three kinds of layers, applied from right to left,

$$\mathbf{c}_{\boldsymbol{\theta}} = \tanh\left(h_\mathrm{l}\right) \circ h_\mathrm{tr} \circ \gamma(h_\mathrm{cnn})\left[\tilde{\mathbf{c}}\right], \tag{4.24}$$

where $\gamma(h_\mathrm{cnn})$ denotes the convolutional layers (4.16) with their activation, h_tr a transformer layer (4.23) and $\tanh(h_\mathrm{l})$ a final layer followed by the hyperbolic tangent, which returns the d^2 numbers of a Cholesky vector. The vectors that enter the attention are the feature vectors $\mathbf{F}^k$, one per kernel, so the index i of (4.17) labels kernels. Each feature vector extends over all positions of the Cholesky vector, and the matrices W_L, W_R and W_V act on entire feature vectors, so one layer relates features computed at all positions of the factor.

4.7 Convolution and attention as noise filters

A convolutional layer and an attention layer both build their output from weighted combinations of their input, and they differ in the range and in the origin of the weights. A convolutional layer combines the entries inside a window of w neighbours, with weights that are fixed after training. An attention layer combines whole vectors, every vector with every other, with weights that are recomputed from every input, while training fixes the rule g that produces them.

An entry of the density matrix is the inner product of two rows of the factor,

$$\rho_{ij} = \sum_l C_{il} C^*_{jl} = \mathbf{r}_i \cdot \mathbf{r}^*_j, \tag{4.25}$$

where $\mathbf{r}_i$ is row i of C. In the flattened Cholesky vector the entries of two rows lie far apart, outside the window of one kernel, and the coherences of an entangled state spread over the whole matrix. Convolutional layers connect such distant entries only after many layers. In the published network the attention acts on feature vectors that extend over all positions, and in our reduced version it acts on the rows themselves, so in both cases one layer relates all parts of the factor.

The best correction also depends on the estimate at hand. A convolution applies the same local average to every input, whereas attention averages over the vectors that the data select. Non-local means, a method of image denoising [39], follows the same principle. It replaces each pixel by an average over all pixels whose surroundings look similar, with Gaussian weights in the distance between the surroundings, and for equal matrices the attention weights (4.22) have this form with a trained distance. These two arguments are an interpretation, and the evidence is the comparison of Ref. [1] with networks of convolutions alone at an equal number of parameters (Section 4.12).

4.8 Two reduced networks

Our reduced version omits the convolutions and uses the structure (4.25) of the factor directly. Every coherence ρ_{ij} couples two rows, so the correction of row i should depend on all rows, and we take the d rows as the vectors that enter attention. Row i, written as the $2d$ real numbers $(\mathrm{Re}\,\mathbf{r}_i, \mathrm{Im}\,\mathbf{r}_i)$, is mapped by a trained matrix W_E to

$$\mathbf{x}_i = W_E(\mathrm{Re}\,\mathbf{r}_i, \mathrm{Im}\,\mathbf{r}_i) + \mathbf{e}_i, \tag{4.26}$$

a vector of length d_{model}, with $d_{\mathrm{model}} = 32$ in our two-qubit network, where the trained vector $\mathbf{e}_i$ marks the position of the row. The position matters, since the first row of a lower-triangular matrix has one nonzero entry and the last row has d.

Both reduced networks return the input factor plus a learned correction,

$$C_{\boldsymbol{\theta}} = \tilde{C} + \Delta_{\boldsymbol{\theta}}(\tilde{C}), \tag{4.27}$$

restricted to the entries that a Cholesky factor may occupy. The last layer starts at zero, so the untrained network returns the projected estimate unchanged and training learns only the deviation. The reduced attention network consists of the map (4.26) from the d rows to d vectors, one transformer layer (4.23) with four heads of width $d_{\mathrm{h}} = d_{\mathrm{model}}/4 = 8$, and a final trained matrix that maps every vector to $2d$ real numbers, the correction of the real and imaginary parts of its row. For comparison, a multilayer perceptron (MLP) treats the sixteen numbers of (4.12) as one vector and maps it through two hidden layers of 80 units with the GELU activation. The MLP receives no information on which entries belong to the same row. Both networks have about 9000 parameters. Figure 36 shows the MLP with its residual connection (4.27).

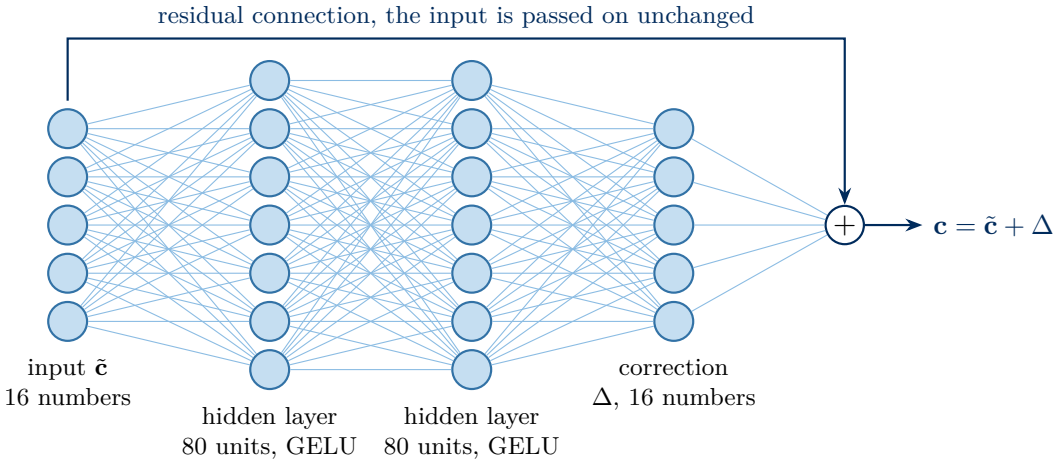


Figure 36: The multilayer perceptron used for comparison. The sixteen numbers of the input Cholesky vector pass through two hidden layers of 80 units with the GELU activation to a correction Δ of the same length, which is added to the input (4.27).

4.9 Training states and training

To correct estimates of arbitrary states, the networks are trained on two families that cover the state space without preferring any direction. Haar-random pure states are distributed uniformly over

the unit sphere of $\mathbb{C}^d$. For a matrix G with independent complex Gaussian entries, writing the QR decomposition $G = QR$ and multiplying each column of Q by the phase of the corresponding diagonal element of R gives a Haar-random unitary [40], whose first column is a Haar-random state. Mixed states are drawn from the Hilbert–Schmidt ensemble (2.3). They have full rank and the mean purity $2d/(d^2+1)$, which is 0.47 for two qubits [7]. The training set contains equal numbers of both, pure states on the boundary of the state space and typical mixed states inside it.

For every shot number $M \in \{10, 100, 1000\}$ we build a training set of 5000 two-qubit states, measure each once in the nine Pauli settings with M shots per setting and use the Cholesky factor of $\tilde{\rho}_{\mathrm{proj}}$ as input and that of the true state as target. Separate test sets of 500 pure and 500 mixed states serve for evaluation, and every M has its own network. We minimise (4.13) with Adam [14] and gradients from automatic differentiation [15], with a step size that starts at 3×10^{-3} and decreases to zero along a cosine, for 800 steps on batches of 250 states. The loss starts at the error of the projected estimate and falls by about a factor of two (Figure 37). The shot noise sets a floor for the loss, because the same estimate can come from many true states, and the short training may end above this floor.

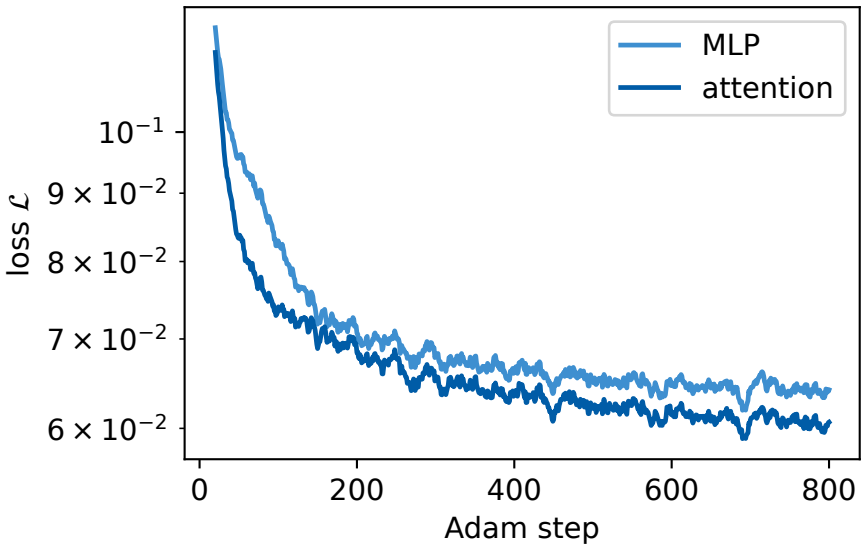


Figure 37: Loss (4.13) on the training batches at $M = 100$, averaged over 20 steps, for the MLP and the attention network. Both start at the error of the projected estimate.

4.10 Two-qubit results

We evaluate four estimates on the same simulated test records, the projected linear inversion, maximum likelihood by (2.23), and the outputs $\tilde{\rho}_{\mathrm{NN}}$ of both networks. Table 3 collects the mean fidelities, and the fraction of unphysical linear-inversion estimates stands in for linear inversion itself, which has no fidelity. For pure test states at $M = 100$ both networks lower the mean infidelity of the projected estimate from about 3.5×10^{-2} to about 1.3×10^{-2}, below the 2.3×10^{-2} of maximum likelihood (Figure 38). For mixed test states the four estimates differ by less than 0.005 in mean fidelity, because a full-rank state far from the boundary has little bias that a network could remove. The MLP and the attention network reach nearly the same fidelities, so for two qubits, with only four rows, attention brings no clear advantage at this training budget.

Over the three budgets (Figure 39) both networks lower the infidelity of the projected estimate for pure states at every M, and at $M = 100$ and 1000 they also improve on maximum likelihood. At $M = 10$, with 90 shots per state, maximum likelihood is the best estimate for pure states, so a short training run does not guarantee an improvement over the best standard estimator. For mixed states the networks gain only at $M = 10$.

Since a network trained on one family may fail on another, we test the networks trained at

M	test states	LI unphysical	LI + projection	MLE	MLP	attention
10	pure	1.00	87.8	90.9	89.5	89.3
10	mixed	0.99	78.8	79.6	81.5	81.5
100	pure	1.00	96.5	97.7	98.6	98.8
100	mixed	0.65	95.7	96.0	95.6	96.0
1000	pure	1.00	98.9	99.3	99.8	99.8
1000	mixed	0.24	99.4	99.4	99.4	99.5

Table 3: Two-qubit tomography with M shots per Pauli setting, our simulation. Fraction of unphysical linear-inversion estimates and mean fidelity in per cent of the projected linear inversion, maximum likelihood (MLE) and the two networks, each over 500 simulated test states.

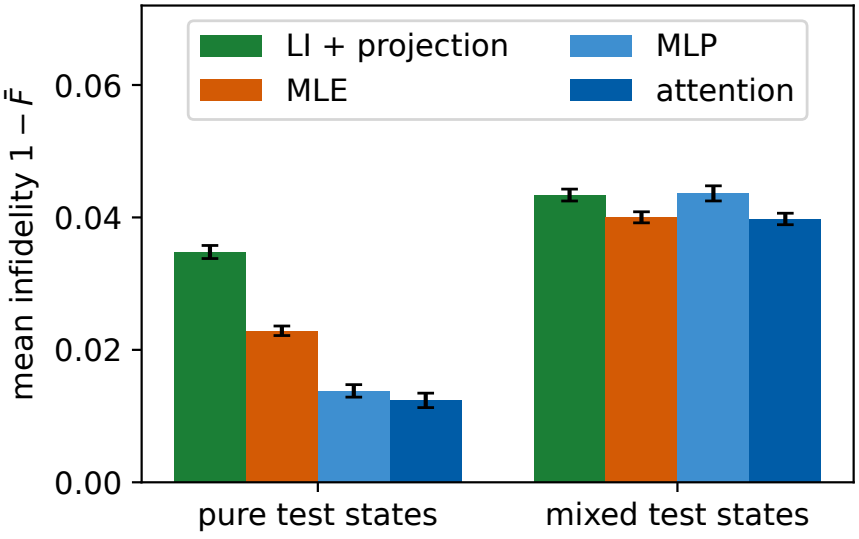


Figure 38: Mean infidelity $1-\bar{F}$ at $M=100$ for pure and mixed two-qubit test states, our simulation, with error bars of one standard error over 500 states. The networks gain on pure states only.

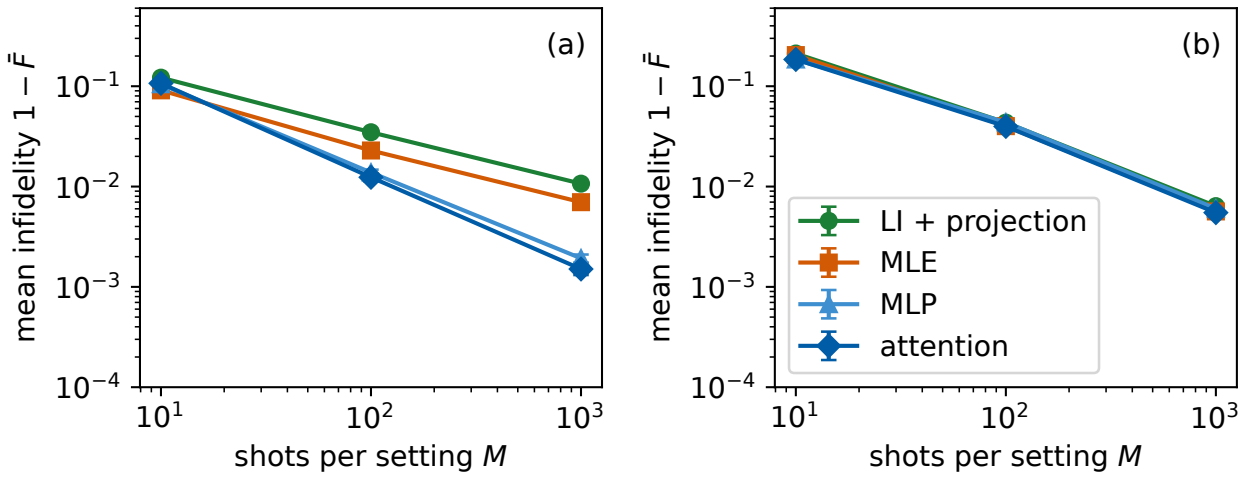


Figure 39: Mean infidelity against the number of shots per setting for (a) pure and (b) mixed two-qubit test states, our simulation, for the projected linear inversion, maximum likelihood and the two networks.

$M = 100$ on Haar-random pure states mixed with white noise,

$$\hat{\rho}_p = (1-p)|\psi\rangle\langle\psi| + p\,\frac{\hat{I}}{d}, \tag{4.28}$$

for p between 0 and 0.5. For $p > 0$ such states have one large eigenvalue and $d-1$ equal small ones, a spectrum that neither training family has. Both networks improve on both standard estimators by three standard errors or more at $p = 0$, 0.2 and 0.3 (Figure 40). At $p = 0.1$ the attention network lies below maximum likelihood by 0.009 in mean fidelity, and at $p = 0.5$ all gains are 0.004 or smaller. At $p = 0.3$, for example, the mean fidelity rises from 93.7% for the projection and 94.0% for maximum likelihood to 95.9% for the attention network. The infidelity of the standard estimators first grows with p, because the small eigenvalues p/d are of the size of the noise, and decreases again as the state approaches $\hat{I}/d$.

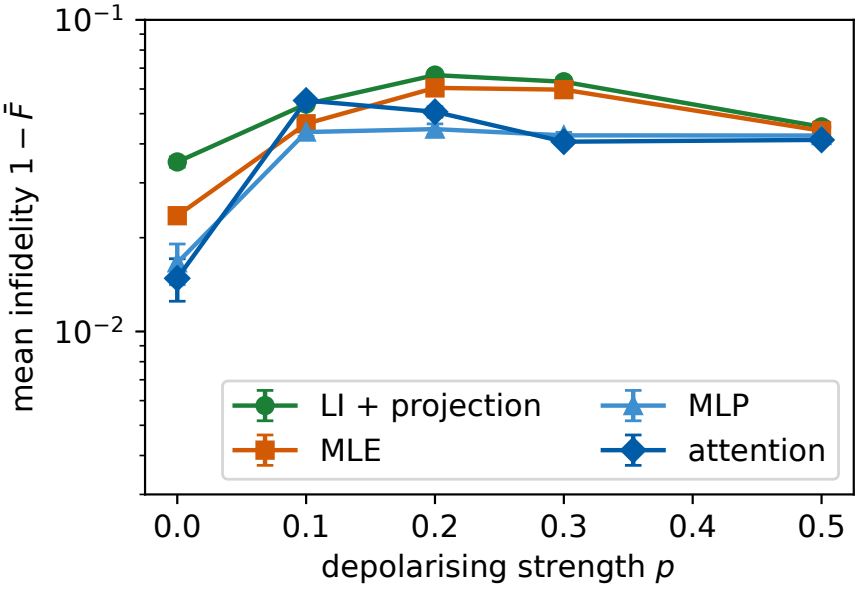


Figure 40: Mean infidelity on depolarised Haar-random states (4.28) against the depolarising strength p, 300 simulated states per point at $M = 100$, with networks trained only on pure and Hilbert–Schmidt states. Error bars are standard errors of the mean.

4.11 Four qubits, one-axis twisting and the quantum Fisher information

The test states of four qubits come from one-axis twisting [41], the evolution of N spins under $\hat{\mathcal{H}} = \hat{J}_z^2$, where $\hat{J}_a = \frac{1}{2}\sum_{q=0}^{N-1} \hat{\sigma}_{a,q}$ is the collective spin along axis a. Starting with all spins along x,

$$|\Psi(t)\rangle = e^{-it\hat{J}_z^2}|+\rangle^{\otimes N}, \qquad |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}. \tag{4.29}$$

The Hamiltonian is diagonal in the computational basis, with eigenvalue j_z^2 on a bit string with magnetisation $j_z = (n_0 - n_1)/2$, where n_0 and n_1 count the bits 0 and 1. Each amplitude only acquires the phase $e^{-itj_z^2}$, and since this phase depends nonlinearly on j_z, the evolution correlates the spins and creates entanglement. For four qubits $e^{-i\pi j_z^2} = (-1)^{j_z}$ up to a global phase, so at $t = \pi$ the state is the product state $|-\rangle^{\otimes 4} = \hat{Z}^{\otimes 4}|+\rangle^{\otimes 4}$. Since $\hat{Z}^{\otimes 4}$ is a product of single-qubit rotations, the entanglement and the QFI of the twisted states are periodic in t with period π. At $t = \pi/2$ the state is a superposition of all spins along $+x$ and all along $-x$, a cat state.

The entanglement of these states shows up in their sensitivity to a collective rotation $e^{-i\varphi\hat{J}_{\mathbf{n}}}$ about an axis $\mathbf{n}$, with $\hat{J}_{\mathbf{n}} = \sum_a n_a \hat{J}_a$. The quantum Fisher information (QFI) F_Q sets the smallest uncertainty of the angle φ that any measurement on μ copies of the state can reach, $\Delta\varphi \geq 1/\sqrt{\mu F_Q}$ [42,43].

It measures how fast the state changes under the rotation, since the squared Bures distance (2.27) between the states rotated by φ and by $\varphi+d\varphi$ is $\frac{1}{4}F_Q\,d\varphi^2$ [42]. For a state with the eigendecomposition $\hat\rho=\sum_\alpha p_\alpha|\alpha\rangle\langle\alpha|$ it reads [42]

$$F_Q[\hat\rho,\hat J_{\mathbf n}]=2\sum_{\alpha,\beta:\,p_\alpha+p_\beta>0}\frac{(p_\alpha-p_\beta)^2}{p_\alpha+p_\beta}\,\bigl|\langle\alpha|\hat J_{\mathbf n}|\beta\rangle\bigr|^2, \tag{4.30}$$

which for a pure state equals $4\,\mathrm{Var}(\hat J_{\mathbf n})$. Equation (4.30) is a quadratic form $\mathbf n^{\mathsf T}\mathcal G\mathbf n$ with a real symmetric 3×3 matrix $\mathcal G$, and we use its largest eigenvalue, the QFI for the best axis, and write it F_Q. A product state of N spins has $F_Q\le N$ [44]. A state that can be prepared from groups of at most m entangled spins is called m-producible, and it obeys [45,46]

$$F_Q\le sm^2+r^2\le mN,\qquad N=sm+r,\quad 0\le r<m. \tag{4.31}$$

The largest number of spins that are entangled with each other is called the entanglement depth, and a value $F_Q/N>m$ certifies a depth of at least $m+1$. For four spins and $m=3$ the first bound gives $F_Q\le 9+1=10$, so $F_Q/N>2.5$ certifies genuine entanglement of all four spins. Ref. [1] uses the looser second bound $F_Q\le mN$, for which the threshold is $F_Q/N>3$. We report both thresholds separately. The cat state reaches the maximum $F_Q=N^2$, which is $F_Q/N=4$. Figure 41 summarises the states and the thresholds.

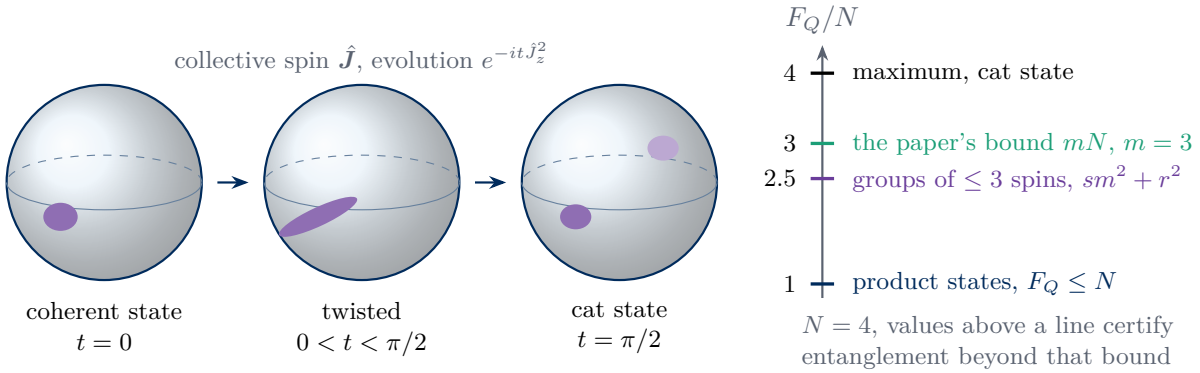


Figure 41: One-axis twisting (4.29) on the sphere of the collective spin, from the coherent state at $t=0$ through twisted states to the cat state at $t=\pi/2$, and the thresholds of the QFI per spin for four spins, $F_Q/N\le1$ for product states, 2.5 for groups of at most three entangled spins (4.31), 3 for the looser bound mN and the maximum 4 of the cat state.

As in Ref. [1], we train an attention network only on Haar-random pure states of four qubits, $d=16$, with $M=10$ shots in each of the 81 settings, a budget at which the projected estimate has a fidelity of about 0.8. The sixteen rows of the factor become sixteen vectors with $d_{\mathrm{model}}=64$, the network has 47072 parameters and is trained on 5000 states for 1500 Adam steps. We test it on 60 twisted states in $0<t<\pi$, each measured in three independent records. The projected estimates lose much of the QFI of the true states, because their noise mixes the state and the factor $(p_\alpha-p_\beta)^2/(p_\alpha+p_\beta)$ in (4.30) penalises mixing. The network raises the fidelity by the same amount as on new Haar states and recovers a large part of the QFI (Table 4 and Figure 42). Maximum likelihood on the same records reaches the same mean fidelity within the statistical error, 91.8% against 91.5%, so at this budget the network gains over the projection and matches maximum likelihood.

The inequality (4.31) certifies entanglement only if the reconstruction does not overestimate the QFI, whereas the network output is a learned estimate without an error bound. In our records the network overestimates the QFI of the true state in a fraction 0.16 of the cases, against 0.01 for maximum likelihood. Its values above the thresholds therefore indicate four-spin entanglement, and a certification needs an independent check, such as a test on states with known QFI under the same noise or a direct measurement of a witness.

	mean fidelity (%)	largest F_Q/N	fraction $F_Q/N > 2.5$	fraction $F_Q/N > 3$
true state	100	4.00	0.73	0.43
LI + projection	78.6	3.22	0.37	0.09
maximum likelihood	91.8	3.97	0.59	0.22
attention network	91.5	3.89	0.66	0.27

Table 4: One-axis-twisted states of four qubits reconstructed from $M = 10$ shots per setting, our simulation, over 60 twisting times with three records each. Mean fidelity, largest QFI per spin and fraction of records above the two thresholds of (4.31). On new Haar-random test states the network raises the fidelity from 78.5% to 90.9%.

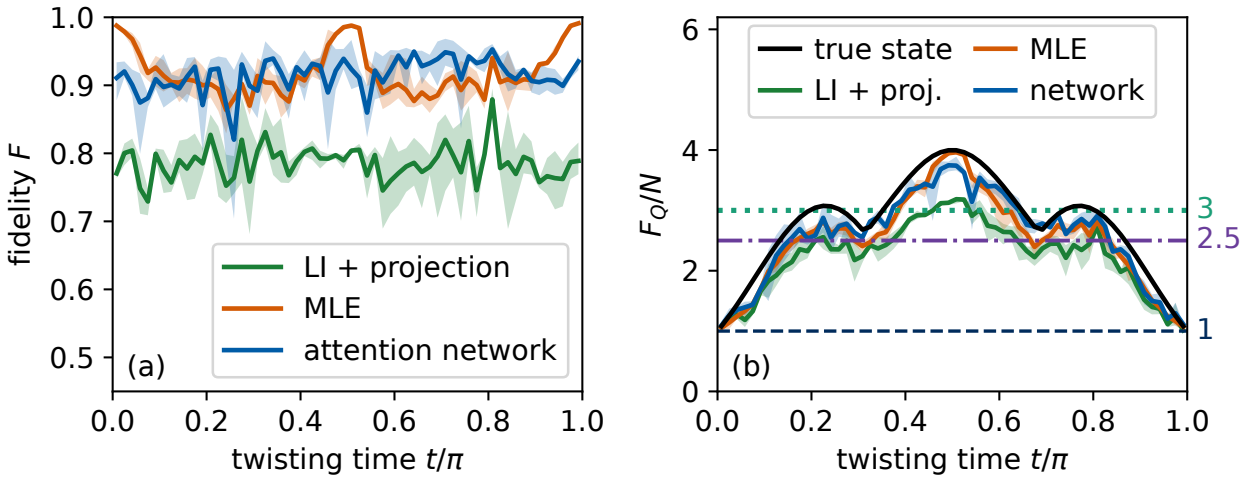


Figure 42: One-axis-twisted states (4.29) of four qubits reconstructed from $M = 10$ shots per setting, our simulation, against the twisting time t/π. (a) Fidelity of the projected estimate, of maximum likelihood (orange) and of the attention network output, lines for the mean of three records and bands between the smallest and largest. (b) Largest QFI per spin of the true state, the projected estimate, maximum likelihood and the network output, with the product-state limit $F_Q/N = 1$ (dashed), the four-spin threshold $F_Q/N = 2.5$ of the first bound in (4.31) (dash-dotted) and the threshold $F_Q/N = 3$ of the looser bound (dotted).

4.12 Published results

The three figures of this subsection are published results of Ref. [1], reproduced without change. The first comes from a pair of qutrits, $d = 9$, measured with one informationally complete POVM of 81 outcomes built from random states. The test states are 1000 Hilbert–Schmidt states, and the error is the mean squared Hilbert–Schmidt distance $\overline{D^2_{\mathrm{HS}}}$. The estimators are linear inversion (LI), maximum likelihood (MLE) and the network applied to either of them (LI-NN and MLE-NN). Panel (a) of Figure 43 plots the error against the number of measured copies N_{trial}, which decreases to the right. The network improves linear inversion at every budget, and applied to maximum likelihood it improves on it for fewer than about 10^3 copies, the regime of our two-qubit example, whereas with many copies maximum likelihood remains the best. In our two-qubit simulation maximum likelihood also beats both networks on pure states at $M = 10$ (Table 3). Panel (b) multiplies N_{trial} by the number of training states N_{train}. The method of Ref. [1] reaches errors similar to those of the network of Ref. [47], which was trained on 8×10^5 states, with far fewer training states, 5000 and 2000, which Ref. [1] summarises as an order of magnitude.

The second figure is the four-qubit test of Ref. [1]. A network trained only on Haar-random states with shot noise reconstructs one-axis-twisted states under additional imperfections. In this test Ref. [1] models the shot noise as Gaussian noise on the Pauli expectation values $\mathrm{Tr}[\hat{\rho}\hat{P}]$, which by itself gives a fidelity of about 0.85. It then depolarises the state by (4.28) with $p = 0.3$ and models a calibration error of the measurement by adding a fixed bias, with random values of order 10^{-4}, to every Pauli expectation value. The two panels of Figure 44 belong to two random draws of this bias.

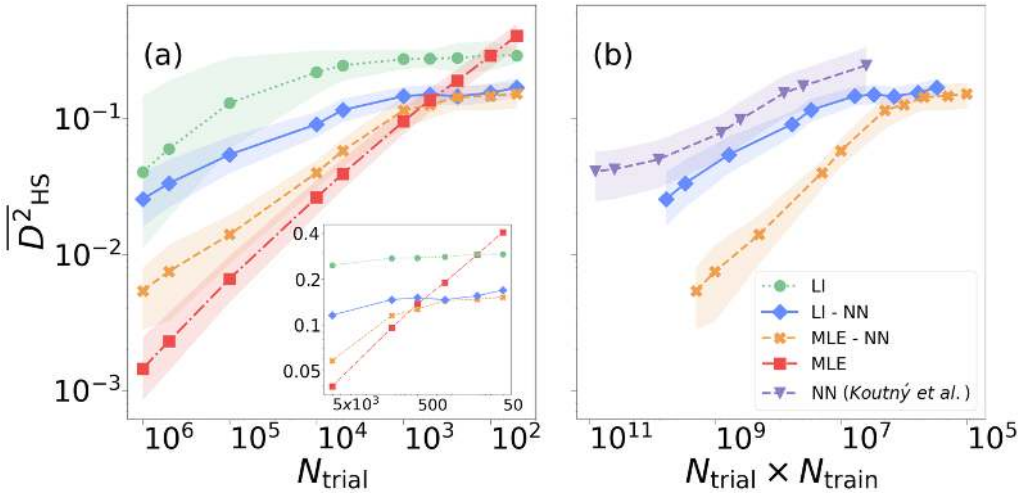


Figure 43: Published result of Ref. [1], not our simulation. Mean squared Hilbert–Schmidt distance for qutrit pairs, $d = 9$, averaged over 1000 Hilbert–Schmidt target states, (a) against the number of measured copies N_{trial} and (b) against $N_{\mathrm{trial}}N_{\mathrm{train}}$, with the network of Ref. [47] for comparison (violet). The networks were trained with 5000 (LI-NN) and 2000 (MLE-NN) states.

Linear inversion reaches a mean fidelity of $(75.4 \pm 1.1)\%$, and the network raises it to $(88.7 \pm 2.3)\%$ and $(91.0 \pm 1.9)\%$ for the two draws [1]. Linear inversion stays below $F_Q/N = 3$ at all times, whereas the network output crosses it near $t = \pi/2$, which Ref. [1] takes as evidence of entanglement of all four spins. Our simulation with shot noise only and a smaller network shows the same trend.

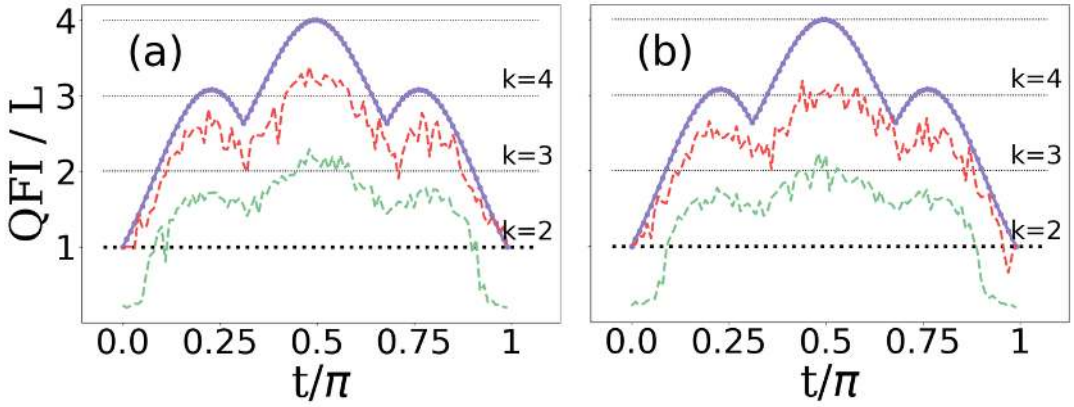


Figure 44: Published result of Ref. [1], not our simulation. Largest QFI per spin, written QFI/L with $L = 4$ spins, of 100 one-axis-twisted states against t/π, under depolarisation and calibration errors. Violet, true states. Green, linear inversion. Red, network applied to linear inversion. Dotted lines mark $F_Q/N = 1, 2, 3$ of the looser bound $F_Q \leq mN$ in (4.31), labelled in the source by $k = m + 1$, the number of spins whose entanglement a value above the line certifies. Panels (a) and (b) are two realisations of the calibration error.

Ref. [1] also tests the transformer layer by replacing it with further convolutional layers at an equal number of parameters. For Haar-random pure states of dimension $d = 16$ the networks of convolutions alone reach lower fidelities, with the largest gap at the smallest number of copies. For one-axis-twisted states their QFI stays close to that of linear inversion, whereas the network with attention follows the true states (Figure 45). For mixed states the three networks are comparable.

4.13 Related approaches

The networks of this section correct a complete estimate and therefore handle its d^2 parameters, which limits them to a few qubits. In neural-network quantum-state tomography [48] a restricted Boltzmann machine represents the wave function itself, its parameters are trained to reproduce the measured bit strings, and expectation values and entanglement entropies are then obtained by sampling the

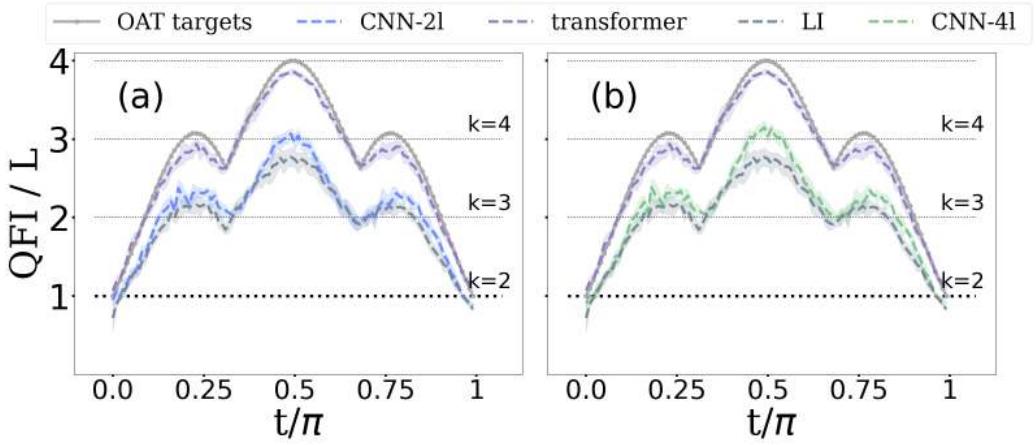


Figure 45: Published result of Ref. [1], not our simulation. Largest QFI per spin of one-axis-twisted states of four qubits against t/π for the true states (grey), linear inversion (dark dashed), the network with the transformer layer (violet) and networks in which convolutional layers replace it, (a) two layers and (b) four layers.

trained network. Such a representation describes structured states, such as ground states of local Hamiltonians, of many more qubits than full tomography, provided the state belongs to the family the network represents. The network of Ref. [47] maps the measured frequencies directly to a state, without an intermediate estimator, and needs many more training states.

5 Hamiltonian learning

A quantum simulator is built to realise a chosen type of interaction between its qubits, for instance nearest-neighbour exchange in a chain. The strengths of the couplings and fields depend on drifting calibrations and on imperfections of the device, so the Hamiltonian is known in its form and unknown in its numbers. Since the device returns only bit strings, we prepare known states, let them evolve for chosen times, measure every qubit in a chosen basis and repeat. Inferring the coefficients of the Hamiltonian from these bit strings is called Hamiltonian learning. Tomography learns the state that a device prepares, whereas Hamiltonian learning learns the law that moves every state.

The quantum graph recurrent neural network of the lecture on variational algorithms [49] solved the same task with a trainable Trotter circuit that had access to the evolved quantum states and compared them with its own output by a SWAP test. Here the device returns only measured bit strings, and the model of the experiment runs on a classical computer. We first fit the coefficients of the assumed Hamiltonian alone, by gradient descent with gradients from automatic differentiation through the simulated evolution. Ref. [50] calls this the vanilla model, and we call it the plain fit. We then add a neural network to the equation of motion during a warm-up, the neural differential equation of Ref. [50].

5.1 The problem and its data

The running model is a chain of $N = 4$ qubits with open ends whose Hamiltonian is assumed to be an anisotropic Heisenberg (XYZ) chain in two uniform fields,

$$\hat{\mathcal{H}}(\boldsymbol{\theta}) = \sum_{i=0}^{N-2} \left(J_x \hat{X}_i \hat{X}_{i+1} + J_y \hat{Y}_i \hat{Y}_{i+1} + J_z \hat{Z}_i \hat{Z}_{i+1}\right) + \sum_{i=0}^{N-1} \left(h_x \hat{X}_i + h_z \hat{Z}_i\right). \tag{5.1}$$

The exchange couplings J_x, J_y and J_z set how strongly neighbouring spins prefer to be aligned or anti-aligned along each axis, and the fields h_x and h_z turn every spin towards the x and z axes. Which Pauli strings appear is known from the design of the device, and the five coefficients $\boldsymbol{\theta} = (J_x, J_y, J_z, h_x, h_z)$

are unknown. Every coefficient multiplies a fixed operator, $\hat{\mathcal{H}}(\boldsymbol{\theta}) = \sum_j \theta_j \hat{P}_j$, where $\hat{P}_j$ is the sum of the Pauli strings that belong to coefficient j, for instance $\hat{P}_{J_x} = \sum_i \hat{X}_i \hat{X}_{i+1}$. We set $\hbar = 1$, so energies are frequencies and times are measured in units of an inverse coupling. The simulated device has $J_x = 1$, $J_y = 0.6$, $J_z = 0.3$, $h_x = 0.5$ and $h_z = 0.2$, values that enter only the simulation of the data and the final comparison.

An experiment prepares a known initial state $|\psi_0\rangle$, called a probe, lets it evolve and measures it. The state at time t is

$$|\psi_{\boldsymbol{\theta}}(t)\rangle = e^{-i\hat{\mathcal{H}}(\boldsymbol{\theta})t}|\psi_0\rangle. \tag{5.2}$$

We use three probes that are easy to prepare, $|0000\rangle$, the Néel state $|0101\rangle$ and $|{+}{+}{+}{+}\rangle$ with $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, and the six times $t_n = 0.25\,n$ with $n = 1, \ldots, 6$. All qubits are measured in the same basis $g \in \{Z, X, Y\}$, by the rotations (1.3), and by the Born rule the probability of the bit string $\mathbf{b}$ is

$$p_{\boldsymbol{\theta}}(\mathbf{b}\,|\,c) = \left|\langle \mathbf{b}|\hat{U}_g|\psi_{\boldsymbol{\theta}}(t_n)\rangle\right|^2, \qquad \hat{U}_Z = \hat{I}, \quad \hat{U}_X = \hat{H}^{\otimes N}, \quad \hat{U}_Y = (\hat{H}\hat{S}^\dagger)^{\otimes N}, \tag{5.3}$$

where $c = (\psi_0, g, t_n)$ labels one configuration of probe, basis and time. There are $3 \times 3 \times 6 = 54$ configurations, and the device returns $M = 1000$ bit strings for each. Because the shots are independent, the counts $n_c(\mathbf{b})$ hold all information of the record (Figure 46).

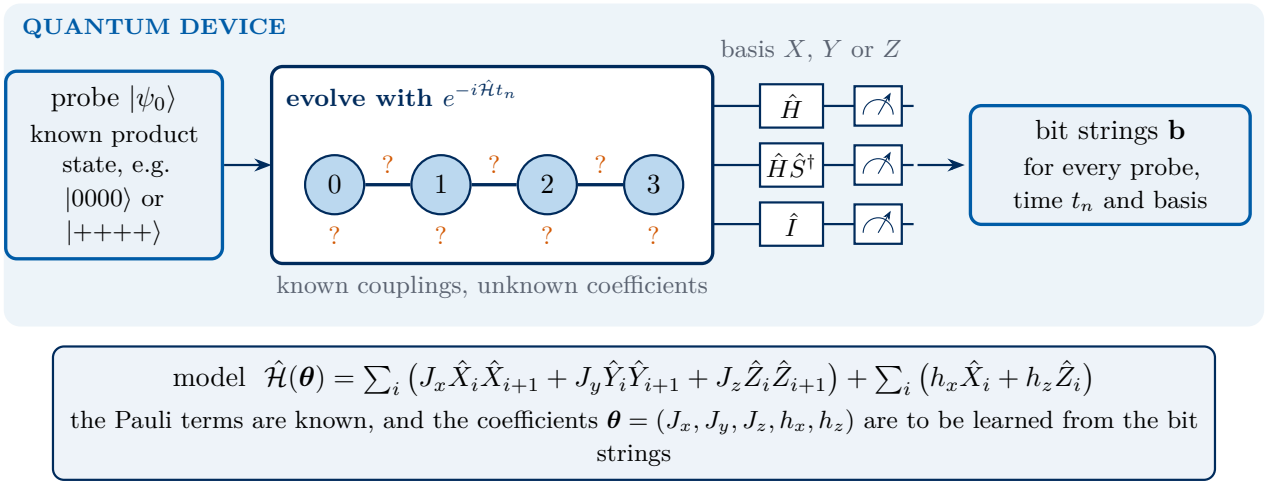


Figure 46: Hamiltonian learning with a known structure. Known product states evolve for a time t_n under the Hamiltonian (5.1) with unknown coefficients, every qubit is rotated into the X, Y or Z basis and measured, and the bit strings are the only output.

In one shot of the Z basis the value of $\hat{Z}_i$ is $(-1)^{b_i}$, and the same rule gives $\hat{X}_i$ in the X basis and $\hat{Y}_i$ in the Y basis. Figure 47 shows the mean magnetisations $m_g(t) = \frac{1}{N}\sum_i \langle \hat{\sigma}_i^g \rangle$ for the probe $|0000\rangle$. All spins start along $+z$, the field h_x rotates them about the x axis and turns m_z into a negative m_y, initially at the rate $dm_y/dt = -2h_x$, and the exchange couplings then reduce and dephase the magnetisation. The magnetisations show only part of the information, since the bit strings also carry all correlations between the qubits.

5.2 Identifiability

Data determine a coefficient only if a change of it changes some probability (5.3). Two Hamiltonians that give identical probabilities in every configuration cannot be distinguished with any number of shots, and such a pair exists for every experiment with only real probes and real bases. Let the probe have real amplitudes, $|\psi_0\rangle^* = |\psi_0\rangle$, where the star denotes complex conjugation in the computational basis. Conjugating the evolved state (5.2) gives

$$|\psi_{\hat{\mathcal{H}}}(t)\rangle^* = e^{+i\hat{\mathcal{H}}^* t}|\psi_0\rangle = e^{-i(-\hat{\mathcal{H}}^*)t}|\psi_0\rangle = |\psi_{-\hat{\mathcal{H}}^*}(t)\rangle, \tag{5.4}$$

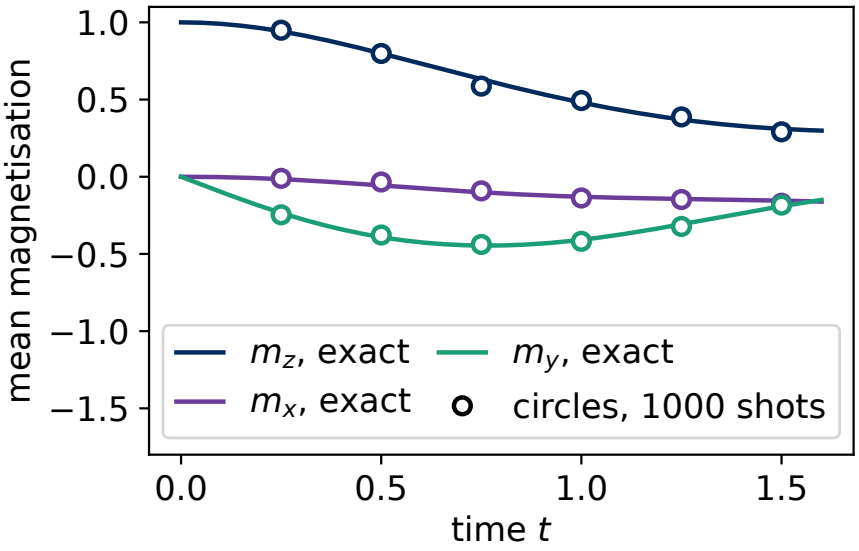


Figure 47: Mean magnetisations m_z, m_x and m_y of the probe $|0000\rangle$ under the chain (5.1). Lines are exact, circles are estimated from 1000 simulated shots at each of the six measurement times. The estimates scatter by about $1/\sqrt{NM}$ around the lines.

so the Hamiltonian $-\hat{\mathcal{H}}^*$ produces the complex conjugate of every state that $\hat{\mathcal{H}}$ produces. When the measured basis has real vectors, the row $\langle \mathbf{b}|\hat{U}_g$ is real and

$$\left|\langle \mathbf{b}|\hat{U}_g|\psi\rangle^*\right|^2 = \left|\langle \mathbf{b}|\hat{U}_g|\psi\rangle\right|^2 \quad \Rightarrow \quad p_{-\hat{\mathcal{H}}^*}(\mathbf{b}\,|\,c) = p_{\hat{\mathcal{H}}}(\mathbf{b}\,|\,c) \quad \text{for } g = Z, X. \tag{5.5}$$

The Z basis, $\hat{U}_Z = \hat{I}$, and the X basis, $\hat{U}_X = \hat{H}^{\otimes N}$, are both real. Data in these two bases from real probes therefore never distinguish $\hat{\mathcal{H}}$ from $-\hat{\mathcal{H}}^*$. The Y basis contains $\hat{S}^\dagger = \mathrm{diag}(1, -i)$, and since $\hat{Y}^* = -\hat{Y}$, conjugation reverses the sign of every y magnetisation, so Y data distinguish the two Hamiltonians. A complex probe such as $|{+i}\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$ does so as well.

For the chain (5.1) every operator $\hat{P}_j$ is real, because $\hat{X}$ and $\hat{Z}$ are real and $\hat{Y}_i\hat{Y}_{i+1} = (-\hat{Y}_i^*)(-\hat{Y}_{i+1}^*)$ is real too. Hence $-\hat{\mathcal{H}}(\boldsymbol{\theta})^* = \hat{\mathcal{H}}(-\boldsymbol{\theta})$, and the indistinguishable partner of any candidate has all five signs flipped (Figure 48). The same holds for every spin model without single $\hat{Y}$ factors, whose Pauli terms are real matrices.

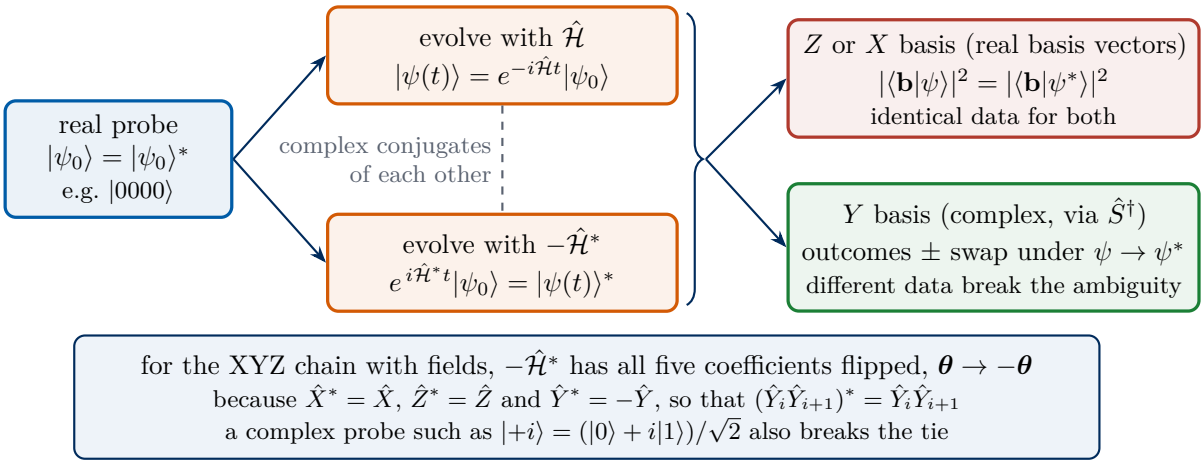


Figure 48: Complex conjugation maps the evolution under $\hat{\mathcal{H}}$ to the evolution under $-\hat{\mathcal{H}}^*$ (5.4). Real bases give identical data for both (5.5), and the Y basis or a complex probe separates them. For the chain (5.1) the partner has all coefficients flipped.

A smaller ambiguity of the same kind comes from symmetries of the probes and the measurement. The transformation $\hat{\mathcal{H}} \to \prod_i \hat{Z}_i\,\hat{\mathcal{H}}\prod_i \hat{Z}_i$ flips $\hat{X}_i$ and $\hat{Y}_i$ and leaves $\hat{Z}_i$, so it maps $\hat{\mathcal{H}}(J_x, J_y, J_z, h_x, h_z)$

to $\hat{\mathcal{H}}(J_x, J_y, J_z, -h_x, h_z)$. Probes in the computational basis and Z-basis data are invariant under this conjugation, so they leave the sign of h_x open, and only the probe $|{+}{+}{+}{+}\rangle$ or the X and Y bases fix it. The choice of probes, bases and times therefore decides which coefficients are learnable.

5.3 The forward model and the likelihood

A candidate $\boldsymbol{\theta}$ predicts, through (5.2) and (5.3), the probability of every bit string in every configuration. The shots are independent, so the probability of the record is a product over all shots. Its negative logarithm divided by the total number of shots $M_{\mathrm{tot}} = \sum_c \sum_{\mathbf{b}} n_c(\mathbf{b})$ is the loss

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{M_{\mathrm{tot}}} \sum_{c} \sum_{\mathbf{b}} n_c(\mathbf{b}) \ln p_{\boldsymbol{\theta}}(\mathbf{b} \,|\, c), \tag{5.6}$$

the negative log-likelihood (NLL) per shot, in nats. Its minimiser is the maximum-likelihood estimate $\tilde{\boldsymbol{\theta}}$. The loss contains only the counts of the device and the probabilities of the model, so it can be evaluated on real data.

The chain from $\boldsymbol{\theta}$ to $\mathcal{L}$ is a differentiable simulation of the experiment, called the forward model (Figure 49). The evolution operator over one interval $\Delta t = 0.25$ is computed once and applied n times to reach t_n. We evaluate the matrix exponential by scaling and squaring, which divides the matrix by 2^s, sums the Taylor series to order $n_{\max}$ and squares the result s times,

$$e^{\hat{A}} \approx \Big[\sum_{n=0}^{n_{\max}} \frac{1}{n!} \Big(\frac{\hat{A}}{2^s} \Big)^n \Big]^{2^s}, \tag{5.7}$$

here with $\hat{A} = -i\hat{\mathcal{H}}\Delta t$, $s = 6$ and $n_{\max} = 12$ [51]. Equation (5.7) consists of matrix products, so automatic differentiation [15] returns the exact gradient $\nabla_{\boldsymbol{\theta}}\mathcal{L}$, which agrees with a central finite difference. The coefficients are then updated by gradient descent,

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \, \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}), \tag{5.8}$$

with a step size η. In practice we use the Adam optimiser [14], which rescales the step of each coefficient by running averages of its past gradients and their squares. Fitting a model to time-resolved measurement data in this way is the trajectory-based approach to Hamiltonian learning [52,53,54], and Ref. [55] makes it scalable for one-dimensional many-body systems.

5.4 Fitting the coefficients from several starts

We start the fit from random coefficients drawn uniformly from $[-1, 1]$, run 1000 Adam steps with step size 0.05 and measure the result by the relative error

$$\epsilon(\tilde{\boldsymbol{\theta}}) = \frac{\sum_j |\tilde{\theta}_j - \theta_j|}{\sum_j |\theta_j|}, \tag{5.9}$$

which only a simulation can compute. The loss (5.6) is not convex, and a single start can end in a local minimum, a point where the gradient vanishes and the loss is higher than at the global minimum. We therefore run 12 independent starts and keep the one with the lowest final loss, a choice that uses only the data. Only 3 of the 12 starts reach the true coefficients, and they end with the lowest loss, slightly below the loss of the true coefficients, as a maximum-likelihood estimate on a finite record must (Figure 50). The best start recovers all five coefficients with a relative error of 0.005.

In this fit the ambiguity of Section 5.2 appears on the Z and X parts of the record, where the losses of $\tilde{\boldsymbol{\theta}}$ and of its partner $-\tilde{\boldsymbol{\theta}}$ agree to numerical precision, whereas the Y data favour $\tilde{\boldsymbol{\theta}}$ by 0.63 nats per shot. Along the line $\boldsymbol{\theta} = s\tilde{\boldsymbol{\theta}}$ the loss of the Z and X data has two equal minima at $s = \pm 1$, and adding the Y data lifts the one at $s = -1$, which remains a local minimum (Figure 51). With only real probes and real bases, an experiment would report two equally likely Hamiltonians with opposite signs of all coefficients.

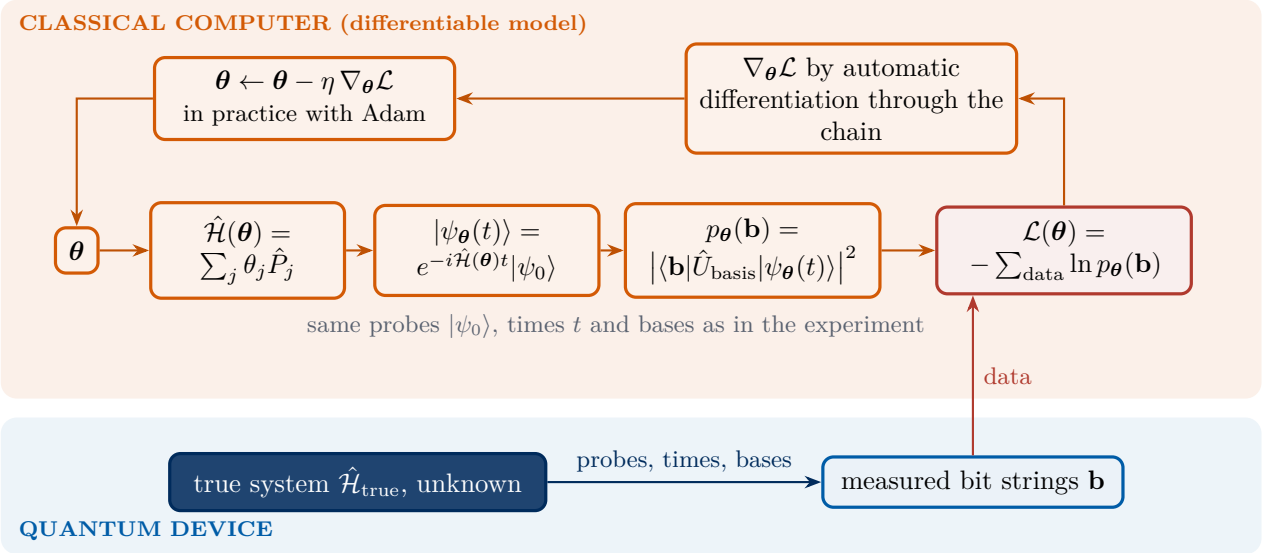


Figure 49: Forward model and training loop. The coefficients define the Hamiltonian, which evolves the probe, the Born rule gives the probabilities of the measured bit strings, and the negative log-likelihood (5.6) compares them with the counts of the device. Automatic differentiation through the chain gives the gradient for the update (5.8).

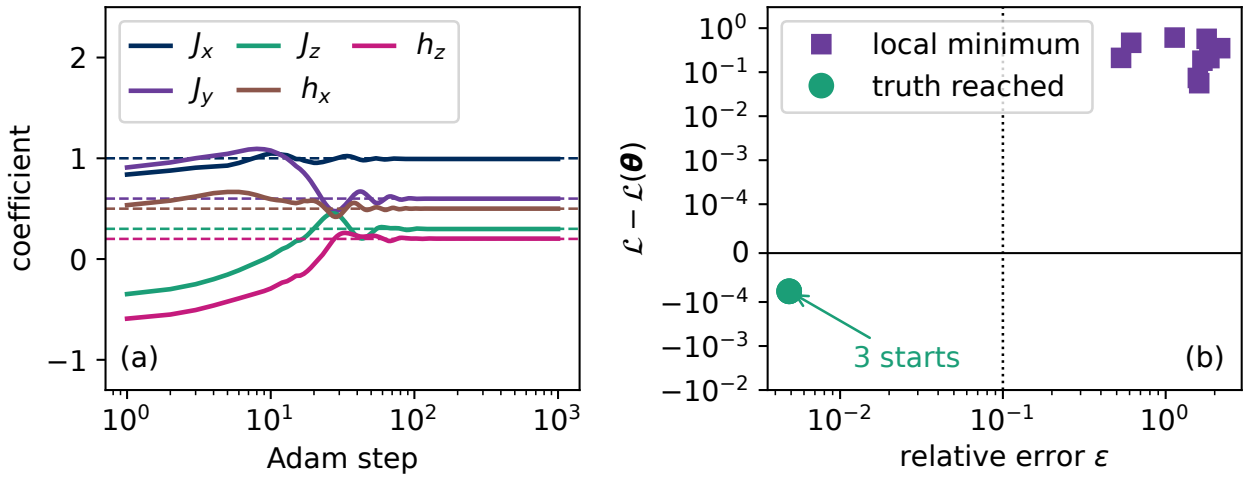


Figure 50: Fit of the chain (5.1) to one simulated record with $M = 1000$ shots per configuration. (a) The five coefficients of the start with the lowest final loss against the Adam step (solid), with the true values dashed. (b) Final loss of all twelve starts relative to the loss of the true coefficients, sorted, coloured by whether the relative error (5.9) is below 0.1.

5.5 Shot noise, Fisher information and the Cramér–Rao bound

The estimate $\tilde{\boldsymbol{\theta}}$ is computed from random bit strings and is itself random. To measure its spread we fit ten independent records for each M between 30 and 10 000 shots per configuration, starting every fit from a previous estimate that lies in the basin of the global minimum. The mean relative error falls with the fitted slope -0.49, from 0.004 at $M = 1000$ to 0.0016 at $M = 10\,000$, the shot-noise scaling $M^{-1/2}$ of independent repetitions.

The Fisher information, which measures how strongly the probabilities change when a coefficient changes, explains both the slope and the size of the error. For the multinomial counts with M shots in every configuration, the log-likelihood is $\ell(\boldsymbol{\theta}) = \sum_{c,\mathbf{b}} n_c(\mathbf{b}) \ln p_{\boldsymbol{\theta}}(\mathbf{b}|c)$, and the covariance of its gradient is M times the Fisher information matrix per shot in every configuration,

$$\frac{1}{M}\,\mathbb{E}\Big[\frac{\partial \ell}{\partial \theta_j}\frac{\partial \ell}{\partial \theta_l}\Big] = \mathcal{I}_{jl}(\boldsymbol{\theta}) = \sum_c \sum_{\mathbf{b}} \frac{1}{p_{\boldsymbol{\theta}}(\mathbf{b}|c)}\,\frac{\partial p_{\boldsymbol{\theta}}(\mathbf{b}|c)}{\partial \theta_j}\,\frac{\partial p_{\boldsymbol{\theta}}(\mathbf{b}|c)}{\partial \theta_l}. \tag{5.10}$$

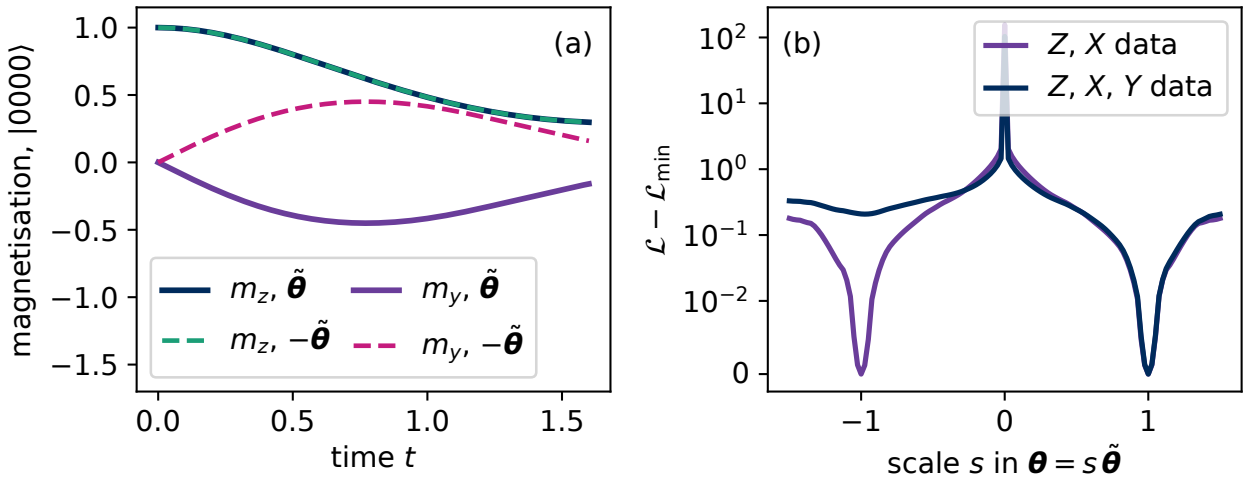


Figure 51: The pair $\hat{\mathcal{H}}(\tilde{\boldsymbol{\theta}})$ and $-\hat{\mathcal{H}}(\tilde{\boldsymbol{\theta}})^* = \hat{\mathcal{H}}(-\tilde{\boldsymbol{\theta}})$, exact evolution. (a) Magnetisations m_z and m_y of the probe $|0000\rangle$ under both Hamiltonians. The z magnetisation coincides and the y magnetisation changes sign. (b) Loss along the line $\boldsymbol{\theta} = s\tilde{\boldsymbol{\theta}}$, evaluated on the Z and X data alone and on all data, relative to its minimum.

A coefficient to which no probability responds gives a zero row of $\mathcal{I}$ and cannot be learned. The Cramér–Rao bound [56,57] states that the covariance matrix of any unbiased estimator obeys $\mathrm{Cov}(\tilde{\boldsymbol{\theta}}) \geq \mathcal{I}^{-1}/M$, in the sense that their difference is positive semidefinite, so in particular

$$\sigma_{\tilde{\theta}_j} \geq \sqrt{\frac{(\mathcal{I}^{-1})_{jj}}{M}} \propto M^{-1/2}. \tag{5.11}$$

The maximum-likelihood estimate reaches this bound when the number of shots is large. Figure 52 compares the measured standard deviation of each coefficient over 10 records with $\sqrt{(\mathcal{I}^{-1})_{jj}/M}$, evaluated with derivatives from automatic differentiation. A standard deviation from ten records is itself uncertain by about 25%, and within this accuracy the measured spread follows the bound at every M, with a mean ratio 1.02 from $M = 300$ on. At $M = 1000$ the bounds range from 2.2×10^{-3} for h_x to 3.4×10^{-3} for J_z, the coupling to which the probabilities respond least. The Fisher information also tells which probes, times and bases reduce a chosen error bar most efficiently, and its quantum counterpart, maximised over all measurements, is the quantum Fisher information (4.30) [42].

5.6 Other routes to the Hamiltonian

If the device prepares an eigenstate $|\psi\rangle$ of the unknown Hamiltonian, for instance its ground state, then $\langle[\hat{\mathcal{H}}, \hat{O}_a]\rangle = 0$ for every operator $\hat{O}_a$, because $\langle\psi|\hat{\mathcal{H}} = E\langle\psi|$ and $\hat{\mathcal{H}}|\psi\rangle = E|\psi\rangle$. With $\hat{\mathcal{H}} = \sum_j \theta_j \hat{P}_j$ these conditions form a homogeneous linear system $B\boldsymbol{\theta} = 0$ with the matrix $B_{aj} = \langle i[\hat{P}_j, \hat{O}_a]\rangle$, whose entries are measurable expectation values, and the coefficients follow as its null vector, up to a common factor [58,59]. A single eigenstate or steady state then suffices, without time evolution. Figure 53 summarises the three routes of this subsection.

A thermal state $e^{-\beta\hat{\mathcal{H}}}/Z$ at known inverse temperature β is fixed by the expectation values of the terms $\hat{P}_j$, and the coefficients follow by matching these measured values with those of a model state, a convex problem in the coefficients. For spatially local Hamiltonians the number of copies of the thermal state needed for a fixed accuracy grows polynomially with the number of qubits [60], and at high temperature only logarithmically [61].

At short times the evolution is linear in the Hamiltonian, $\langle\hat{O}\rangle_t = \langle\hat{O}\rangle_0 + it\langle[\hat{\mathcal{H}}, \hat{O}]\rangle_0 + O(t^2)$, so the initial time derivatives of measured expectation values give linear equations for the coefficients. Ref. [62] estimates these derivatives by polynomial interpolation from product-state probes and single-qubit measurements. Interleaving the evolution with quantum control allows errors that decrease as the inverse of the total evolution time, faster than the inverse square root of independent repetitions [63].

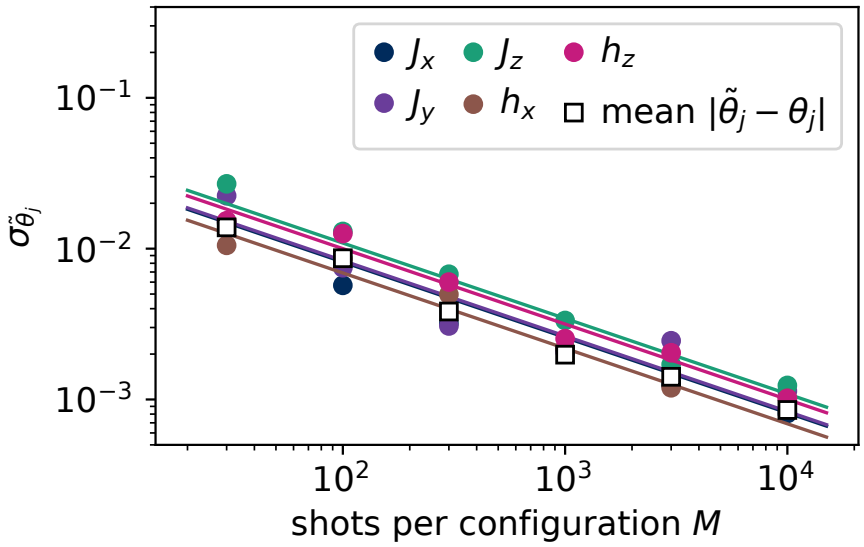


Figure 52: Standard deviation of the five fitted coefficients over ten simulated records (markers) against the number of shots per configuration, with the Cramér–Rao bounds (5.11) as lines of the same colour. Open squares show the mean absolute error over the coefficients.

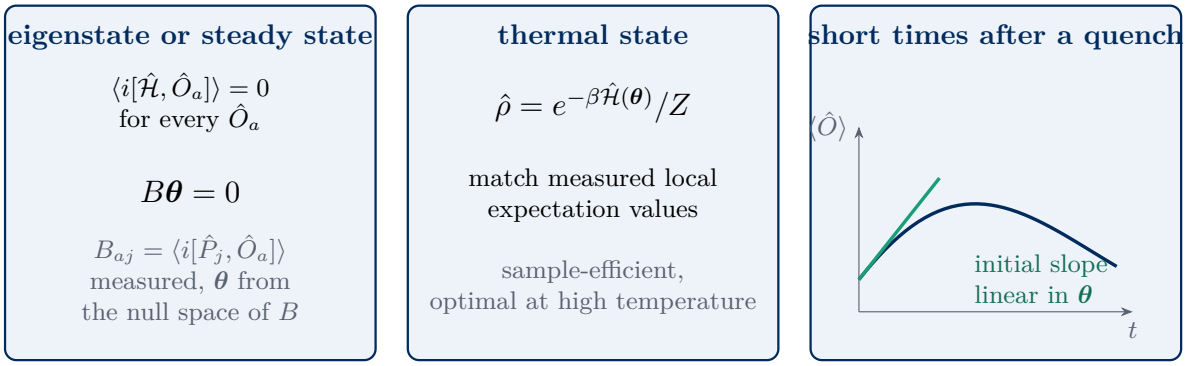


Figure 53: Routes to a Hamiltonian besides fitting trajectories. An eigenstate or steady state gives the linear system $B\boldsymbol{\theta} = 0$. A thermal state $e^{-\beta\hat{\mathcal{H}}(\boldsymbol{\theta})}/Z$ is matched through measured local expectation values. At short times after a quench the initial slopes $i\langle[\hat{\mathcal{H}}, \hat{O}]\rangle$ are linear in $\boldsymbol{\theta}$.

5.7 Rugged landscapes and the PXP model

The fit of Section 5.4 succeeds only from starts in the basin of the global minimum, and for strongly interacting models this basin becomes small. Neutral atoms excited to high-lying Rydberg states realise such a model [64]. Each atom is a qubit with the ground state $|0\rangle = |g\rangle$ and the Rydberg state $|1\rangle = |r\rangle$, driven by a laser with Rabi frequency Ω_i, the angular frequency at which the laser alone rotates the atom between $|g\rangle$ and $|r\rangle$. Two neighbouring atoms in the Rydberg state interact so strongly that the laser cannot excite them both, an effect called the Rydberg blockade. In the limit of perfect blockade the dynamics is described by the PXP model [65,66,67] (Figure 54),

$$\hat{\mathcal{H}}_{\mathrm{PXP}}(\boldsymbol{\Omega}) = \sum_{i=0}^{N-1} \Omega_i \, \hat{P}^{(0)}_{i-1} \hat{X}_i \hat{P}^{(0)}_{i+1}, \qquad \hat{P}^{(0)}_i = |0\rangle\langle 0|_i, \tag{5.12}$$

where $\hat{X}_i$ flips atom i between the ground and the Rydberg state and the projectors onto the ground state allow the flip only when both neighbours are in it. At the ends of the chain the missing neighbour is dropped, $\hat{P}^{(0)}_{-1} = \hat{P}^{(0)}_N = \hat{I}$. From a state without neighbouring excitations the evolution never creates a pair 11 on neighbouring atoms. The unknowns are the N Rabi frequencies $\boldsymbol{\Omega} = (\Omega_0, \ldots, \Omega_{N-1})$, and every coefficient multiplies a known three-qubit operator as in (5.1). The transformation $\hat{\mathcal{H}} \to \hat{Z}_i \hat{\mathcal{H}} \hat{Z}_i$ flips the sign of Ω_i alone and leaves computational basis states and Z-basis data unchanged, so the Z data cannot tell the 2^N sign patterns apart and only the X and Y data select one of them.

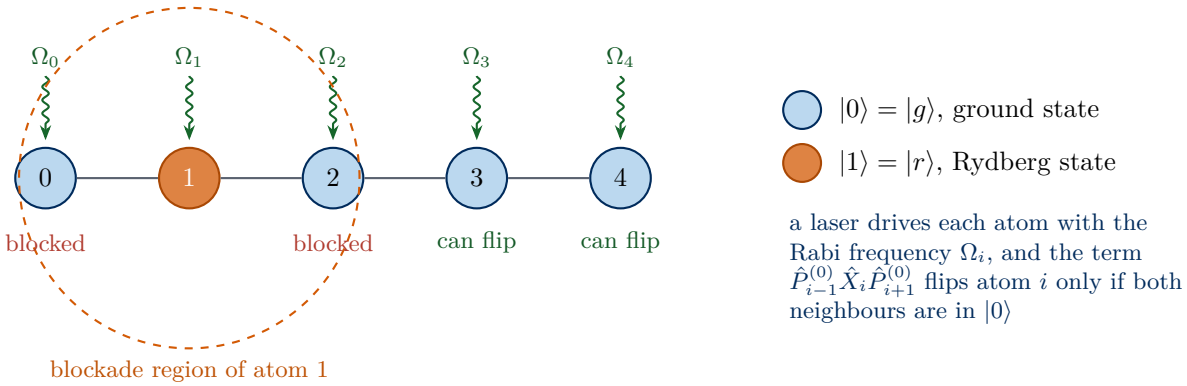


Figure 54: A chain of Rydberg atoms described by the PXP model (5.12). A laser drives each atom with its own Rabi frequency Ω_i. An atom next to an excited neighbour is blocked, and only atoms whose neighbours are both in the ground state can flip.

We generate data as before with the Z, X and Y bases, the five times $t_n = 0.2\,n$, $M = 1000$ shots and three random computational basis states as probes, without neighbouring Rydberg atoms for the PXP model, and draw the coefficients uniformly from $[-1, 1]$ as in Ref. [50]. A run succeeds when its relative error (5.9) is below 0.1, the criterion of Ref. [50]. To see the shape of the loss, we cut it along five random unit vectors $\mathbf{u}$ through the true coefficients and plot $\mathcal{L}(\boldsymbol{\theta} + \alpha\mathbf{u}) - \mathcal{L}(\boldsymbol{\theta})$ for $|\alpha| \leq 2$, the range at which random starts lie (Figure 56). Counting as a local minimum every grid point lower than its two neighbours, a cut of the XYZ loss contains a median of 6 minima and a cut of the PXP loss a median of 12. These counts describe the roughness of one-dimensional cuts and do not predict the success of a fit in all coefficients. Single starts rarely succeed for either model in our small sample of 16 random problems, 3 for the XYZ chain with these computational probes and 0, 2 and 0 for the PXP model with $N = 4$, 5 and 6, where the fits for $N = 6$ use 600 Adam steps and the others 1000 (Figure 55). These counts are too small to separate the two models, and the contrast between them rests on the published success rates of the plain fit for the PXP model, measured with 200 random Pauli bases per time and five probes, which are equally low.

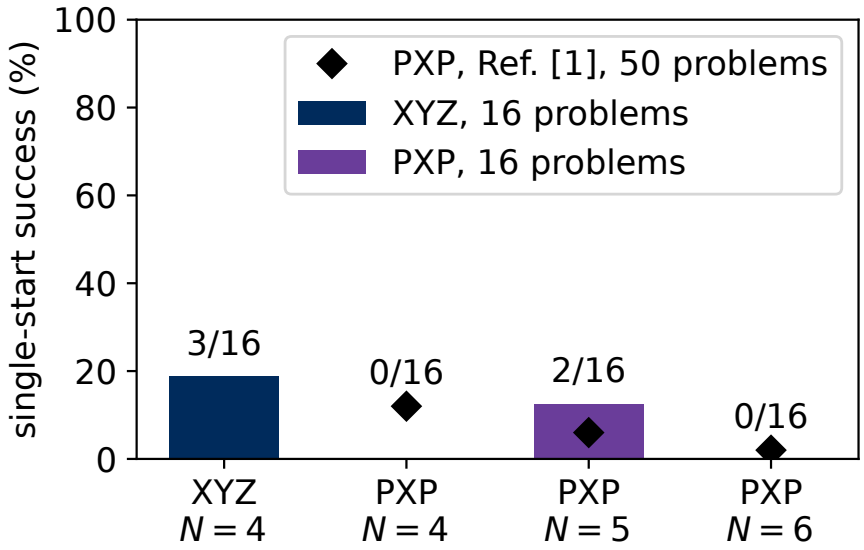


Figure 55: Single-start success rates of the plain fit, relative error below 0.1, in 16 random simulated problems for the XYZ chain with $N = 4$ and the PXP model with $N = 4, 5, 6$. Diamonds mark the published success rates of the same fit for the PXP model from Table 1 of the preprint version of Ref. [50], obtained with a different data set. The reference number in the legend is that of the notebook.

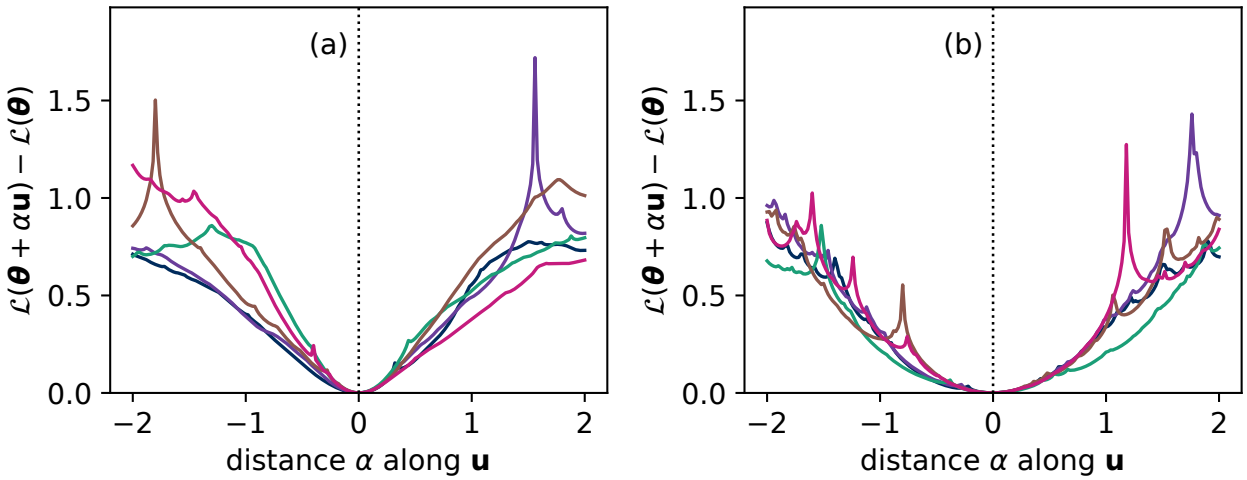


Figure 56: Loss along five random directions through the true coefficients, each from one simulated record, for (a) the XYZ chain (5.1) and (b) the PXP model (5.12) with four atoms. Both rise steeply around $\alpha = 0$, and the PXP cuts contain many more side minima.

5.8 Neural differential equations

A neural ordinary differential equation [68] is a differential equation whose right-hand side contains a neural network, trained by differentiating through the numerical solver. Ref. [50] adds such a network to the Schrödinger equation of the assumed Hamiltonian $\hat{\mathcal{H}}_A(\boldsymbol{\theta})$,

$$i\frac{d}{dt}|\psi(t)\rangle = \hat{\mathcal{H}}_A(\boldsymbol{\theta})|\psi(t)\rangle + \mathrm{NN}_{\boldsymbol{\phi}}(|\psi(t)\rangle), \tag{5.13}$$

where $\mathrm{NN}_{\boldsymbol{\phi}}$ maps the state vector to a correction vector and has its own parameters $\boldsymbol{\phi}$. The coefficients $\boldsymbol{\theta}$ keep their physical meaning, and the network can represent dynamics that the assumed form misses or that a coefficient far from its true value fails to produce. Our network takes the $2 \cdot 2^N$ real and imaginary parts of the amplitudes, passes them through one hidden layer of 32 units with tanh activation and returns $2 \cdot 2^N$ numbers that form a complex vector. The correction need not conserve the norm, so the state is normalised before the Born rule (5.3). The network makes the right-hand side nonlinear in $|\psi\rangle$, so (5.13) is integrated with the fourth-order Runge–Kutta method [69], which for $\dot{\psi} = f(\psi)$ and a step h reads

$$\begin{aligned} \psi(t+h) &= \psi(t) + \frac{h}{6}(\kappa_1 + 2\kappa_2 + 2\kappa_3 + \kappa_4), \\ \kappa_1 = f(\psi), \quad \kappa_2 &= f(\psi + \tfrac{h}{2}\kappa_1), \quad \kappa_3 = f(\psi + \tfrac{h}{2}\kappa_2), \quad \kappa_4 = f(\psi + h\kappa_3), \end{aligned} \tag{5.14}$$

with $h = 0.05$, four steps per interval of 0.2. The loss is the NLL (5.6) plus a penalty $\lambda \sum \phi^2$ on the size of the network parameters, with $\lambda = 10^{-4}$, which keeps the correction small.

Training follows the curriculum of Ref. [50] (Figure 57). A warm-up trains $\boldsymbol{\theta}$ and $\boldsymbol{\phi}$ together. Ref. [50] reports that the plain fit converges only from particular starts, because of local minima and flat regions of the loss, that the warm-up removes this dependence, and that the loss of the combined model is smoother. The network is then switched off in the forward and backward pass, which turns (5.13) back into the plain model, and only $\boldsymbol{\theta}$ is trained, so that the reported coefficients are those of a Hamiltonian. Finally $\boldsymbol{\theta}$ is frozen and only the network is trained, so that it absorbs what the assumed Hamiltonian cannot describe.

On the same 16 problems of the PXP model with four atoms, we test whether the warm-up repairs the fit. From the same start we run the plain model for 1500 Adam steps, the warm-up for 500 steps followed by 1000 steps of the plain model, and a control with the schedule of the curriculum but without the network, 500 plain steps, a restart of Adam and 1000 further plain steps. All three use the Runge–Kutta integrator. The plain model succeeds in 0 of the 16 problems, the control in 0 and the warm-up with the network in 16 (Figure 58). The network is switched off before the final fit, so

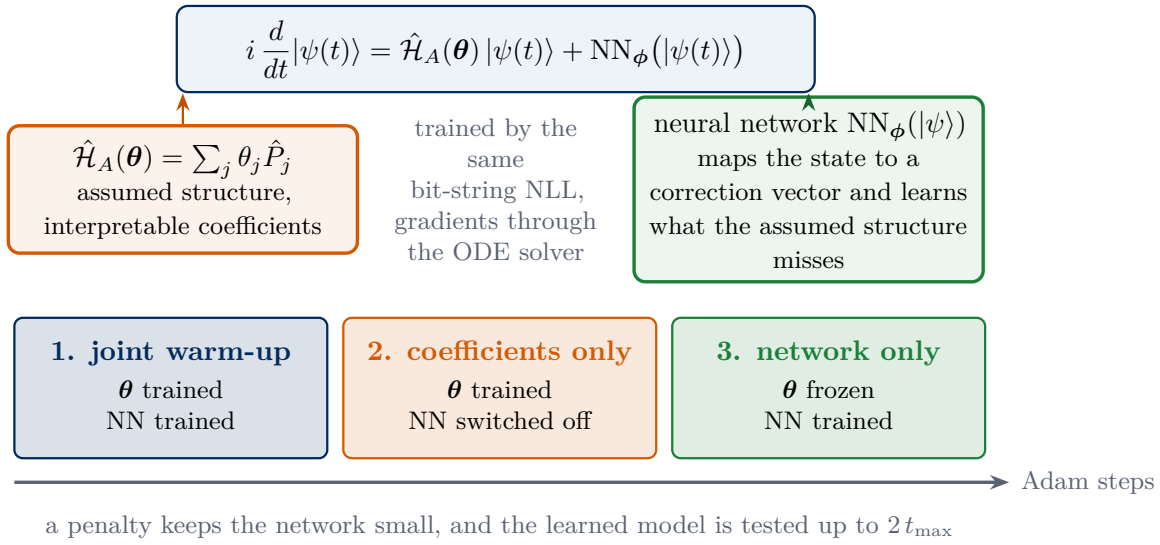


Figure 57: Neural differential equation (5.13) for Hamiltonian learning. The assumed Hamiltonian with interpretable coefficients and a neural network share the right-hand side of the Schrödinger equation and are trained on the same NLL. Training proceeds in three phases, joint warm-up, coefficients only and network only.

the reported coefficients come from the plain model alone. Since the control has the same restart and number of steps, the improvement comes from the network during the warm-up, in line with the published loss landscapes of Ref. [50] in Figure 60. Their top row, the plain fit, shows the rugged PXP landscape of Figure 56 in two dimensions, and their bottom row, the neural differential equation, is smooth around the global minimum. The published success rates (Figure 59) show the same contrast up to eight spins.

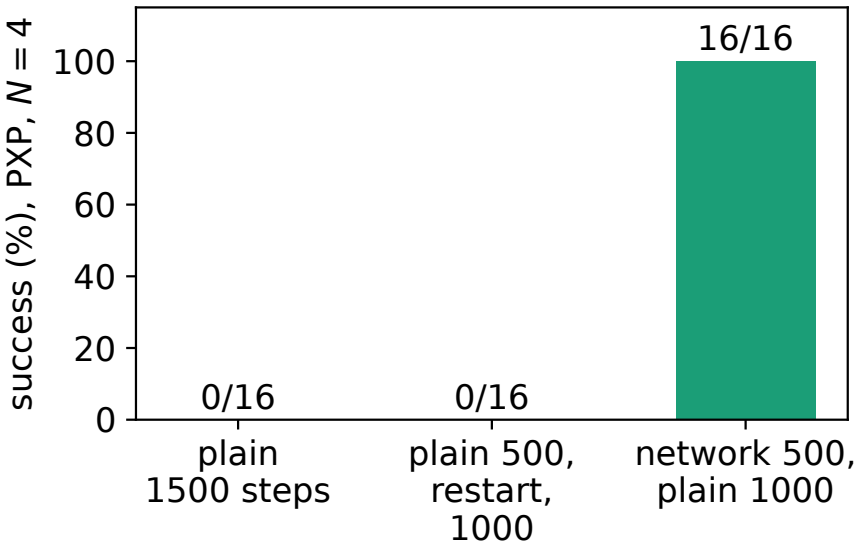


Figure 58: Our simulation. Success of the fit of the PXP model with four atoms in 16 random problems for the plain model, the control without network and the warm-up with the network followed by the plain model.

5.9 A term missing from the model and extrapolation

A real Rydberg chain also has a weaker interaction between next-nearest neighbours, which the PXP model (5.12) leaves out. We simulate the device with

$$\hat{\mathcal{H}}_{\text{true}} = \hat{\mathcal{H}}_{\text{PXP}}(\boldsymbol{\Omega}) + V_2 \sum_{i=0}^{N-3} \hat{n}_i \hat{n}_{i+2}, \qquad \hat{n} = |1\rangle\langle 1|, \tag{5.15}$$

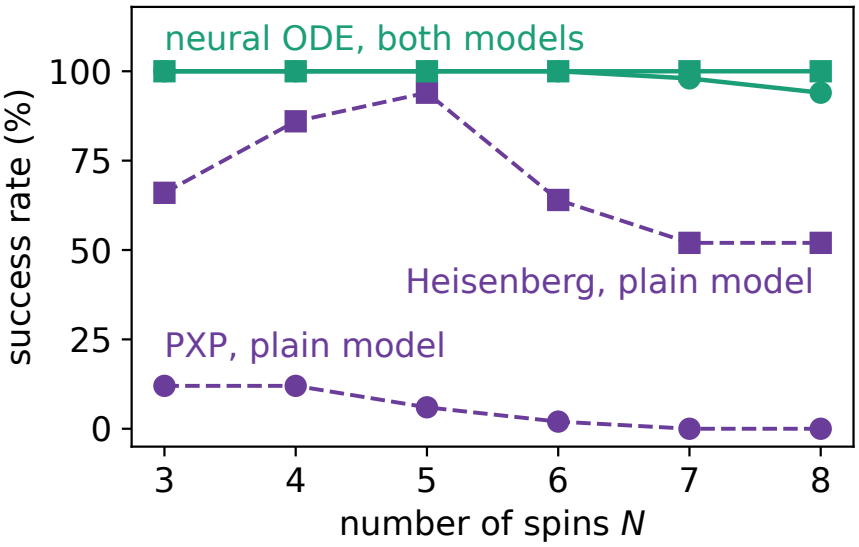


Figure 59: Published result of Ref. [50] (preprint version, Table 1), not our simulation. Success rates of the plain fit (grey) and of the neural differential equation (green) for the PXP model and a Heisenberg chain in a field against the number of spins.

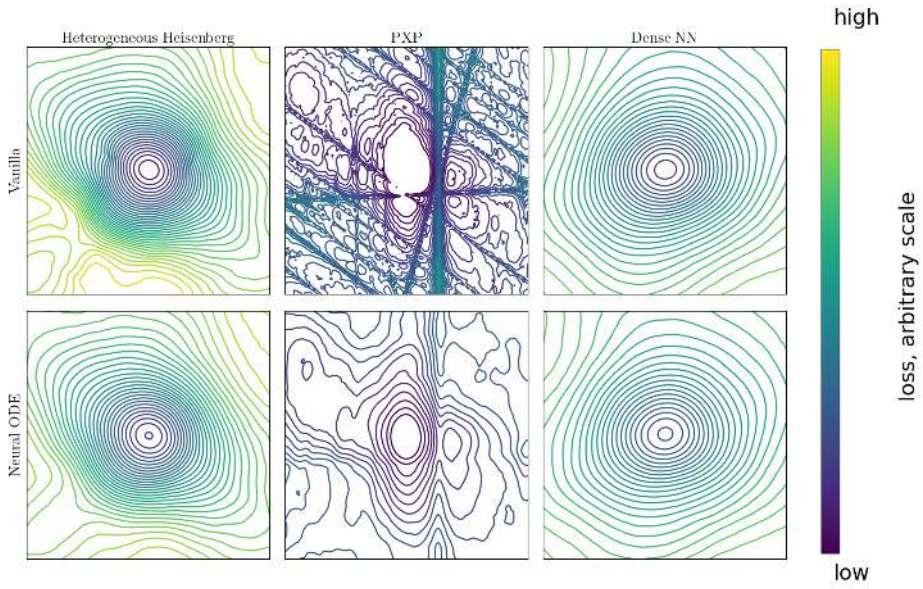


Figure 60: Published result of Ref. [50] (preprint version, Fig. 4), not our simulation. Loss in a plane spanned by two random directions through the global minimum (centre) for an anisotropic Heisenberg chain with site-dependent couplings, the PXP model and a model with all two-body nearest-neighbour terms (columns). Top row, the plain fit. Bottom row, the neural differential equation (5.13). Contours run from low loss (dark) to high loss (light), and the colour scale carries no values in the source.

with $V_2 = 0.4$ and four atoms, and fit the PXP model with the network of (5.13). The training data cover times up to $t_{\max} = 1$, and the learned models are tested on fresh bit strings up to $2t_{\max}$. The three phases have 500, 800 and 1500 Adam steps, and of four random starts we keep the one whose plain model reaches the lowest loss after the second phase. The fitted Rabi frequencies have a relative error of 0.013 although the model lacks the V_2 term. During training (Figure 61) switching the network off raises the loss above that of the true Hamiltonian, which only the simulation knows. The third phase lowers it slightly below that reference, so the network also fits part of the shot noise of the training record. The mean size of the correction along the trajectories, 0.23 compared with Rabi frequencies of order one, does not by itself reveal the missing term. The same curriculum on a record of the complete model, without the V_2 term, gives a correction of similar size, 0.19 against 0.23. The likelihood of fresh data separates the two cases in our single pair of records, since the network

lowers the test NLL when the term is missing and raises it when the model is complete.

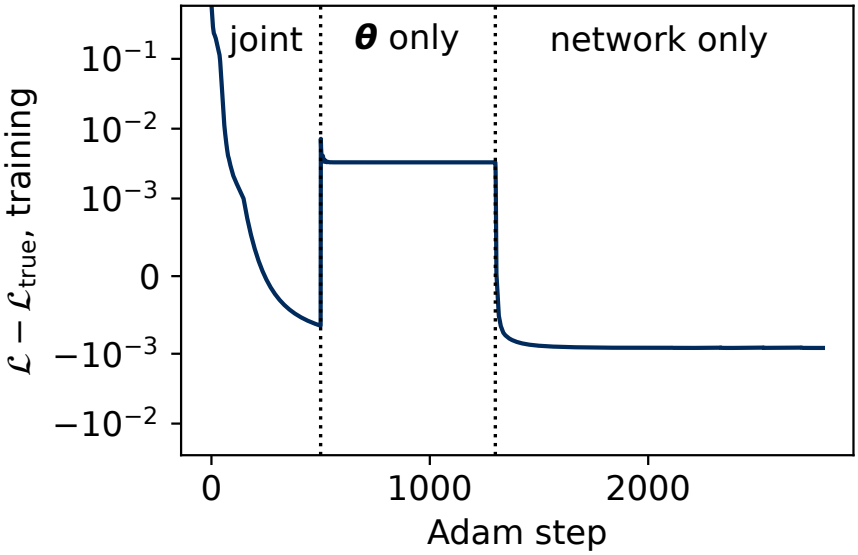


Figure 61: Training loss of the neural differential equation for the PXP model with the missing term (5.15), relative to the loss of the true Hamiltonian, through the three phases of the curriculum (dotted lines), our simulation. The vertical axis is linear near zero and logarithmic beyond.

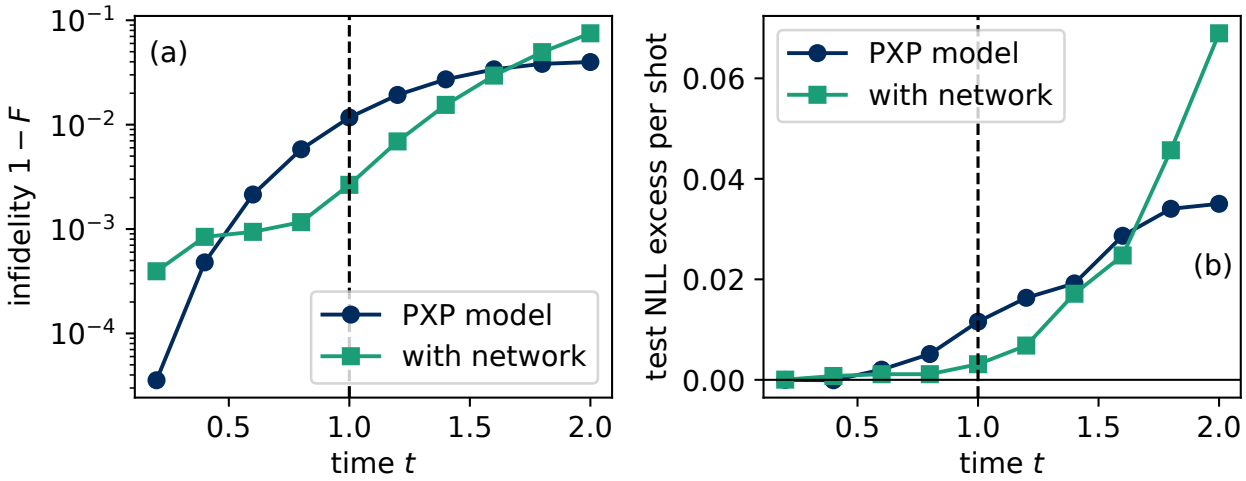


Figure 62: Test of the learned models on fresh simulated data up to $2t_{\max}$, with $t_{\max}$ dashed. (a) Infidelity with the state of the device (5.15), averaged over the three probes, computable only in simulation. (b) Excess of the test NLL per shot over that of the true Hamiltonian, computable from bit strings. Circles, the fitted PXP model. Squares, the same model with the trained network.

Between about $0.6\,t_{\max}$ and $1.6\,t_{\max}$ the network improves the model, and at $t_{\max}$ it lowers the infidelity with the state of the device from 1.2×10^{-2} to 2.7×10^{-3} (Figure 62). At the shortest times and beyond $1.6\,t_{\max}$ it is worse, and at $2t_{\max}$ its infidelity, 7.5×10^{-2}, is about twice that of the plain model, 4.0×10^{-2}. The test NLL, which needs only bit strings, shows the same crossing. The learned correction is accurate along the training trajectories and harmful beyond them, whereas the error of the plain model, whose dynamics is generated by a Hamiltonian, grows smoothly. The network therefore detects and absorbs a missing term within the time window of the data, and extrapolation requires the missing term itself, for instance V_2 added as a further coefficient of the assumed model.

5.10 Learning decay and dephasing rates

A real qubit also exchanges energy and phase with its environment, and its density matrix obeys the Lindblad equation [70,71]

$$\frac{d\hat{\rho}}{dt} = -i[\hat{\mathcal{H}}, \hat{\rho}] + \sum_{\mu} \Big(\hat{L}_{\mu} \hat{\rho} \hat{L}_{\mu}^{\dagger} - \frac{1}{2} \{ \hat{L}_{\mu}^{\dagger} \hat{L}_{\mu}, \hat{\rho} \} \Big), \tag{5.16}$$

where the jump operators $\hat{L}_{\mu}$ describe the processes caused by the environment and $\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$. We take one qubit with two processes. Energy decay has the jump operator $\hat{L}_1 = \sqrt{\Gamma}\, \hat{S}^{+}$ with $\hat{S}^{+} = |0\rangle\langle 1|$, which relaxes the qubit to the ground state $|g\rangle = |0\rangle$ of the bare qubit. Pure dephasing has $\hat{L}_2 = \sqrt{\gamma/2}\, \hat{Z}$, which randomises the relative phase of $|0\rangle$ and $|1\rangle$ without changing the populations. Without a Hamiltonian, the matrix elements of (5.16) follow from $\hat{S}^{+\dagger}\hat{S}^{+} = |1\rangle\langle 1|$ and $\hat{Z}\hat{\rho}\hat{Z} - \hat{\rho}$, whose diagonal vanishes and whose off-diagonal entries are $-2\rho_{01}$,

$$\frac{d\rho_{11}}{dt} = -\Gamma \rho_{11}, \qquad \frac{d\rho_{01}}{dt} = -\Big(\frac{\Gamma}{2} + \gamma \Big) \rho_{01}. \tag{5.17}$$

The population of $|1\rangle$ thus relaxes with the time $T_1 = 1/\Gamma$, and the coherence decays with $1/T_2 = \Gamma/2 + \gamma$, where the decay contributes half its rate because it removes amplitude from only one of the two states. The qubit is also driven by the known Hamiltonian $\hat{\mathcal{H}} = (\Omega/2)\hat{X}$ with $\Omega = 1$, which rotates populations into coherences, so that both rates affect both kinds of data. The unknowns are Γ and γ, with true values 0.3 and 0.2.

Equation (5.16) is linear in $\hat{\rho}$. Writing the 2×2 matrix as the vector $\mathrm{vec}(\hat{\rho})$ of its stacked rows and using $\mathrm{vec}(\hat{A}\hat{\rho}\hat{B}) = (\hat{A} \otimes \hat{B}^{\mathsf{T}})\, \mathrm{vec}(\hat{\rho})$ turns it into $d\,\mathrm{vec}(\hat{\rho})/dt = \mathbb{L}\, \mathrm{vec}(\hat{\rho})$ with the 4×4 Liouvillian

$$\mathbb{L} = -i(\hat{\mathcal{H}} \otimes \hat{I} - \hat{I} \otimes \hat{\mathcal{H}}^{\mathsf{T}}) + \sum_{\mu} \Big(\hat{L}_{\mu} \otimes \hat{L}_{\mu}^{*} - \frac{1}{2} \hat{L}_{\mu}^{\dagger} \hat{L}_{\mu} \otimes \hat{I} - \frac{1}{2} \hat{I} \otimes (\hat{L}_{\mu}^{\dagger} \hat{L}_{\mu})^{\mathsf{T}} \Big). \tag{5.18}$$

The state at time t is $e^{\mathbb{L}t}\mathrm{vec}(\hat{\rho}_0)$, computed with (5.7) and checked against a Runge–Kutta integration of (5.16). The probes are $|1\rangle$ and $|+\rangle$, the bases Z and X, the times $t_n = 0.5\, n$ with $n = 1, \ldots, 8$, and the probability of outcome b is $\langle b|\hat{U}_g \hat{\rho}(t_n) \hat{U}_g^{\dagger}|b\rangle$. The rates must be positive, and the model writes them through the softplus function,

$$\Gamma = \ln\big(1 + e^{a_\Gamma}\big), \qquad \gamma = \ln\big(1 + e^{a_\gamma}\big), \tag{5.19}$$

so that Adam updates the unconstrained numbers a_Γ and a_γ. The loss is again the NLL (5.6) (Figure 63).

From one record with $M = 1000$ shots per configuration the fit gives $\tilde{\Gamma} = 0.297$ and $\tilde{\gamma} = 0.199$, within one standard error of the true 0.3 and 0.2 (Figure 64). The probe $|1\rangle$ relaxes towards the ground state while the drive rotates it, so $\langle \hat{Z} \rangle$ oscillates and approaches a steady value, and the coherence of the probe $|+\rangle$ decays. We judge the fit successful when the estimates agree with the true rates within their statistical errors and when these errors reach the Cramér–Rao bound, both checks that only a simulation can make. Over 20 records per budget the standard deviations of both rates fall as $M^{-1/2}$ along their Cramér–Rao bounds, with a mean ratio 1.03 from $M = 300$ on. Only the forward model changed, as it now propagates a density matrix with the Liouvillian (5.18), and the likelihood, the gradient and the statistics of the estimate stay the same.

6 Summary of the lecture

1 What bit strings can teach

A device returns bit strings, and the Born rule (1.2) connects their frequencies to the density matrix. One expectation value needs a number of shots proportional to $1/\varepsilon^2$ for an accuracy ε. The full state, many properties at once and the Hamiltonian are increasingly ambitious questions answered from the same kind of data (Figure 2).

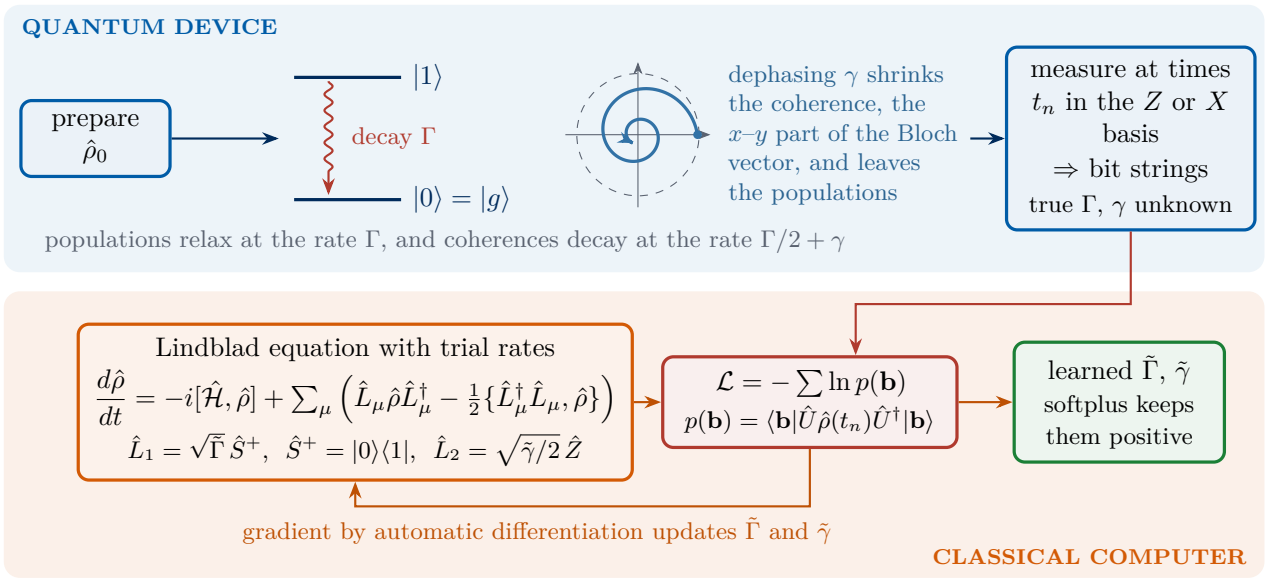


Figure 63: Learning two rates of an open qubit. Probes evolve under the Lindblad equation (5.16) with unknown Γ and γ, bit strings in the Z and X bases feed the same negative log-likelihood, and automatic differentiation through the evolution updates the rates, kept positive by (5.19).

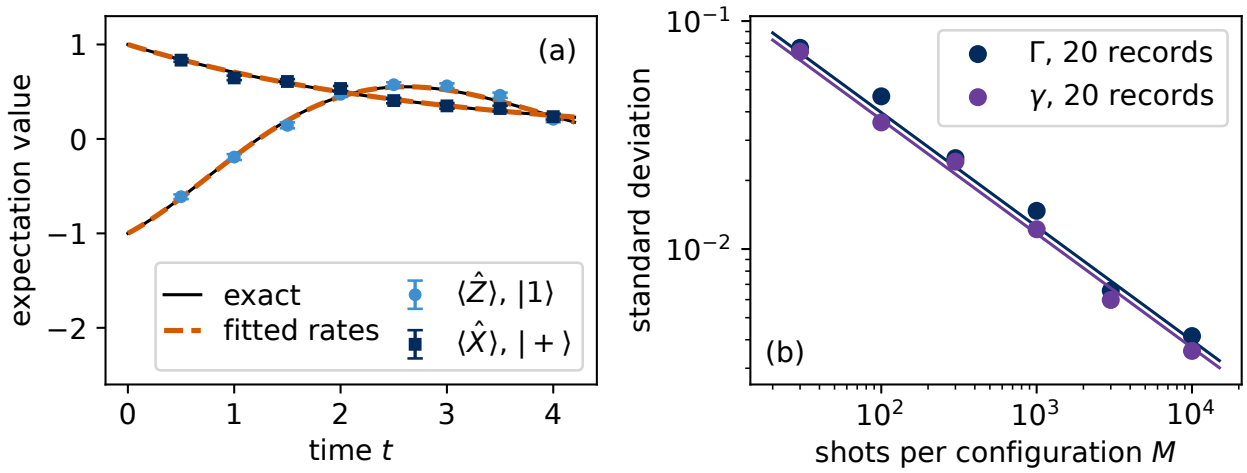


Figure 64: Learning the decay rate Γ and the dephasing rate γ of a driven qubit from simulated bit strings. (a) $\langle \hat{Z} \rangle$ of the probe $|1\rangle$ and $\langle \hat{X} \rangle$ of the probe $|+\rangle$, exact (solid), estimated from 1000 shots (markers) and from the model with the fitted rates (dashed). (b) Standard deviation of $\tilde{\Gamma}$ and $\tilde{\gamma}$ over 20 records against the number of shots per configuration, with the Cramér–Rao bounds as lines.

2 Quantum state tomography

A state of N qubits has $4^N - 1$ real parameters, the Pauli expectation values of (2.6), and the 3^N Pauli settings measure all of them. Linear inversion (2.11) solves the Born rule for these parameters and is unbiased, with an error proportional to $M^{-1/2}$, but for a nearly pure state it is unphysical in almost every data set. The projection (2.18) repairs the spectrum, and maximum likelihood (2.20) searches for the most probable physical state, by the $R\rho R$ iteration (2.23) or by gradient descent on a Cholesky factor (2.24). The squared error of linear inversion (2.30) grows by a factor of about five per qubit at a fixed total number of shots, which together with the 3^N settings limits full tomography to a few qubits.

3 Classical shadows

Measuring every qubit in a random Pauli basis and inverting the average measurement channel (3.1) turns every record into a snapshot (3.4) whose mean is the state. A Pauli string of weight k has the

single-run variance $3^k - \langle \hat{P} \rangle^2$ (3.7), independent of N, so one data set predicts many local observables with the sample bound (3.10). Strings on all qubits still need 3^N runs, and for two qubits shadows offer no advantage over tomography. Their advantage lies in local properties of many qubits.

4 Neural networks for tomography

A pure state is an extreme point of the set of states, so physical estimates from few shots are biased towards mixed states. A network trained on simulated pairs of an estimate and its true state approximates the conditional mean (4.1) and corrects this bias as far as the training states match the states of the experiment. The Bures distance is the smallest Euclidean distance over all pairs of amplitudes (4.3), and the Cholesky factors are one fixed pair, so the squared error of Cholesky vectors bounds the squared Bures distance (4.14) at the cost of one sum. Attention compares its input vectors with a trained bilinear form $g = W_{\mathrm{L}}^{\mathsf{T}} W_{\mathrm{R}}$ (4.17), turns the scores into Boltzmann weights (4.18) and averages the content of all vectors with them (4.19). The training data fix g, and the present input fixes the weights. In our two-qubit simulation the networks improve on the projection for pure states and on maximum likelihood from 100 shots per setting on, mixed states gain little, and an MLP performs as well as attention. For four qubits a network trained only on Haar-random states improves on the projection for one-axis-twisted states and matches maximum likelihood. It recovers much of their quantum Fisher information (4.30), which indicates their entanglement without certifying it. The published results of Ref. [1] show the same trends and an advantage of attention over convolutions alone.

5 Hamiltonian learning

For a Hamiltonian of known form $\hat{\mathcal{H}}(\boldsymbol{\theta}) = \sum_j \theta_j \hat{P}_j$, a differentiable model of the experiment gives the probability of every bit string, and gradient descent on the negative log-likelihood (5.6) fits the coefficients. The design of the experiment limits what can be learned, since real probes measured in real bases cannot distinguish $\hat{\mathcal{H}}$ from $-\hat{\mathcal{H}}^*$ (5.5), which Y-basis data or a complex probe resolve. The error falls as $M^{-1/2}$ and reaches the Cramér–Rao bound of the Fisher matrix (5.10). The loss has local minima, and single starts of the plain fit often fail. A neural differential equation (5.13) adds a network to the Schrödinger equation during a warm-up, after which the network is switched off, so the reported coefficients belong to a Hamiltonian. In our PXP problems this warm-up succeeds in 16 of 16 cases, against 0 for the same schedule without the network. The same likelihood learns the decay and dephasing rates of an open qubit with a Lindblad forward model (5.16).

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